

Endpoint pointwise convergence for the Schrödinger equation in higher dimensions

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Abstract

We resolve the Sobolev endpoint of Carleson’s pointwise convergence problem in every dimension $n \geq 3$. For initial data in $H^{n/(2(n+1))}(\mathbb{R}^n)$, the free Schrödinger evolution converges almost everywhere to the initial data as $t \downarrow 0$. The evolution is defined by removing Gaussian regularization on one full-measure spatial set, uniformly over the interval $0 < t < 1$.

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1 Introduction

The pointwise convergence problem for the free Schrödinger equation asks for the Sobolev regularity that guarantees convergence to the initial data almost everywhere. We establish the critical Sobolev case in every spatial dimension at least three. The representative of the evolution is specified by Gaussian regularization; its exceptional spatial set is independent of time.

For $n \geq 3$, put

$$s_n = \frac{n}{2(n+1)}.$$

We use the Fourier transform $\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x) \, dx$, extended from integrable functions to L^2 , and the inhomogeneous norm

$$\|f\|_{H^s}^2 = (2\pi)^{-n} \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 \, d\xi.$$

For $a > 0$, let

$$U_a f(x, t) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \xi - it|\xi|^2 - a|\xi|^2} \widehat{f}(\xi) \, d\xi. \quad (1.1)$$

The integral is absolutely convergent for $f \in L^2$, by Cauchy–Schwarz. At a Lebesgue point of f , write

$$f^*(x) = \lim_{r \downarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) \, dy.$$

Theorem 1.1 (Endpoint convergence). *For every fixed integer $n \geq 3$ and every complex-valued $f \in H^{s_n}(\mathbb{R}^n)$, there is a full-measure set E_f of Lebesgue points of f and a function $v(x, t)$, $x \in E_f$, $0 \leq t < 1$, such that $v(x, \cdot)$ is continuous, $v(x, 0) = f^*(x)$, and*

$$\lim_{a \downarrow 0} \sup_{0 < t < 1} |U_a f(x, t) - v(x, t)| = 0 \quad (x \in E_f). \quad (1.2)$$

In particular the finite Gaussian limit

$$u_f(x, t) = \lim_{a \downarrow 0} U_a f(x, t)$$

exists for every real $0 < t < 1$ on this one set, and $\lim_{t \downarrow 0} u_f(x, t) = f^(x)$.*

The Gaussian limit is taken first. No Fourier support, radially, logarithmic strengthening of H^{s_n} , or additional regularity is assumed. The theorem concerns all real times approaching zero, not a chosen sequence.

1.1 History and significance

Carleson proved the one-dimensional convergence theorem for $H^{1/4}(\mathbb{R})$, and Dahlberg and Kenig proved failure below that regularity [9, 10]. The one-dimensional threshold therefore includes its endpoint. Sjölin and Vega independently obtained convergence for $s > 1/2$ in higher dimensions [28, 32]. At $s > 1/2$, Sjögren and Sjölin also constructed representatives continuous on whole time lines outside one exceptional spatial set, using spatial or spacetime averages [27, Theorem 2]. In the plane, restriction and bilinear advances by Bourgain, Moyua–Vargas–Vega, Tao–Vargas and Tao [3, 23, 29, 31] preceded Lee’s $s > 3/8$ theorem [22]. In dimensions $n \geq 3$, Bourgain obtained the sufficient range $s > 1/2 - 1/(4n)$ [5].

Bourgain’s later counterexample gives the necessary condition $s \geq n/(2(n+1))$ in dimensions $n \geq 2$ [6, Proposition 1]. That obstruction does not assert failure at equality. Du, Guth and Li proved convergence for $s > 1/3$ in dimension two [13, Theorems 1.1–1.2]. Du, Guth, Li and Zhang then proved higher-dimensional convergence for $s > (n+1)/(2(n+2))$ and multilinear refined Strichartz estimates [14, Theorems 1.1 and 4.2]. Du and Zhang proved convergence for $s > n/(2(n+1))$ in higher dimensions [12, Theorem 1.1]. Recent work improves higher-dimensional maximal bounds in other L^p ranges [11, Theorems 1.1–1.2]; a Bessel-potential extension still assumes $s > n/(2(n+1))$ at $p = 2$ [25, Theorems 1.3–1.4].

Theorem 1.1 resolves the critical Sobolev case of Carleson’s pointwise convergence problem in higher dimensions. The distinction between equality and a strict Sobolev inequality is decisive: an arbitrarily small uncompensated frequency loss does not prove the endpoint theorem. We use the planar companion, *Endpoint convergence for the planar Schrödinger equation* [24, Sections 3 and 9], specifically its *Limiting information calculus* and *Block comparison and telescoping costs* lemmas and its time-tree mechanism. Its planar conclusion is not invoked as a higher-dimensional result. The dimension-dependent frame, projection and curvature arguments needed here are established below, with the hypotheses of the reusable lemmas stated at their points of use.

The analytic background includes positive-curvature decoupling [7], multilinear refined Strichartz estimates [14, Theorem 4.2], and the refined-packet line of Guth–Iosevich–Ou–Wang and Du–Iosevich–Ou–Wang–Zhang [15, 20]. Singh and Singh Parmar identified a support-versus-counting collar

mismatch in the normalized printed arbitrary-packet formulation and proved a buffered correction in ambient dimension two [26, Theorems 1.1–1.2 and Appendix A]. Theorem 2.5 establishes the needed buffered activity-count estimate in every ambient dimension used below, following that induction structure and using decoupling; neither the printed arbitrary-packet estimate nor the planar correction is imported as an all-dimensional input. Other inputs are multilinear Keakeya and restriction [2, 16, 30], discretized projections [21], and fractal Schrödinger estimates [12]. The multilinear restriction estimate is used away from its endpoint, and decoupling losses are removed through strict induction gains; an endpoint estimate without such losses is not assumed.

1.2 The new estimates and the proof

There are three principal steps.

A transverse frame estimate. In k spatial dimensions, Theorem 3.1 counts unit cubes where $k + 1$ transverse packet fields are simultaneously large relative to their square envelopes, allowing a separate witness for each field. Its constants depend polynomially on inverse transversality and on the fixed geometric parameters. The estimate applies to compact packet profiles and is therefore stable under the later geometric reductions. Its proof combines weighted packet inequalities with a finite hierarchy of amplitudes and occupancies. A positive smoothing argument handles gaps that are subpower in the original scale, while a robust projection theorem forces the limiting amplitude-ratio profiles. The explicit restricted-path construction keeps the probability laws compatible with cancellations and direction-dependent deletions.

A geometric induction in full dimension. The fractal estimates and the full-dimensional transverse saving first yield a sparse gain. An induction on packet scale and sparsity then reduces a large level set to a polynomial wall. Codimension-two angular narrowing alone does not exclude a curved ruled wall. We also prove a quadratic-sublevel narrow estimate and use it to detect the second fundamental form. Coefficient-uniform graph estimates and paths inside the small-Hessian set convert the relevant wall pieces into affine plates. Flattening along these plates makes Theorem 3.1 applicable in dimension $n - 1$. Two energy budgets, one per time bin and one global with density weights, ensure that summing the geometric pieces loses no growing factor.

A common time tree. The induction produces a weak L^p maximal bound for some $p > 2$ on time intervals of length $1/N$ at frequency N , with precisely the factor N^{s_n} . Write $S(t)f = e^{it\Delta}f$ for the L^2 evolution with multiplier $e^{-it|\xi|^2}$. Using one measurable time graph for all frequencies, the time-tree argument gives frequency square summation and the local bound

$$\int_{\Omega} \sup_{0 \leq t < 1} |S(t)f(x)| \, dx \lesssim \|f\|_{H^{s_n}}, \quad (1.3)$$

initially for compact Fourier support, uniformly on translated unit balls Ω . Uniform-in-time approximation then produces a common representative. Applying the Gaussian approximate identity to countably many error envelopes proves (1.2) without an uncountable intersection of exceptional sets. This also proves Theorem 1.1; the details appear in Section 8.3.

The packet estimates lead to the frame theorem, Theorem 3.1; separately, the full-dimensional fractal and transverse estimates, including Theorem 5.1, lead to the sparse estimate, Theorem 6.1. These two inputs meet in Theorem 7.1: the wall geometry and flattening developed there make the frame theorem applicable with $k = n - 1$. The final section proves the time-tree and Gaussian

representative assertions. All constants may depend on the fixed dimension. Whenever a sufficiently large finite profile order is required, it is chosen once for the relevant array class; no induction increases that order.

2 Packet estimates and a near-Cauchy bound

We first specify the packet class and the estimates that will be used through all subsequent decompositions. In particular, the auxiliary square weights in the estimates below need not be the squares of the packet amplitudes. Keeping this distinction is necessary when some branches of a decomposition have been removed.

2.1 Conventions and external estimates

We write $\langle z \rangle = (1 + |z|^2)^{1/2}$. Fix an integer $k \geq 1$, and write

$$q_k = \frac{2(k+2)}{k}, \quad p_k = \frac{2(k+1)}{k}. \quad (2.1)$$

All frequency patches lie in a fixed bounded set in $\mathbb{R}^k$. The phase is $x \cdot v - t|v|^2$, so its normal vector is $(2v, 1)$. Frequency sets $U_1, \dots, U_{k+1}$ are δ -transverse if

$$|\det((2v_1, 1), \dots, (2v_{k+1}, 1))| \geq \delta \quad (v_j \in U_j). \quad (2.2)$$

Changing the sign of time changes the sign of the paraboloid and changes none of the estimates. A positive definite quadratic form with fixed upper and lower eigenvalue bounds is handled by a fixed linear change of variables. No estimate for an indefinite quadratic form is being invoked.

Convention 2.1 (Packets and counting). A packet of scale $N \geq 1$ has a frequency label v in a grid of mesh N^{-1} and a unit lattice cell T in the coordinates

$$X_{N,v}(x, t) = \left(\frac{x - 2vt}{N}, \frac{t}{N^2} \right). \quad (2.3)$$

The center of its physical cell is denoted by z_T , and $d_T(z)$ is the distance from $X_{N,v}(z)$ to its native cell. Thus $|T| = N^{k+2}$. Choose fixed, sufficiently large exponents P and $P' > P$, and put $w_T(z) = (1 + d_T(z))^{-P}$. A band-limited packet of amplitude s_T is

$$f_T(x, t) = e^{i(x \cdot v - t|v|^2)} F_T(X_{N,v}(x, t)), \quad |F_T(X)| \leq s_T(1 + \text{dist}(X, T))^{-P'},$$

where the Fourier transform of F_T is supported in a fixed box. Alternatively, a smooth packet has a native profile supported in a fixed enlargement of its associated cell T , with a fixed sufficiently large number of derivatives bounded by s_T . The Fourier box, the profile bounds, and the exponents are uniform throughout a packet family.

The phase-space multiplicity is bounded as follows. On each slab of length N^2 containing packet centers, there are boundedly many packets per frequency slot and per spatial intercept cell of side N , after transporting their centers to the central time of that slab. The bound is relative to each slab; there is no restriction on the number of slabs. Frequencies may move within their slots and centers may be recentered. Moving all native centers by a fixed lattice displacement changes this multiplicity by at most a fixed polynomial in the displacement; such multiplicities can also be split into bounded-multiplicity layers. In particular,

$$\#\{T : d_T(z) \leq a\} \leq CN^k(1+a)^{C_k}, \quad a \geq 1. \quad (2.4)$$

The energy parameter of the family is

$$E = N^{k+2} \sum_T s_T^2. \quad (2.5)$$

It is a parameter of the decomposition, not an assertion that the packets are orthogonal in spacetime.

A unit-cube count means a count in a fixed lattice, allowing a bounded number of translates or bounded enlargements. There is one witness per counted cube; the square-envelope and field witnesses are specified explicitly when they need not coincide. Estimates are first proved for finite families and finite sets of test cubes. Absolutely convergent decompositions, or truncations with arbitrarily small error on those test cubes, then give the stated generality. Constants may depend on the fixed profile and multiplicity bounds, never on the size of the configuration.

We record the exact forms of two external geometric inputs. They will also be used after the packet estimates have been proved.

Proposition 2.2 (Weighted endpoint multilinear Kakeya). *Let $d \geq 2$. For $j = 1, \dots, d$, let $\mathcal{T}_j$ be a family of infinite cylinders of radius h in $\mathbb{R}^d$, with unit direction vectors $e(T)$. Suppose*

$$|\det(e(T_1), \dots, e(T_d))| \geq \vartheta > 0 \quad (T_j \in \mathcal{T}_j).$$

For nonnegative weights $c_j(T)$,

$$\int_{\mathbb{R}^d} \prod_{j=1}^d \left(\sum_{T \in \mathcal{T}_j} c_j(T) \mathbf{1}_T \right)^{1/(d-1)} \leq C_d \vartheta^{-1/(d-1)} h^d \prod_{j=1}^d \left(\sum_{T \in \mathcal{T}_j} c_j(T) \right)^{1/(d-1)}. \quad (2.6)$$

For lattice cubes of side $a \leq h$, the same right-hand side bounds a^d times the corresponding sum of cube suprema, after a fixed dimensional enlargement of every cylinder.

Proof. The unweighted unit-radius assertion with arbitrary transverse directions is Guth [16, Theorem 2]; the near-coordinate-axis multilinear formulation originates in Bennett et al. [2]. Integral weights are obtained by repeating cylinders. Dividing by a common denominator gives rational weights, and monotone approximation gives arbitrary nonnegative weights. Dilation gives h^d . Every side- a cube meeting a cylinder lies in its enlargement of radius $h + C_d a \lesssim h$. Integration over each cube, of volume a^d , proves the lattice version, also with separate suprema for the factors. For paraboloid normals in a bounded patch, normalization to unit vectors changes the determinant by fixed factors, so (2.2) gives an inverse power of δ . $\square$

For a set A , write $\mathcal{N}_\Delta(A)$ for its covering number by balls of radius Δ . This notation is interchangeable with the number of occupied Δ -cubes up to dimensional constants.

The projection input is He's higher-rank extension of Bourgain's discretized projection theorem [4, 21]. Its conclusion is simultaneous for every sufficiently large subset, so the successful position set may depend on the direction.

Proposition 2.3 (Robust hyperplane projection input). *Fix $d \geq 2$, $0 < \alpha < d$, and $\rho > 0$. There are positive constants η_0, γ and Δ_0 with the following consequence. If $0 < \eta \leq \eta_0$, $0 < \Delta < \Delta_0$, A lies in a fixed unit ball,*

$$\Delta^{-\alpha+\eta} \leq \mathcal{N}_\Delta(A) \leq \Delta^{-\alpha-\eta}, \quad \mathcal{N}_\Delta(A \cap B(x, r)) \leq \Delta^{-\eta} r^\rho \mathcal{N}_\Delta(A) \quad (\Delta \leq r \leq 1),$$

and ν is a probability law on unit kernel directions satisfying

$$\nu\{v : |v \cdot w| < r\} \leq \Delta^{-\eta} r^\rho \quad (|w| = 1, \Delta \leq r \leq 1), \quad (2.7)$$

then some direction in the support of ν has

$$\mathcal{N}_\Delta(\pi_{v^\perp} A') \geq \Delta^{-(d-1)\alpha/d-\gamma} \quad \text{for every } A' \subset A \text{ with } \mathcal{N}_\Delta(A') \geq \Delta^\eta \mathcal{N}_\Delta(A). \quad (2.8)$$

Fixed multiplicative constants and bounded affine normalizations in the hypotheses are allowed after reducing η_0, γ .

Proof. This is He [21, Theorem 1], with $m = d - 1$, after reducing its error parameter and, if necessary, replacing ρ by a smaller positive number. Here is the match to its Grassmannian hypothesis. The competing subspaces have dimension $d - m = 1$; if one is spanned by a unit vector w , its Schubert angle to the projection plane $v^\perp$ is $|v \cdot w|$. Thus (2.7) is exactly the required nonconcentration near Schubert cycles. The theorem excludes a direction set of small ν -mass and is simultaneous for all sufficiently large subsets of A . Its size and nonconcentration errors can be made larger than those displayed above, leaving a fixed positive projection gain γ . Shrinking this gain and the error tolerance absorbs the fixed comparison constants. Its theorem tolerance is chosen larger than η : our lower size bound then implies its required lower bound, and our subset threshold is stronger than its subset threshold. No choice of a new size exponent is necessary. $\square$

2.2 Localization with fixed original regularity

Lemma 2.4 (Reproduction, tails, and unit suprema). *For the band-limited class in Theorem 2.1, the following operations are uniform in the size of the packet family.*

1. *A polynomially decaying packet is an absolutely convergent sum of packets with the same fixed enlarged native Fourier support and arbitrarily rapid Schwartz decay about translated native centers. The coefficient at a translation $j \in \mathbb{Z}^{k+1}$ is at most $C_{A,S_T}(1 + |j|)^{-P'}$, where A is the requested output decay order.*
2. *All fixed native derivative orders satisfy the corresponding envelope bounds. If F has Fourier support in a fixed bounded set and $r > 0$, then, for every sufficiently large fixed L ,*

$$\sup_{|u-z| \leq C} |F(u)|^r \leq C_{r,L} \int |F(y)|^r (1 + |y - z|)^{-L} dy. \quad (2.9)$$

Consequently integral packet estimates below also hold with unit-cube suprema, or with separate unit suprema for factors of a product. Output Schwartz orders may be chosen after a small loss parameter; the original exponent P' and original smooth-profile derivative order stay fixed.

Proof. Choose a smooth partition $\sum_j \psi_j = 1$ on native space, where ψ_j is supported within fixed distance of j . Choose a compactly supported smooth Fourier multiplier equal to one on the original Fourier box, and let K be its Schwartz inverse Fourier transform. Then

$$F_T = \sum_j K * (\psi_j F_T), \quad |\partial^\beta K * (\psi_j F_T)(X)| \leq C_{A,\beta} s_T (1 + |j|)^{-P'} (1 + |X - j|)^{-A}.$$

These inequalities follow by integrating the Schwartz bounds of K over the fixed support of ψ_j ; the decay at j comes from the original pointwise bound. They prove absolute convergence and every asserted output derivative bound. Convolution without the partition also proves fixed derivative bounds for the original packet.

For completeness, the maximal inequality is valid also for $r < 1$. Let $M = \sup_y |F(y)|(1 + |y - z|)^{-a}$, choosing a so large that $ar > L + k + 1$, and choose a point y_0 realizing at least $M/2$.

Reproduction and its derivative version give $|\nabla F(y)| \leq CM(1 + |y - z|)^a$. On a fixed sufficiently small ball about y_0 , this and the choice of y_0 imply $|F(y)| \geq cM(1 + |y_0 - z|)^a$. Integrating on that ball proves the weighted maximal estimate with weight $(1 + |y - z|)^{-ar}$, and hence (2.9) after choosing ar large enough. Truncation of the weighted supremum justifies the argument when its finiteness has not yet been established.

Translating a native core by j multiplies its weight comparisons and phase-space multiplicity by a fixed polynomial in $1 + |j|$. For example, $w_{T+j}(z) \leq C(1 + |j|)^P w_T(z)$. The sum over j therefore converges once P' exceeds those fixed powers and $k + 1$. Crucially, A occurs in the constant C_A , not in that required original power P' . For packet estimates use (2.9) and the polynomial comparison of envelopes at translated points. For products apply it separately to each factor and use Tonelli. The exponent in each convolution is fixed after the product exponents have been fixed. $\square$

2.3 Weighted decoupling

The refined estimate needed here counts where a packet can actually be active. In the arbitrary-packet formulations printed in Guth et al. [20, Theorem 4.2] and Du et al. [15, Theorem 4.2], the packet may occupy $2T$ while the count is made on T . The collar counterexample of Singh and Singh Parmar [26, Theorem 1.1] rules out that literal formulation in ambient dimension two. Their Appendix A gives a buffered proof in that dimension. We give the corresponding calculation in dimension $k + 1$, with a strict support-to-count margin and a common longitudinal slab. Its only analytic input is the positive-definite ℓ^2 decoupling theorem of Bourgain and Demeter [7].

Lemma 2.5 (Buffered refined decoupling). *Put $d = k + 1$, $R \geq 2$, and $2 \leq p \leq q_k$. For every $\varepsilon > 0$ there is $\vartheta_0 = \vartheta_0(d, p, \varepsilon) > 0$ such that the following holds for $0 < \vartheta \leq \vartheta_0$ and $\Lambda = R^\vartheta$, with a constant allowed to depend on $(d, p, \varepsilon, \vartheta)$. Work in a fixed $O(R)$ test box and one common padded time slab S of length $O(R)$. Every tube core meets a fixed $O(R)$ enlargement of the test box; in particular the family has polynomial cardinality. Fine frequency labels lie in a bounded patch of a translated, bounded-overlap lattice of mesh $R^{-1/2}$; the allowed class of such lattices is closed under parabolic rescaling. A fixed $O_d(1)$ number of cap grids may be separated into colors. A labelled tube is the intersection with S of*

$$\{(x, t) : |x - 2vt - u|_\infty \leq \Lambda R^{1/2}\}.$$

For $c \geq 1$, cT dilates the transverse width only, in the same common slab. Tubes of a fixed label have bounded overlap before clipping by a spatial box; equivalently, their intercepts are globally separated at scale $\Lambda R^{1/2}$ up to fixed multiplicity. Fix $1 \leq a \leq 10$ and $a + 1/2 \leq b \leq 20$. Suppose the nonzero F_T have comparable L^p norms,

$$\text{supp } \widehat{F}_T \subset \{(\xi, \tau) : |\xi - v| \lesssim R^{-1/2}, |\tau + |\xi|^2| \lesssim \Lambda/R\}, \quad \|F_T\|_{L^p(\mathbb{R}^d \setminus aT)} \leq R^{-B} \|F_T\|_p,$$

where B is a sufficiently large fixed number. Let U be a union of $R^{1/2}$ -cubes in the interior of the box and slab, each meeting at most $M \geq 1$ of the counted tubes bT . Then

$$\left\| \sum_T F_T \right\|_{L^p(U)} \leq C_{\varepsilon, \vartheta, d, p} R^\varepsilon M^{1/2-1/p} \left(\sum_T \|F_T\|_p^p \right)^{1/p}. \quad (2.10)$$

The same assertion holds for finitely many fixed cap-grid colors. The spatial boxes and tube strips may be translated; all overlap assumptions refer to the unclipped strips.

Proof. We track the dimension-dependent bookkeeping in the induction of Singh and Singh Parmar [26, Appendix A]; no refined estimate is used as an input. Weighted local Bourgain–Demeter decoupling says that, for any fixed large L , $s \geq 1$, $2 \leq p \leq q_k$, and a cube Q_s of side s ,

$$\left\| \sum_J H_J \right\|_{L^p(Q_s)} \leq D_{\delta, L, d, p} s^\delta \left(\sum_J \|H_J\|_{L^p(w_{Q_s, L})}^2 \right)^{1/2}, \quad w_{Q_s, L}(z) = \left(1 + \frac{\text{dist}(z, Q_s)}{s} \right)^{-L}, \quad (2.11)$$

for $s^{-1/2}$ tangential caps with $O(s^{-1})$ normal thickness. The weighted version follows by summing the usual local decoupling estimate over translates of Q_s with rapidly decreasing coefficients. Its constant is uniform under the fixed positive-definite affine normalizations used below.

Let $\mathfrak{A}_d(R, \Lambda, a, b)$ be the least constant in (2.10) without R^ε , uniformly for the stated configurations but with the full domain $1 \leq a < b \leq 20$. The theorem starts with $b - a \geq 1/2$; smaller positive margins occur only inside the induction. Define it for every $\Lambda \geq 1$ using the trivial estimate when $\Lambda \gtrsim R^{1/4}$; it is not defined only on the curve $\Lambda = R^\vartheta$, since Λ stays fixed when R is square-rooted. For the one-step estimate assume $R^\vartheta \leq \Lambda \ll R^{1/4}$; this lower bound persists at each scale reached from the initial $\Lambda = R^\vartheta$. Group fine labels into coarse $R^{-1/4}$ -cubes J , with fixed finite coloring at cap boundaries. Translate the midpoint of S to $t = 0$, adjusting tube intercepts accordingly. If v_0 is the center of J , assign each fine tube to one parent box

$$\square_{J, u} = \{(x, t) \in S : |x - 2v_0t - u|_\infty \leq C_d R^{3/4}\}, \quad u \in c_d R^{3/4} \mathbb{Z}^k.$$

The entire supporting aT lies in its assigned parent box. This uses $|v - v_0| = O_d(R^{-1/4})$, the common $O(R)$ time slab, and $\Lambda R^{1/2} \ll R^{3/4}$. The boxes have bounded global overlap for each J , and packet families assigned to distinct boxes are disjoint. On a parent box, remove the carrier and set

$$x' = R^{-1/4}(x - 2v_0t - u), \quad t' = R^{-1/2}t, \quad v' = R^{1/4}(v - v_0), \quad R' = R^{1/2}.$$

Then $x - 2vt - u_T = R^{1/4}(x' - 2v't' - u'_T)$ and $|\det \Phi| = R^{-(k+2)/4}$. A fine cap and its Λ/R normal thickness become a cap of width $R'^{-1/2}$ and thickness Λ/R' ; an aT or bT becomes the corresponding dilate at radius R' with the same Λ . The transformed fine labels form an allowed translated lattice. The Jacobian occurs on both sides of the desired estimate and cancels. A lower-scale $R'^{1/2}$ -cube pulls back to a cell C of transverse width $O_d(R^{1/2})$ and longitudinal length $O(R^{3/4})$.

Apply (2.11) with $s = R^{1/2}$ on an original $R^{1/2}$ -cube Q . Choose its neighborhood radius $h = \Lambda^{1/4} R^{1/2}$ and take $L \geq C(B + d)/\vartheta$. Outside that neighborhood the decoupling weight is $O(R^{-B-d-10})$. The relative packet tails make boxes missing the neighborhood negligible too. There are at most $C_d \Lambda^{d/4}$ neighboring $R^{1/2}$ -cubes Q'' , only $O_d(1)$ parent boxes of a fixed J meet the neighborhood, and each Q'' meets only $O_d(1)$ rescaling cells C in such a box. Thus, writing $F_\square = \sum_{T \mapsto \square} F_T$,

$$\|F\|_{L^p(Q)} \leq C_d D R^{\delta/2} \left(\sum_{(\square, Q'', C)} \|F_\square\|_{L^p(Q'' \cap C)}^2 \right)^{1/2} + R^{-B/2} \mathcal{R}, \quad (2.12)$$

where $\mathcal{R}$ denotes the right side of the defining refined inequality without its constant. Finite truncation and the polynomial number of packets and cubes in the local box justify the displayed negligible error after increasing B .

Put $b' = b - C_d \Lambda^{-3/4}$. If $b'T$ meets a cell C that meets a neighbor Q'' of Q , compare the point of intersection with the center of Q in the adapted coordinates $y = x - 2v_0t - u$. The change in y is $O_d(\Lambda^{1/4} R^{1/2})$; the change in t is $O_d(R^{3/4} + \Lambda^{1/4} R^{1/2})$. Since $|v - v_0| = O_d(R^{-1/4})$, the

transverse residual for the fine tube changes by at most $C_d \Lambda^{1/4} R^{1/2}$. Dividing by its width $\Lambda R^{1/2}$ gives $C_d \Lambda^{-3/4}$. All strips use the same longitudinal slab, so this implies bT meets Q ; no individual tube endpoint is crossed.

For completeness, here is the counting step. Pigeonhole the cube norms, parent packet counts $W_\square$, lower-scale cell counts $m_{\square, b'}(C) \asymp M'$, and numbers $n(Q) \asymp n$ of retained triples. The polynomial configuration size makes the cost $(\log R)^{C_d}$. By (2.12) and $(\sum_{i=1}^n a_i^2)^{p/2} \leq n^{p/2-1} \sum_i a_i^p$, a retained class satisfies

$$\|F\|_{L^p(U)}^p \lesssim (\log R)^{C_d} (DR^{\delta/2})^p n^{p/2-1} \Lambda^{d/4} \sum_{\square} \|F_\square\|_{L^p(U_{\square, M'})}^p + \text{negligible}. \quad (2.13)$$

A fixed triple is associated with at most $C_d \Lambda^{d/4}$ original test cubes. Rescaling and applying $\mathfrak{A}_d(R', \Lambda, a, b')$ gives, since the original packet norms are comparable and the assigned families are disjoint,

$$\sum_{\square} \|F_\square\|_{L^p(U_{\square, M'})}^p \lesssim \mathfrak{A}_d(R', \Lambda, a, b')^p \left(\frac{M'}{W} \right)^{p/2-1} \left(\sum_T \|F_T\|_p^2 \right)^{p/2}, \quad W = \#\{T\}.$$

For a fixed Q , one parent box contributes at most $C_d \Lambda^{d/4}$ triples. Choose one retained triple from each of at least $c_d n \Lambda^{-d/4}$ distinct boxes. Each contributes $\asymp M'$ different fine packets whose bT meets Q by the preceding geometric transfer. Disjoint assignment to boxes therefore gives $nM' \lesssim \Lambda^{d/4} M$. Substitution into (2.13), followed by a p th root, proves

$$\mathfrak{A}_d(R, \Lambda, a, b) \lesssim D(\log R)^{C_d} R^{\delta/2} \Lambda^{d/8} \mathfrak{A}_d(R^{1/2}, \Lambda, a, b - C_d \Lambda^{-3/4}) + 1. \quad (2.14)$$

The two neighborhood counts give $\Lambda^{d/(4p)} \Lambda^{(d/4)(1/2-1/p)} = \Lambda^{d/8}$; there is no dimension-dependent power of R .

Stop at the first $R_j = R^{2^{-j}}$ with $\Lambda \gtrsim R_j^{1/4}$. Bounded *unclipped* overlap per label and $O_d(R_j^{k/2})$ labels give the direct estimate $\mathfrak{A}_d(R_j, \Lambda, a, b_j) \lesssim R_j^{k/4} \lesssim \Lambda^k$. There are $j = O(\log(1/\vartheta))$ steps. For R sufficiently large, their total count-dilation drift $C_d j \Lambda^{-3/4}$ is less than $1/4$, so the initial margin $b - a \geq 1/2$ remains positive. Iterating (2.14) yields

$$\mathfrak{A}_d(R, \Lambda, a, b) \lesssim_{d,p,\delta,\vartheta} (\log R)^{C_d j} R^{C_d \delta} \Lambda^{k+dj/8}.$$

Choose ϑ first so that $\vartheta O_d(\log(1/\vartheta)) < \varepsilon/3$, then δ so that the decoupling factors cost less than $R^{\varepsilon/3}$, and absorb the logarithms into the remaining loss. The weight order L and its fixed constant are chosen after ϑ ; no original packet decay order is being chosen as a function of ε . This proves (2.10). $\square$

Proposition 2.6 (Arbitrary auxiliary weights). *Let $2 < p \leq q_k$, and let f_T be band-limited packets of scale N . For $b_T \geq 0$, require $f_T = 0$ if $b_T = 0$, and set $B_b(z)^2 = \sum_T b_T^2 w_T(z)$. Then, for every $\varepsilon > 0$,*

$$\int B_b(z)^{2-p} \left| \sum_T f_T(z) \right|^p dz \leq C_\varepsilon N^\varepsilon N^{k+2} \sum_T s_T^p b_T^{2-p}. \quad (2.15)$$

The integrand is defined as zero where both the field and B_b vanish. The estimate admits unit suprema. If all frequency labels lie within C/N of a κ -dimensional affine plane, $1 \leq \kappa \leq k$, the range extends to $2 < p \leq q_\kappa$. In the zero-dimensional case there are only boundedly many frequency slots and the assertion holds for every fixed finite $p > 2$.

Proof. We derive the weighted conclusion from Theorem 2.5, applying that lemma to relative-tail atoms, not to the unsplit polynomial-tail f_T with a single original-tube count. Fix $\varepsilon > 0$, put $R = N^2$, and choose $\vartheta > 0$ small enough, after the fixed packet exponents, to pay every power of $\Lambda = R^\vartheta$ below. The original profile has fixed native Fourier support. Choose a reciprocal sampling lattice with fixed spacing γ in the k native spatial variables and spacing γ/Λ in native time. Smooth Fourier cutoffs equal to one on the original Fourier box and supported inside the reciprocal fundamental domains give, by Poisson summation, fixed Schwartz kernels K_x, K_t and the exact expansion

$$F_T(X) = \sum_{j \in \mathbb{Z}^k, m \in \mathbb{Z}} c_{T,j,m} K_x(X_x - u_{T,x} - \gamma j) K_t(\Lambda(X_t - u_{T,t}) - \gamma m), \quad (2.16)$$

where u_T is the native cell center and, with a fixed normalization,

$$c_{T,j,m} = C_\gamma F_T(u_{T,x} + \gamma j, u_{T,t} + \gamma m/\Lambda), \quad |c_{T,j,m}| \leq C s_T (1 + |j| + |m|/\Lambda)^{-P'}.$$

The temporal Fourier support of an atom is $O(\Lambda)$ in native coordinates. In physical coordinates the Fourier map is

$$\xi = v + \eta/N, \quad \tau = -|v|^2 - 2v \cdot \eta/N + \sigma/N^2, \quad \tau + |\xi|^2 = (\sigma + |\eta|^2)/N^2.$$

Thus an atom has tangential cap width $O(R^{-1/2})$ and normal thickness $O(\Lambda/R)$, exactly the range of Theorem 2.5. Its L^p norm is comparable to $|c_{T,j,m}| N^{(k+2)/p} \Lambda^{-1/p}$, with no lower bound on $|c_{T,j,m}|$ required. A fixed-kernel sampling in time would not give the required relative R^{-B} tail outside an R -long tube; the Λ^{-1} native-time compression in (2.16) does.

Fix one sample index (j, m) and write g_T for its atom. Let $d_{T,j,m}$ be native distance to its shifted center in the *original* coordinates $X_{N,v}$ of (2.3), and set

$$w_{T,j,m} = (1 + d_{T,j,m})^{-P}, \quad B_{b,j,m}^2 = \sum_T b_T^2 w_{T,j,m}.$$

Partition spacetime into test boxes with time cores of length R and bounded spatial overlap. For a core centered at t_0 , retain atoms whose sample time centers lie in $[t_0 - 2R, t_0 + 2R]$, and use the *common* support slab $S = [t_0 - 5R, t_0 + 5R]$ for all retained atoms. Test cubes lie in $[t_0 - R/2, t_0 + R/2]$. Thus retained centers have at least a $3R$ margin to ∂S , while omitted centers have at least a fixed R separation from the test cubes. The retained atom has R^{-B} relative time tail outside S ; the omitted atoms gain $C_A \Lambda^{-A}$ on the core, with a summable further factor in their time-slab distance. Retain spatially only trajectories that can meet a fixed padded box; the other atoms have an arbitrarily small Schwartz spatial tail. The phase-space count (2.4) sums both discarded classes: with the full shifted envelope kept in the denominator, three-factor Hölder as below and annular counting give an arbitrary Λ^{-A_1} or N^{-A_1} gain in their weighted L^p norms, respectively. This charge is made before applying the local lemma. A retained atom is concentrated, relative to its own norm, on the common-slab strip of transverse radius ΛN , with $C_A \Lambda^{-A}$ tail outside a fixed support dilate aT , say $a = 2$. Count a larger dilate, say bT with $b = 3$. Since A may be chosen after ϑ , this is R^{-B} for the B of Theorem 2.5. Each atom belongs to only $\Lambda^{O_d(1)}$ padded test boxes.

The actual v may move within its N^{-1} frequency slot. Relabel it by the slot center and use one of a fixed number of cap-grid colors. Over an $O(R)$ slab, the two trajectories differ by $O(N)$, which is $O(\Lambda^{-1})$ of the inflated transverse radius; the fixed $a = 2$, $b = 3$ collar absorbs this displacement. For this fixed sample index, the shifted phase-space multiplicity is bounded by a polynomial in $J = 1 + |j| + |m|/\Lambda$. Color intercepts on the *unclipped* strips into at most $C \Lambda^{C_d} J^{C_d}$ families of

bounded same-label overlap. There are only $O(1)$ relevant time slabs per test cube for each such family. In every local box there are polynomially many packets. Set $a_T = |c_{T,j,m}| b_T^{2/p-1}$ for $b_T > 0$ and $a_* = \max_T a_T$ in that box. Discard only the classes with $a_T < N^{-C_0} a_*$, where C_0 is chosen after the polynomial local packet count. This is a cutoff in the *weighted* amplitude, not in $|c_{T,j,m}|$. Indeed, writing $h_T = |g_T|/|c_{T,j,m}|$ and taking a Schwartz majorant $h_T \lesssim w_{T,j,m}$, three-factor Hölder gives

$$B_{b,j,m}^{2/p-1} \sum_{a_T < N^{-C_0} a_*} |g_T| \lesssim \left(\sum_{a_T < N^{-C_0} a_*} a_T^p h_T \right)^{1/p} \left(\sum_T h_T \right)^{1/2}.$$

The last sum and the number of local packets are polynomially bounded; $\int h_T \lesssim N^{k+2}/\Lambda$. Thus this discarded contribution is $N^{-C_0+O_d(1)} \Lambda^{O_d(1)} J^{O_d(1)}$ times $(N^{k+2}/\Lambda)^{1/p} (\sum_T a_T^p)^{1/p}$. The retained a_T occupy $O(\log N)$ dyadic classes. Within each $b_T \asymp \alpha$ class, comparability of a_T also makes the atom L^p norms comparable, so Theorem 2.5 applies. No comparison between the multiplicity of shifted atom tubes and the multiplicity of the original tubes is used.

To see the weighted reduction quantitatively, sort N -cubes by $B_{b,j,m} \asymp \beta$ and atoms by $b_T \asymp \alpha$, then by retained a_T . The slot-center relabeling above changes only the local counted strips; the shifted envelope here keeps v and therefore compares globally with B_b . No absolute cutoff on α or β is needed: the cube classes are disjoint, and for each T the moderate relative range below contains only $O(\log N)$ β -classes. On a cube meeting a counted bT , its atom center is within $O(\Lambda)$ in native space; hence the atom contributes at least $\alpha^2 \Lambda^{-P}$ to $B_{b,j,m}^2$. The count $M_{j,m}$ for that class obeys

$$M_{j,m} \lesssim \Lambda^P \beta^2 / \alpha^2, \quad \beta^{2-p} M_{j,m}^{p/2-1} \lesssim \Lambda^{P(p/2-1)} \alpha^{2-p}.$$

Use Theorem 2.5 on the moderate ratios $N^{-C} \leq \alpha/\beta \leq C\Lambda^{P/2}$ and sum their $O(\log N)$ relative ranges. For smaller ratios the identity

$$B_{b,j,m}^{2/p-1} |c_{T,j,m}| = |c_{T,j,m}| b_T^{2/p-1} (b_T/B_{b,j,m})^{1-2/p} \quad (2.17)$$

gives $N^{-C(1-2/p)}$; choose C to dominate the polynomial local packet count and use pointwise Hölder. For the complement of the counted activity strips, retain the *anisotropic* atom envelope

$$h_{A,\Lambda}(z) = (1 + |X_x - u_{T,x} - \gamma j|)^{-A} (1 + \Lambda |X_t - u_{T,t} - \gamma m/\Lambda|)^{-A}.$$

It gains Λ^{-A_1} outside the strip in either transverse or temporal direction, while $\int h_{A,\Lambda}^p \lesssim N^{k+2}/\Lambda$. By (2.4), annular counting, and pointwise Hölder, the off-activity p th power is bounded by $C_A N^{C_p} J^{C_p} \Lambda^{-A_1} (N^{k+2}/\Lambda) \sum_T |c_{T,j,m}|^p b_T^{2-p}$. Choose A after ϑ to absorb the polynomial factor. Keeping the temporal factor in $h_{A,\Lambda}$ is essential to retain Λ^{-1} in this bound. The fixed-index conclusion, including local-box and color summation, is

$$\int B_{b,j,m}^{2-p} \left| \sum_T g_T \right|^p \leq C_\varepsilon N^{\varepsilon/2} \Lambda^{C(d,p,P)} J^{C(d,p,P)} \frac{N^{k+2}}{\Lambda} \sum_T |c_{T,j,m}|^p b_T^{2-p}. \quad (2.18)$$

Here and below the exponents $C(d,p,P)$ are fixed before ϑ .

Native-distance comparison gives $B_{b,j,m} \leq C J^{P/2} B_b$. Since $2-p < 0$, $B_b^{2-p} \leq C J^{P(p-2)/2} B_{b,j,m}^{2-p}$. Apply Minkowski in the original weighted L^p norm and use (2.18) and the sampled coefficient bound. The m -sum of $\Lambda^{-1/p} (1 + |m|/\Lambda)^{-P'+C}$ is $O(\Lambda^{1-1/p})$; the j -sum converges if the original fixed P' is chosen greater than $C(d,p,P) + k + 1$. Consequently

$$\left\| B_b^{2/p-1} \sum_T f_T \right\|_p \leq C_\varepsilon N^{\varepsilon/(2p)} \Lambda^{C(d,p,P)+1-1/p} N^{(k+2)/p} \left(\sum_T s_T^p b_T^{2-p} \right)^{1/p}.$$

Choose ϑ sufficiently small that the displayed Λ power, all preceding color and logarithmic losses, and the deliberately smaller buffered loss fit below $N^{\varepsilon/p}$. Taking the p th power proves (2.15). The original P' is chosen once, independently of ε ; only the output Schwartz order and weighted decoupling order depend on the chosen ϑ . Unit suprema follow from Theorem 2.4 and fixed-scale envelope comparability.

Finally rotate an affine κ -plane to the first κ coordinates and remove its constant and linear normal terms by modulation and shear. The normal frequencies then have size $O(N^{-1})$, and their quadratic contribution is $O(N^{-2})$. Put $m = k - \kappa$ and write $x = (u, y)$ in tangential and normal coordinates. For a packet T , let (y_T, t_T) be the normal position and time of its physical center and, on a fixed normal slice y , put $j_T(y) = \lfloor (y - y_T)/N \rfloor \in \mathbb{Z}^m$ coordinatewise. Write $d_{T,\kappa}$ and $w_{T,\kappa} = (1 + d_{T,\kappa})^{-P}$ for the projected tangential-time native cell. Since the normal frequency is $O(N^{-1})$, its native displacement at time t is $(y - y_T)/N - 2(Nv_T^\perp)(t - t_T)/N^2$. Consequently, uniformly in t ,

$$1 + d_T \asymp 1 + d_{T,\kappa} + |j_T(y)|, \quad w_T \gtrsim (1 + |j_T(y)|)^{-P} w_{T,\kappa}.$$

Choose fixed A, P'' with $A + P'' \leq P'$ and P'' sufficiently large for the κ -dimensional estimate. On the slice y , the packet has projected amplitude at most $Cs_T(1 + |j_T(y)|)^{-A}$ and decay order P'' . Its projected native Fourier box is fixed: substitution in the normal native variable shifts time frequency by $O(Nv_T^\perp) = O(1)$, while the normal quadratic carrier shifts it by $N^2|v_T^\perp|^2 = O(1)$.

Group the sliced packets by $j_T(y) = j$, and write V_j for their sum. For fixed y, j and each N^2 time slab, this condition restricts the normal intercept to $O(1)$ cells for every projected frequency/intercept cell; there are also only $O(1)$ normal frequency slots. Hence each j -family has bounded projected phase-space multiplicity. Its tangential frequency slots are split into $O(1)$ colors and snapped to a κ -dimensional N^{-1} grid. Use the dimension- κ estimate already obtained from the preceding full-dimensional argument (or induct on k). If $B_j^2 = \sum_{j_T(y)=j} b_T^2 w_{T,\kappa}$, then $B_b^2 \gtrsim (1 + |j|)^{-P} B_j^2$. As $2 - p < 0$, applying the κ -dimensional estimate to that family gives

$$\int_{u,t} B_b^{2-p} |V_j|^p \lesssim N^{\varepsilon+\kappa+2} (1 + |j|)^{-Ap+P(p-2)/2} \sum_{j_T(y)=j} s_T^p b_T^{2-p}.$$

For each T, j , the set of y with $j_T(y) = j$ has volume N^m . Integrating in y supplies exactly N^m , and Minkowski sums the $j \in \mathbb{Z}^m$ contributions provided $A > m + P(p-2)/(2p)$. This is possible with the fixed sufficiently large original P' , and yields (2.15) in the range $2 < p \leq q_\kappa$. If $\kappa = 0$, put $\phi_T = (1 + d_T)^{-P'}$. The bounded number of slots and phase-space counting give $\sum_T \phi_T \leq C$. Three-factor Hölder gives

$$\left(\sum_T s_T \phi_T \right)^p \leq \left(\sum_T s_T^p b_T^{2-p} \phi_T \right) \left(\sum_T b_T^2 \phi_T \right)^{(p-2)/2} \left(\sum_T \phi_T \right)^{p/2}.$$

Since $\phi_T \leq w_T$, multiplication by B_b^{2-p} and integration, using $\int \phi_T \leq CN^{k+2}$, prove the assertion for every finite $p > 2$. $\square$

2.4 Local estimates without a scale loss

Proposition 2.7 (Local linear estimate). *Let $V = \sum_T f_T$ be either a band-limited or a smooth packet field of scale N , and let $B^2 = \sum_T s_T^2 w_T$. For a ball D of radius N and any $z_0 \in D$,*

$$\|V\|_{L^{q_k}(D)} \leq CN^{k/2} B(z_0). \quad (2.19)$$

The same bound holds for the unit-supremum norm on D .

Proposition 2.8 (Local multilinear estimate). *Let V_j and B_j be as in Theorem 2.7, with frequency labels in sets satisfying (2.2). For fixed $p > p_k$, a ball D of radius N , and arbitrary points $z_j \in D$,*

$$\left\| \prod_{j=1}^{k+1} |V_j|^{1/(k+1)} \right\|_{L^p(D)} \leq C_p \delta^{-C_p} N^{k/2} \prod_{j=1}^{k+1} B_j(z_j)^{1/(k+1)}. \quad (2.20)$$

Separate unit suprema may be taken for the $k+1$ factors. In particular the constant has no N^ε factor.

Proof of Theorems 2.7 and 2.8. Write $Eg(x, t) = \int e^{i(x \cdot \xi - t|\xi|^2)} g(\xi) d\xi$. We reduce the packets on D to exact extensions, keeping track of the norm. Translate the center of D to (x_0, t_0) . For a frequency label v , take an integral-normalized smooth bump g_v of width c/N , centered at v , and multiply it by $e^{-i(x_0 \cdot \xi - t_0|\xi|^2)}$. On a fixed enlargement of D its extension is its carrier times

$$H_v(x, t) = \int \varphi(\eta) e^{i(c/N)(x - x_0 - 2v(t - t_0)) \cdot \eta - i(c/N)^2(t - t_0)|\eta|^2} d\eta.$$

For fixed sufficiently small c , this differs from 1 by less than $1/2$. Both H_v and H_v^{-1} have bounded derivatives in the variables $(x - x_0, t - t_0)/N$. Also $\|g_v\|_2 \leq CN^{k/2}$. In the multilinear case c can be shrunk by a fixed power of δ so that enlarged supports retain determinant at least $\delta/2$; all resulting constants are polynomial in δ^{-1} .

For a packet T with label v , divide its demodulated profile by H_v on this enlarged ball, multiply by a fixed cutoff, and expand in a Fourier series in $(x - x_0, t - t_0)/N$. The resulting identity on D has the form

$$f_T(z) = \sum_{m \in \mathbb{Z}^{k+1}} c_{T,m} e^{i\ell_m \cdot (z - z_0)/N} Eg_v(z), \quad |c_{T,m}| \leq C_L s_T (1 + d_T(z_0))^{-P''} (1 + |m|)^{-L},$$

where $P'' > P/2 + k + 1$ can be fixed below the available original decay margin. Only finitely many derivatives are needed for a chosen L . For each mode, combine the coefficients with the same frequency slot; Cauchy–Schwarz in the spatial and time labels and their summable native tails gives

$$\left\| \sum_T c_{T,m} g_{v(T)} \right\|_2^2 \leq C_L N^k (1 + |m|)^{-2L} \sum_T s_T^2 w_T(z_0).$$

Here one first uses bounded overlap of the bump supports in distinct frequency slots; within a slot the native-center lattice has a summable weight. This explains why the square bound has no factor equal to the number of spatial labels.

The exact paraboloid Strichartz inequality $\|Eg\|_{q_k} \leq C\|g\|_2$ follows here directly from the dispersive kernel and the TT^* argument. Indeed the Gaussian oscillatory integral gives $\|e^{it\Delta}\|_{1 \rightarrow \infty} \leq C|t|^{-k/2}$, while Plancherel gives its L^2 norm equal to one. Interpolation at spatial exponent q_k gives decay $|t|^{-k(1/2 - 1/q_k)} = |t|^{-2/q_k}$. The one-dimensional Hardy–Littlewood–Sobolev inequality in time consequently bounds $TT^* : L^{q'_k}(\mathbb{R}^{k+1}) \rightarrow L^{q_k}(\mathbb{R}^{k+1})$, and duality gives the stated extension estimate. Applying it to each mode gives (2.19), after summing the modes. For the multilinear assertion use Tao [30, Theorem 1.7]. In ambient dimension $d = k + 1$, its product exponent $2a$, $a > 1/(d - 1)$, becomes geometric-mean exponent $p = 2da > p_k$. Its quantitative constant is polynomial in a parameter controlling domain size, finitely many graph derivatives, inverse determinant and inverse interior support distance. On bounded paraboloid patches these quantities are bounded by a fixed power of δ^{-1} : enlarge each relevant domain by $c\delta$, and, if necessary, subdivide into patches of polynomially small width in δ . The Hessian is constant, all required higher derivatives are bounded,

and the bump supports lie a polynomial distance inside the domains. Thus the theorem applies globally with the displayed $C_p \delta^{-C_p}$ and no scale loss. The external modulation belonging to a fixed Fourier mode has modulus one. Expand the product over modes using subadditivity of the $1/(k+1)$ power, and then Minkowski when $p \geq 1$. Taking L larger than the dimension times $k+1$ makes this sum converge.

Finally apply (2.9) separately to the exact extension factors. A spacetime translation of Eg is another extension with input multiplied by a unimodular phase, so it has the same input norm and frequency support. Tonelli and the global estimates bound each translated product. The convolution exponents can be chosen to sum all mode and translation weights. This proves the separate-supremum assertions and permits the independent choices of z_j . $\square$

2.5 Hierarchies, masks, and positive weights

Lemma 2.9 (Sampling preserves the envelope exponent). *Let $K' \geq K \geq 1$, let a K' -cap lie in a K -cap, and let T' be a child cell of scale K' . If $\mathcal{G}_K$ is the parent grid in that cap, then, for $P > k+2$ sufficiently large,*

$$\sum_{T \in \mathcal{G}_K} w_T(z) w_{T'}(z_T) \leq C_P w_{T'}(z). \quad (2.21)$$

The same exponent P occurs on both sides, independently of K'/K . If the sum is restricted to $d_T(z) > L$, $L \geq 2$, it is at most

$$C_P \left(L^{-(P-k-1)} + (K/K')^{P-k-2} \right) w_{T'}(z). \quad (2.22)$$

When $d_{T'}(z)$ is bounded, the second term can be omitted.

Proof. Put $r = K/K'$. In parent coordinates the linear map to child coordinates has spatial coefficients $O(r)$ and time coefficient r^2 , determinant r^{k+2} ; the possible velocity shear has bounded normalized coefficients because the child frequency lies in the parent cap. Let $a = 1 + d_{T'}(z)$. Parent centers whose child distance is at least $a/2$ contribute at most $C a^{-P} \sum_T w_T(z) \leq C a^{-P}$. With $d_T(z) > L$, this sum has the additional lattice-tail factor $L^{-(P-k-1)}$. For the other centers, necessarily $a \gg 1$ and $d_T(z) \geq ca/r$. Their contribution is at most

$$C(a/r)^{-P} \sum_{T \in \mathcal{G}_K} w_{T'}(z_T) \leq C a^{-P} r^P r^{-(k+2)}.$$

The last lattice sum is bounded by its integral at the child scale plus a fixed boundary term, with volume ratio $r^{-(k+2)}$. This proves both conclusions and shows explicitly why no envelope power is lost in repeated convolutions. $\square$

Proposition 2.10 (Weighted transport through a hierarchy). *Take nested frequency grids at scales $K < K'$, rounded within fixed factors. Assemble all child fields in each parent cap, and localize them on its parent physical grid. The localizers can be chosen band-limited, with summable Schwartz tails and bounded sums, so that the resulting parent packets have amplitudes a_T and satisfy*

$$b_T^2 \geq c \sum_{T' \text{ in the parent cap}} b_{T'}^2 w_{T'}(z_T) \quad (2.23)$$

whenever the parent weights are chosen to dominate the displayed sum. For $2 < p \leq q_k$,

$$\sum_T K^{k+2} a_T^p b_T^{2-p} \leq C_\varepsilon (K'/K)^\varepsilon \sum_{T'} (K')^{k+2} a_{T'}^p b_{T'}^{2-p}. \quad (2.24)$$

It remains valid after retaining any subset of parent cells, at every step of a finite hierarchy. After modifying a child field, use its fresh amplitudes in the same fixed parent masks. On subtrees lying in a codimension-one strip at their final frequency resolution one may use $p = q_{k-1}$ at every transition; for $k = 1$ any fixed finite $p > 2$ is available. Products of the transition estimates give the complete finite-hierarchy estimate.

Fixed polynomial weights on macroscopic boxes, and any fixed additional edge decay, are allowed. Their orders are chosen once, before the finite grid is chosen. Constants may depend on the finite grid; the required original profile regularity and decay order do not grow with its depth.

Proof. A concrete band-limited partition is obtained by taking $\chi = |\phi|^2$, where $\hat{\phi}$ is smooth with sufficiently small support. Poisson summation makes $\sum_{j \in \mathbb{Z}^{k+1}} \chi(X - j)$ a positive constant after normalization; alternatively a bounded lattice average gives the same identity. These nonnegative localizers have arbitrarily rapid tails, compact Fourier support, and bounded sums. If U is the pre-cutoff field in a parent cap, choose

$$a_T = C \sup_z |U(z)| (1 + d_T(z))^{-C_0}, \quad (2.25)$$

where C_0 exceeds the required output decay order by a fixed margin. Multiplying by the localizer makes the resulting normalized packet satisfy any prescribed fixed decay order. Its native Fourier support stays in a fixed enlarged box.

In parent coordinates the child packets have scale K'/K . Apply Theorem 2.6 there, and then its unit-supremum version to (2.25). The child square sum varies by at most a fixed polynomial over parent distance, since velocity differences are $O(K^{-1})$. The sufficiently large C_0 absorbs this variation, while (2.23), whose exponent $2 - p$ is negative, gives the required upper bound for the parent weighted norm. Changing coordinates supplies K^{k+2} and yields (2.24). Deleting parent outputs decreases its left side. For modified inputs the same proof uses the same localizers and recomputes (2.25); it never assumes that cancellation or an old amplitude survives deletion. The affine part of Theorem 2.6 gives the larger exponent on strip subtrees: a strip thin enough at the final precision is thin enough at each intervening child precision.

We explain the two uniformity assertions needed in iteration. The new square background formed by sampling child squares satisfies

$$\sum_T w_T(z) \sum_{T'} b_{T', w_{T'}}^2(z_T) \leq C \sum_{T'} b_{T', w_{T'}}^2(z)$$

by Theorem 2.9; the outer exponent has not decreased. Additional decay on an edge only improves this bound. A large displacement along a finite path forces a large displacement on an edge after normalization to the largest relevant cell. Keeping a fixed reserve of decay on every edge therefore bounds paths with a large displacement by the full-background convolution times a negative power of that displacement. The choice of which edge is large costs at most the number of edges, rather than a new decay order at each edge.

For a positive macro weight Ψ , suppose both ratios of its values at an edge's endpoints are bounded by $C(1 + d)^J$, where $d = d_{T'}(z_T)$ is the child distance. The constants and J must be independent of the main scale; this holds for polynomial weights on macro boxes whose size dominates the affected cells. Conjugate the given denominators by

$$\tilde{b}_T = b_T \Psi(z_T)^{-1/(p-2)}, \quad Q = P + \frac{2J}{p-2}.$$

Writing $w_{T'}^{(Q)} = (1 + d_{T'})^{-Q}$, one has

$$\begin{aligned} \sum_{T'} \tilde{b}_{T'}^2 w_{T'}^{(Q)}(z_T) &\leq C \Psi(z_T)^{-2/(p-2)} \sum_{T'} b_{T'}^2 w_{T'}^{(P)}(z_T) \\ &\leq C \tilde{b}_T^2. \end{aligned}$$

Apply the one-step proof with this stronger fixed child envelope exponent and its fixed supremum-decay reserve. Since $\tilde{b}_T^{2-p} = b_T^{2-p} \Psi(z_T)$, it proves the explicit estimate

$$\sum_T \Psi(z_T) K^{k+2} a_T^p b_T^{2-p} \leq C_\varepsilon (K'/K)^\varepsilon \sum_{T'} \Psi(z_{T'}) (K')^{k+2} a_{T'}^p b_{T'}^{2-p}.$$

The original denominators have been retained. The same value of Q works at each edge; the reserve does not accumulate with depth. For a partition of macro space with summable polynomial weights, summing over macros costs their bounded total overlap. This gives the version for restricted subtrees and frozen masks, without using discontinuous physical cutoffs.

Finally, for separated nested scales, inverse widths added by localization form a geometric series, including the bounded velocity shears. Thus the fixed Fourier enlargement can be chosen once. A block of comparable scales is treated as one transition; finitely many extra masks change constants. Repeated constants can grow with the fixed grid, but not with the main scale. In a later slow diagonal those constants may be absorbed only after the grid is fixed. Arbitrarily high Schwartz output decay is supplied by Theorem 2.4; it imposes no increasing regularity condition on the original packets. $\square$

2.6 A near-Cauchy estimate

The following estimate has no loss depending on N . Its threshold is measured against the pointwise Cauchy bound $|V| \leq CN^{k/2}B$; the allowed loss depends instead on the distance R below that bound.

Theorem 2.11 (Near-Cauchy large values). *Let $V = \sum_T f_T$ be a band-limited field of scale N , and let $B^2 = \sum_T s_T^2 w_T$. Suppose a finite set Y of unit cubes has witnesses $z_Q \in Q$ such that*

$$B(z_Q) \leq 1, \quad |V(z_Q)| \geq N^{k/2}/R, \quad R \geq 1.$$

Then, for every $\varepsilon > 0$,

$$|Y| \leq C_\varepsilon R^{q_k + \varepsilon} \sum_T s_T^2. \quad (2.26)$$

The conclusion holds with unit suprema and fixed enlargements of the witness cubes. The Fourier enlargement and original decay orders can be fixed before ε is chosen.

We first provide a finite exponent from which the improvement can start. This also makes the exponent-infimum argument below noncircular.

Lemma 2.12 (A polynomial starting bound). *Under the hypotheses of Theorem 2.11, there is a fixed finite M_k such that*

$$|Y| \leq C(2 + R)^{M_k} \sum_T s_T^2. \quad (2.27)$$

Proof. Write $S = \sum_T s_T^2$. By Theorem 2.6 with $b_T = s_T$ and $p = q_k$,

$$|Y| \leq C_\varepsilon N^\varepsilon R^{q_k} S. \quad (2.28)$$

At a witness, $B \leq 1$ and $2 - q_k < 0$; the unit-supremum version of the weighted estimate supplies the inequality for arbitrary witness positions. Consequently any range $N \leq (2 + R)^D$, with D fixed, already satisfies (2.27). We may assume N exceeds a sufficiently large fixed power of $2 + R$.

Discard the packets with $s_T > CRN^{-k/2}$. Their total field at a witness has modulus at most

$$\sum_{s_T > CRN^{-k/2}} s_T (1 + d_T)^{-P'} \leq (CRN^{-k/2})^{-1} \sum_T s_T^2 w_T \leq C^{-1} N^{k/2} / R.$$

Choose C large, so the remaining field retains half the threshold. Its coefficients have maximum $s_0 = CRN^{-k/2}$. Put $P_1 = (C(2 + R))^{10}$, partition the frequency patch into bins of side P_1/N , and write the remaining field as $\sum_b V_b$. There are $O((N/P_1)^k)$ bins. Cauchy–Schwarz therefore gives

$$\sum_b |V_b(z_Q)|^2 \geq c P_1^k / R^2. \quad (2.29)$$

In this square sum remove all ordered pairs of frequency labels whose separation is at most C'/N , where C' is a sufficiently large fixed constant compared with the native Fourier box. The removed expression is pointwise bounded by CB^2 : each frequency slot has only boundedly many such neighbors, and the sum over the spatial and temporal labels is bounded by Cauchy–Schwarz with the summable packet envelopes. It is therefore $O(1)$ at the witnesses. Let $G = \sum_b G_b$ be the sum of the remaining products. Choosing the constant in P_1 large makes (2.29) imply $|G(z_Q)| \geq 1$.

We verify the required spacetime almost orthogonality. If v_b is the center of a bin, the Fourier support of G_b is contained in

$$c/N \leq |\zeta| \leq CP_1/N, \quad |\sigma + 2v_b \cdot \zeta| \leq CP_1^2/N^2. \quad (2.30)$$

To see this, subtract the two cap Fourier coordinates used in the proof of Theorem 2.6. Removing the C'/N pairs guarantees the lower bound on $|\zeta|$. The deviation from the tangent plane at v_b is the product of two $O(P_1/N)$ quantities, together with the $O(N^{-2})$ normal errors. For a fixed (ζ, σ) , the bin centers lie in a slab of thickness $CP_1^2/(N^2|\zeta|)$. Counting their grid of mesh P_1/N gives overlap at most $CN^{k-1}P_1^{C_k}$. Plancherel yields

$$\|G\|_2^2 \leq CN^{k-1}P_1^{C_k} \sum_b \|G_b\|_2^2.$$

Within a bin, the effective pointwise packet multiplicity is $O(P_1^k)$. If $A_b = \sum_{T \in b} s_T (1 + d_T)^{-P'}$, then $|G_b| \leq A_b^2$, $A_b \leq CP_1^k s_0$, and $A_b^2 \leq CP_1^k \sum_{T \in b} s_T^2 (1 + d_T)^{-P'}$. Integrating these inequalities proves

$$\|G\|_2^2 \leq CN^{k-1}P_1^{C_k} (RN^{-k/2})^2 E = CP_1^{C_k} R^2 N^{k+1} S. \quad (2.31)$$

All Fourier frequencies of G have size at most CP_1/N in the $k + 1$ spacetime variables. Cover spacetime by cubes of side $L = N/P_1^{C_2} \geq 1$, choosing C_2 fixed and sufficiently large. The rescaled version of (2.9), summed over this lattice, shows that the number of these cubes with $\sup |G| \geq 1$ is at most

$$CL^{-(k+1)} \|G\|_2^2 \leq (2 + R)^{C_k} S.$$

Each such cube is contained in an N -ball. If it contains a witness, use that witness as the envelope reference in Theorem 2.7; its unit-supremum estimate bounds the number of high unit cubes in the ball by CR^{q_k} . Multiplying the last two bounds proves the lemma. $\square$

Proof of Theorem 2.11. Fix the frequency box, Fourier enlargement, phase-space bound and original regularity class once and for all. Call an exponent p admissible if $|Y| \leq C_p R^p S$ for all configurations in this class. Fixed changes of multiplicity, amplitude normalization, or cell enlargement change constants and do not change the infimum of admissible exponents. By Theorem 2.12 this infimum p_* is finite. We prove $p_* \leq q$, where $q = q_k$.

Suppose instead that $p_* > q$. There are configurations and parameters $R \rightarrow \infty$ with

$$|Y| \geq R^{p_* - o(1)} S. \quad (2.32)$$

Indeed use a failing exponent $p_* - \epsilon_j$ and let $\epsilon_j \downarrow 0$, with R large enough to absorb all constants for the finitely many estimates already chosen. Bounded R cannot give such failure by Theorem 2.12. Equation (2.28), used with each fixed positive ϵ , now implies

$$\frac{\log N}{\log R} \rightarrow \infty. \quad (2.33)$$

All subsequent $R^{o(1)}$ statements have this finite-parameter meaning: first fix positive exponent tolerances and finitely many tests; then take R sufficiently large, and finally let the tolerances tend to zero. In particular an admissible exponent just above p_* , rather than the possibly inadmissible exponent p_* itself, is used in every application below.

Removing excessively occupied fine packets. Fix C_1 large. Delete a fine packet if its CR native enlargement contains more than R^{C_1} witness points. Double counting with $B(z_Q) \leq 1$ gives

$$\sum_{T \text{ deleted}} s_T^2 \leq CR^{P-C_1} |Y|.$$

Apply any fixed admissible estimate of exponent $p > p_*$ to the deleted subfield, with threshold a small fixed multiple of $N^{k/2}/R$; its square envelope is bounded by the original one. The number of points where this subfield reaches that threshold is at most $CR^{p+P-C_1} |Y|$. Choose $C_1 > p + P + 10$. Except on a vanishing fraction of Y , the remaining field still reaches a fixed multiple of the old threshold. Constants of this sort are absorbed by changing R by a fixed factor. Every surviving fine packet now has occupancy at most R^{C_1} in its CR enlargement.

Coarse assembly and saturation of a ratio class. Choose a fixed A much larger than C_1 and all preceding fixed exponents, and put

$$K = N/R^A, \quad L = N/K = R^A.$$

Round to nested grids within bounded factors. Equation (2.33) ensures $K \rightarrow \infty$ faster than every fixed power of R . Assemble the surviving fine fields into K -packets by Theorem 2.10. In the sampling definition of their backgrounds use one slightly larger, fixed fine weight exponent $P + \sigma$, $\sigma > 0$:

$$b_T^2 = C \sum_{T' \text{ in its cap}} s_{T'}^2 (1 + d_{T'}(z_T))^{-P-\sigma}. \quad (2.34)$$

The original decay margin permits this choice. Apply Theorems 2.6 and 2.10 with this exponent for the fine estimate and with the original exponent P for the outer parent envelope. By Theorem 2.9,

$$\sum_T b_T^2 w_T(z) \leq CB(z)^2, \quad \sum_T K^{k+2} a_T^q b_T^{2-q} \leq R^{o(1)} E. \quad (2.35)$$

The ratio satisfies $a_T/b_T \leq CL^{k/2}$ by pointwise Cauchy–Schwarz in the L^k fine frequency slots and summability over their native centers. The global contribution of ratios less than $L^{k/2}R^{-2}$ is at most $CK^{k/2}L^{k/2}R^{-2}B$, again by Cauchy–Schwarz, so is negligible at the witnesses. Only $O(\log R)$ dyadic ratio classes remain. By the triangle inequality one class

$$a_T \asymp rb_T$$

retains threshold $N^{k/2}/R^{1+o(1)}$ on at least $R^{-o(1)}|Y|$ points. Use this class henceforth. Its amplitude-square envelope is $O(r^2)$, and (2.35) bounds its energy by $R^{o(1)}r^{2-q}E$. After dividing its field by Cr , its effective threshold parameter at scale K is $R^{1+o(1)}rL^{-k/2}$. This parameter is bounded below by a fixed constant whenever the threshold is attained, by the pointwise Cauchy bound. Apply an admissible exponent tending to p_* . Since r and L are fixed powers of R , its small exponent error is absorbed in $R^{o(1)}$, giving

$$\begin{aligned} |Y| &\leq R^{o(1)}(RrL^{-k/2})^{p_*}EK^{-(k+2)}r^{-q} \\ &= R^{p_*+o(1)}S(rL^{-k/2})^{p_*-q}. \end{aligned} \quad (2.36)$$

Here $kq/2 = k + 2$. Comparison with (2.32), and the upper bound $r \leq CL^{k/2}$, force

$$r = L^{k/2}R^{o(1)}, \quad \sum_{T \text{ selected}} b_T^2 \leq R^{o(1)}EK^{-(k+2)}r^{-q} \leq R^{o(1)}S. \quad (2.37)$$

The same fixed packet class remains available at this coarser scale: the inherited Fourier enlargement shrinks by L^{-1} in parent coordinates, whereas the parent localizer adds a fixed amount. A sufficiently large initial enlargement is consequently preserved as $L \rightarrow \infty$. Its Schwartz profiles have the originally fixed required bounds after constant renormalization. This verifies the uniformity needed to use the same p_* .

Localizing the selected square mass. We detail the restrictions used to draw a packet through a point. First discard globally all selected labels with $b_T > R^3K^{-k/2}$. Since the packet decay exceeds P ,

$$\left| \sum_{b_T > R^3K^{-k/2}} f_T(z) \right| \leq CrK^{k/2}R^{-3} \sum_T b_T^2 w_T(z) \leq CrK^{k/2}R^{-3}$$

at the witnesses. By (2.37) this is negligible relative to $N^{k/2}/R$. This global estimate, made before restricting distances, avoids any unsupported estimate of field tails from a square bound alone.

The remaining selected square envelope is at least $R^{-o(1)}$ on $R^{-o(1)}|Y|$ retained points. To verify this, fix $\eta > 0$ and consider points where that envelope squared is at most $R^{-\eta}$. Normalize the field there by $rR^{-\eta/2}$ and apply an admissible near-Cauchy exponent. Compared with (2.36), the threshold parameter gains $R^{-\eta/2}$ and the coefficient-square sum costs R^η . The count of these points is therefore at most

$$R^{p_*+o(1)}S R^{-\eta(p_*/2-1)}.$$

Since $p_* > q > 2$, this is negligible compared with (2.32). Letting η tend to zero along the finite-test diagonal proves the asserted lower bound.

For any fixed small $\eta > 0$, Theorem 2.9 gives, term by term in the original fine square sum,

$$\sum_{T: d_T(z) > R^\eta} b_T^2 w_T(z) \leq C(R^{-\eta(P-k-1)} + R^{-A(P-k-2)})B(z)^2.$$

For fine distance $a \gg 1$, the second term comes only from parent centers moving to within $a/2$ of the fine core; they require parent distance $\gtrsim aN/K$ and cost $R^{-A(P-k-2)}a^{-P}$. For bounded a only the first term occurs. Thus, choosing the square-mass tolerance smaller than these fixed gains, we may restrict to $d_T(z) \leq R^{o(1)}$ and retain square mass $R^{-o(1)}$.

We may also require each such selected T to have a surviving fine packet of its cap within $R^{o(1)}$ fine native distance of z . Indeed, if all its fine packets have distance greater than $R^{c\eta}$, then parent centers at distance at most R^η from z still have those fine distances greater than $\frac{1}{2}R^{c\eta}$, since the displacement in fine coordinates is $O(L^{-1}R^\eta)$. The extra fixed σ in (2.34) supplies $R^{-c\eta\sigma}$ when it is dropped back to exponent P . Sum against the parent weights and use Theorem 2.9. These unattached labels have negligible square mass. This is why one larger exponent was chosen in the sampling definition; no stronger square-envelope hypothesis at the original witnesses is needed.

Let $I_0(z, T)$ denote the resulting incidence relation. With

$$m_0(z) = \sum_T b_T^2 I_0(z, T),$$

we have, on the retained set still called Y ,

$$R^{-o(1)} \leq m_0(z) \leq R^{o(1)}, \quad b_T \leq R^{3+o(1)}K^{-k/2}, \quad \sum_T b_T^2 \leq R^{o(1)}S. \quad (2.38)$$

The upper bound follows by removing w_T at distance $R^{o(1)}$ in (2.35); the lower bound follows from the localized square mass. Draw T at z with probability $b_T^2 I_0(z, T)/m_0(z)$ and attach one of its nearby surviving fine packets. At a fixed z there are only $R^{o(1)}$ local parent cells in each angular $1/K$ slot. Consequently its angular law μ_z satisfies

$$\mu_z(\text{one } 1/K \text{ slot}) \leq R^{6+o(1)}K^{-k}. \quad (2.39)$$

Typical directions and resampling. Restrict the draw to directions v satisfying

$$\mu_z(B(v, s)) \geq R^{-1-o(1)}s^k \quad (K^{-1} \leq s \leq 1). \quad (2.40)$$

This removes $O(R^{-1+o(1)})$ probability, without a logarithm of the number of scales. Here is a finite covering proof. In a fixed finite collection of shifted dyadic grids, restrict attention to cells meeting the bounded frequency box, with side between c/K and a fixed constant. Select the maximal cells in this finite collection for which $\mu_z(Q) < cR^{-1}|Q|$. Within each grid these cells are disjoint, so the sum of their masses is at most CR^{-1} . A ball failing (2.40) contains a cell of comparable side through its center in one of these grids; that cell belongs to a maximal bad cell. The bounded number of grids and fixed comparison factors are absorbed by the indicated subpower tolerance. Renormalizing the restricted law costs a bounded factor. Let $I(z, T)$ be the resulting incidence relation and $m(z) = \sum_T b_T^2 I(z, T)$; (2.38) remains true.

Fix $z \in Y$. For another point $y \in Y$ write $\tau = |t_y - t_z|$. If $\tau \geq N$, the probability, under the restricted draw, that y is in the same $R^{o(1)}$ -enlarged K -cell as z is at most

$$R^{6+o(1)}(K/\tau)^k. \quad (2.41)$$

Indeed its directions lie in a ball of radius $KR^{o(1)}/\tau$; (2.39) counts the slots in that ball. A nonempty incidence also requires $\tau \leq K^2R^{o(1)}$, so the lowest angular precision $1/K$ costs only a further $R^{o(1)}$ at the endpoint of this range.

If this probability is nonzero, choose a typical direction v that sees y . In the original draw, the ball of radius cN/τ about v has mass at least

$$R^{-1-o(1)}(N/\tau)^k \quad (2.42)$$

by (2.40). The radius lies between $1/K$ and a fixed constant: $\tau \geq N$ gives the upper bound, and $\tau \leq K^2 R^{o(1)}$ together with $N/K = R^A$ gives the lower one. Every draw in that ball has an attached fine packet which sees both z and y in its CR enlargement. To check this last statement, its intercept at z differs by at most $NR^{o(1)}$; the change of angular direction over time τ costs at most cN ; and the discrepancy between a parent label and a fine label is $O(1/K)$, costing at most $\tau/K \leq KR^{o(1)} \ll N$. The time separation is also much less than N^2 , whereas z was already at fine native distance $R^{o(1)}$. All these errors are smaller than the available CR enlargement.

Let $P_f(y)$ be the probability that the attached fine packet in the original draw sees y in that enlargement. Comparing (2.41) and (2.42) bounds the expected number of far points in the sampled coarse cell by

$$\sum_{\tau \geq N} R^{7+o(1)}(K/N)^k P_f(y) \leq R^{C_1+7-Ak+o(1)}. \quad (2.43)$$

The last inequality uses the fine occupancy cutoff: every attached fine packet sees at most R^{C_1} points, so $\sum_y P_f(y) \leq R^{C_1}$. This is a comparison with the original draw, not an independence assertion about a direction and a point.

For $\tau < N$, a point in the same enlarged coarse cell lies in a ball of radius CN about z , since the velocities are bounded and $KR^{o(1)} \ll N$. The local linear estimate at z , or a bounded cover by N -balls and envelope comparability, bounds the number of such high points by CR^q . Choose A so large that $Ak > C_1 + q + 10$. Combining the near and far parts, the expected occupancy of the sampled enlarged coarse cell, for every retained z , is at most

$$R^{q+o(1)}. \quad (2.44)$$

The contradictory lower occupancy. Put $n_T = \#\{z \in Y : I(z, T) = 1\}$. This is no larger than the geometric occupancy of the enlarged coarse cell. Average that occupancy first over the draw at z , then uniformly over $z \in Y$. By (2.38) and Cauchy–Schwarz, the result is at least

$$\begin{aligned} \frac{R^{-o(1)}}{|Y|} \sum_T b_T^2 n_T^2 &\geq \frac{R^{-o(1)}}{|Y|} \frac{(\sum_T b_T^2 n_T)^2}{\sum_T b_T^2} \\ &\geq R^{-o(1)} \frac{|Y|}{S} \geq R^{p_*-o(1)}. \end{aligned}$$

The middle inequality uses $\sum_T b_T^2 n_T = \sum_{z \in Y} m(z) \geq R^{-o(1)}|Y|$. All selections retained a subpower fraction, so the last inequality still follows from (2.32). This contradicts (2.44) because $p_* > q$. Thus $p_* \leq q$, and every $q + \varepsilon$ is admissible, proving (2.26). The unit-supremum and fixed-enlargement versions used in the proof follow from Theorem 2.4 and bounded changes of the lattice. $\square$

3 The transverse frame estimate

Write

$$q = \frac{2(k+2)}{k}, \quad p_k = \frac{2(k+1)}{k}, \quad m = k+1.$$

All packet coordinates, envelopes, and weighted packet sizes in this section are those of Section 2. In particular, an inverse frequency width r corresponds to spatial width r , time length r^2 , and

spacetime volume r^{k+2} . A packet mask means multiplication by a selected sum of the smooth packet cutoffs at its own scale. Such sums have bounded absolute value. A restriction of a mask is allowed to keep any subset of its terms.

Theorem 3.1 (Frame estimate). *Fix $k \geq 2$, $c > 0$, and fixed packet bounds C_0 . There is a finite integer P_0 such that, for every fixed $P \geq P_0$, there are a finite integer J_0 and a constant C , depending only on k, c, C_0, P and the packet conventions, with the following property. Let $h \geq 2$, let $V_1, \dots, V_m$ be scale- h packet sums on one time interval of length at most $C_0 h^2$, and suppose their normalized compact profiles have derivatives through order J_0 bounded by C_0 . Let Ξ_i be their assigned frequency-label sets, with the fixed cap enlargements included. Assume bounded phase-space counting and*

$$|\det((2\xi_1, 1), \dots, (2\xi_m, 1))| \geq \delta > 0 \quad (\xi_i \in \Xi_i).$$

Use P for the packet envelope exponent. Let Y be a set of distinct unit cubes and, for every $Q \in Y$ and i , let $z_{i,Q} \in Q$ satisfy

$$|V_i(z_{i,Q})| \geq A_i \geq h^c, \quad B_i(z_{i,Q})^2 = \sum_T s_{i,T}^2 w_T(z_{i,Q}) \leq 1.$$

The points $z_{i,Q}$ need not agree for different i . Set

$$H_i = A_i^{2/k}, \quad H_g = \left(\prod_{i=1}^m H_i \right)^{1/m}, \quad D_i = h^{k+2} \sum_T s_{i,T}^2.$$

Then

$$|Y| \leq C \delta^{-C} H_g^{-2} \sum_{i=1}^m \frac{D_i}{A_i^2}. \quad (3.1)$$

Only the finite number J_0 of profile derivatives is required.

Unit translations change the square envelopes by a fixed factor. Consequently we use the cube centers for counting and retain the individual witnesses when testing the fields. Every local norm estimate below uses a weighted supremum on the relevant cube; thus none of the arguments replaces separate witnesses by an unsupported common witness.

The proof is by contradiction. A failing configuration determines transition scales and an excess packet amplitude. We encode the amplitude ratios along a finite hierarchy by logarithmic profiles. Projection estimates force each profile to have only a zero slope or the full Cauchy slope, and an incidence argument locates the interval on which the full slope occurs. The resulting transition ratio is subpower, contradicting the original excess amplitude and its energy budget. Smoothing is needed only when that excess grows subpolynomially in the original packet scale.

3.1 Exact packets and a normalized countersequence

Lemma 3.2 (Reduction to exact packets). *It suffices to prove Theorem 3.1 for translated exact paraboloid extensions, with smooth data supported in bounded native frequency boxes, multiplied by a common Schwartz function of t/h^2 which is positive on the test interval and has compact Fourier support. All required seminorms of these exact packets can be fixed in advance.*

Proof. Use a fixed enlargement of the test interval and a cutoff equal to one there. Fourier transform a compact native profile in its spatial variables and insert a smooth partition into unit frequency boxes centered at $a \in \mathbb{Z}^k$. In the a th box remove the quadratic evolution factor, and expand the

remaining smooth function of native time in a Fourier series on the enlarged interval. This writes the profile as a sum, indexed by $(a, n) \in \mathbb{Z}^{k+1}$, of exact extensions times common time modulations. Integration by parts in the Fourier coefficient formulas shows that, for prescribed finite J, M ,

$$\|\text{data}_{a,n}\|_{C^J} \leq C_{J,M}(1 + |a| + |n|)^{-M},$$

provided finitely many original derivatives are bounded. The physical frequency center changes by a/h . All intercepts are specified at one common reference time. Recentring about the shifted trajectory on the test interval changes a polynomial envelope by at most a fixed polynomial in $1 + |a|$.

For completeness, normalize each layer by $c_{a,n} = (1 + |a| + |n|)^{-M/2}$ and allocate threshold fractions proportional to $(1 + |a| + |n|)^{-M/4}$. Their sum is finite when M is large. If a sum has magnitude at least A_i , one of its normalized layers exceeds the corresponding allocated threshold divided by $c_{a,n}$. Apply the exact packet estimate to the tuples of these layers. In its i th energy term the threshold dependence is

$$D_i A_i^{-2} \prod_{j=1}^m A_j^{-4/(km)}.$$

Every threshold has a strictly negative power; hence, after taking M large, the resulting mode sums converge, including the polynomial envelope losses. Modes $|a| > h^{1/2}$ contribute less than any specified negative power of h by weighted Cauchy and the same decay. For the remaining modes transversality is preserved when $\delta \geq h^{-\varepsilon_0}$ and $\varepsilon_0 > 0$ is sufficiently small. If $\delta < h^{-\varepsilon_0}$, the polynomial bound in h from the elementary packet estimate is absorbed by δ^{-C} . This also disposes of the discarded tail.

An exact packet outside its central time bin has only polynomial growth in native time in each of the finitely many spatial localization seminorms: differentiate the compactly supported frequency integral and integrate by parts. Multiplication by a common positive-on-the-slab Schwartz cutoff makes all relevant spacetime integrals finite. Choose its decay after those seminorms. The whole argument therefore uses a fixed finite derivative order. The initial native-time cutoff used for the Fourier expansion is distinct from the final compact-Fourier multiplier $\chi_h(t) = \chi((t - t_0)/h^2)$. The latter is bounded above and below by positive fixed constants on the test slab. Multiplying the exact-layer sum by it therefore changes the witness thresholds and packet bounds by fixed factors, which suffices for the reduction. $\square$

Suppose now that the exact-packet assertion fails. By choosing the successive counterexamples to violate every fixed constant and every fixed inverse power of δ , we obtain $h \rightarrow \infty$ with $\delta = h^{-o(1)}$. Absolute Cauchy gives $H_i \lesssim h$. Proposition 2.6 gives

$$|Y| \leq h^{o(1)} D_i H_i^{-k-2}.$$

Comparing this for all i with the failed estimate proves

$$H_i/H_g = h^{o(1)}, \quad |Y| H_i^{k+2}/D_i = h^{o(1)}. \quad (3.2)$$

Here and below a positive quantity equal to a base to the power $o(1)$ is bounded both above and below in that sense.

Write the original exact-packet field as $V_i = \chi_h U_i$, where U_i is its uncut exact extension, with any common temporal modulation removed. For the smooth inverse-width- H_i cap projection P_b , put $U_b = P_b U_i$ and $V_b = \chi_h U_b$. Thus U_b always denotes the full original cap extension before any

transition masks. Localize each actual cap field V_b at inverse width H_i , and select one dyadic packet amplitude class in each field:

$$a_T \asymp S_i H_i^{-k/2}, \quad S_i \gtrsim 1. \quad (3.3)$$

These are the *original transition masks*. They will remain in every later hierarchy, even though the fields inside them change. The selected sum has high threshold $A_i(1 + |\log S_i|)^{-C}$ on a subset, again denoted Y , satisfying

$$|Y| \geq (\log(2 + S))^{-C} H_g^{-2} \sum_i D_i H_i^{-k}, \quad S = \max_i S_i, \quad (3.4)$$

with an arbitrarily large additional constant and inverse power of δ along the countersequence. This countable pigeonhole is legitimate: classes below a fixed small value of S_i have a summably small absolute contribution, and the threshold fractions above are summable in the dyadic index. At a transition cell, local Cauchy gives $a_T \lesssim (h/H_i)^{k/2} b_T$, where b_T is its full descendant square weight. The weighted square sum at witnesses is bounded. Consequently classes larger than a fixed polynomial in h have a negligible root contribution, so that $\log S = O(\log h)$ for the selected tuple.

Spatial almost orthogonality, followed by the local weighted Bernstein bound, gives

$$H_i^{k+2} \sum_{T \text{ selected}} a_T^2 \lesssim D_i. \quad (3.5)$$

Bounded counting and (3.3) also give square background $O(S_i)$ for the selected packets without using the original B_i . Theorem 2.11, applied to these packets, yields

$$H_i/H_g + H_g/H_i + D_i/(|Y|H_i^{k+2}) + |Y|H_i^{k+2}/D_i \lesssim S^C. \quad (3.6)$$

It follows that $S \rightarrow \infty$ and $\delta = S^{-o(1)}$. Indeed, bounded S , or a fixed positive value of $\log_S \delta^{-1}$, would make this estimate contradict the choice of the countersequence in (3.4).

Pass to a subsequence in one of two cases. In the *power case*, $\log S \gtrsim \log h$ and the logarithmic base is $\Lambda = h$. In the *subpower case*, $S = h^{o(1)}$ and the base is $\Lambda = S$. The next subsection is needed only in the latter case.

The transition cells chosen in (3.3) will stay fixed, even when we replace the fields localized by them. We retain their original amplitudes as data, writing

$$a_T^{\text{orig}} \asymp S_i H_i^{-k/2}.$$

Later assemblies have their own actual amplitudes; these need not remain comparable to a_T^{orig} . The original energy bound (3.5), however, continues to charge every surviving transition cell.

Here is the count we seek. We will show that the high set requires at least

$$|Y| H_i^k \Lambda^{-o(1)}$$

distinct surviving transition cells in each field. Applied to a field with $S_i = S$, their original amplitudes would give

$$D_i \gtrsim |Y| H_i^{k+2} S^2 \Lambda^{-o(1)}.$$

The scale balances will exclude this inequality in both cases. To obtain the count, we first build finite scale hierarchies through the fixed transition masks. Their selected amplitude ratios define logarithmic profiles. The projection and incidence arguments will force the ratio at the transition scale to be subpower and bound the number of visits to each cell. Cauchy–Schwarz gives many incident cells at each high point, and double counting yields the displayed global bound. In the subpower case, we first need a square background controlled in the base S ; this is the purpose of the smoothing argument that follows.

3.2 Smoothing and clipping on the subpower scale

Fix a field and abbreviate $H = H_i$, $D = D_i$. First remove original transition packets visited by more than S^{C_1} witnesses in a fixed large power enlargement. Their squared-amplitude sum is at most $|Y|S^{-C_1+C_2}$: count visits, using the fixed amplitude class and bounded counting, and put the enlargement costs into C_2 . Theorem 2.11, with square background $O(S)$, shows that their root contribution is negligible when C_1 is sufficiently large. Each remaining high point needs at least H^k/S^C nearby cap slots, by absolute summation and (3.3). Packet tails outside the chosen enlargement are negligible by the fixed rapid cutoff decay. Counting these incidences in a ball gives

$$\#(Y \cap B_r) \lesssim S^C (r/H)^k (1 + r/H^2), \quad r \geq H. \quad (3.7)$$

For each of the $O(H^k)$ cap directions, at most $O((r/H)^k (1 + r/H^2))$ packet cells meet the ball, up to the fixed power enlargement. Their occupancy is at most S^{C_1} . Division by the H^k/S^C incidences required at each point proves the formula.

Lemma 3.3 (Clipped smoothing). *Fix C_3 sufficiently large. There is a fixed L such that, with $K_0 = HS^L \leq h$, each cap field V_b admits a scale- K_0 packet representation with integrated square energy at most CD and envelopes $e_b^2 = \sum_{\gamma \in b} s_\gamma^2 w_\gamma^{\text{work}}$ satisfying*

$$\sum_b \min(e_b^2(z), M_c) \leq S^{o(1)}, \quad M_c = H^{-k} S^{C_3}, \quad (3.8)$$

outside $o(|Y|)$ witnesses. The work weight $w_\gamma^{\text{work}} = (1 + d_\gamma)^{-P_{\text{work}}}$ has a fixed order $P_{\text{work}} > P$, chosen in the proof; all subsequent levels of this subpower tree use that order. The envelope and profile orders are finite and independent of any subsequent scale-grid length. If $K_0 > h$, the tree instead ends at the unchanged original scale- h packets with their original order P ; their square envelope is already bounded at the witnesses.

Proof. The positive smoothing functional. We give separately the positive smoothing estimate and its comparison with packet sizes. The original selections and deletions are left unchanged and applied externally, and χ_h remains in every actual field. Only the auxiliary positive density below uses the uncut original cap field U_b . Fix a sufficiently large integer $J \geq 1$, independent of all later grids, and put

$$\rho = H^{k+2} \sum_{z \in Y} \delta_z, \quad F_b(x, t) = w_h(t) |U_b(x, t)|^2, \quad w_h(t) = \left(1 + \frac{(t - t_0)^2}{h^4}\right)^{-J}.$$

Spatial unitarity and the original fine-packet Gram bound give

$$\sum_b \int F_b = c_J h^2 \sum_b \|U_b(t_0)\|_2^2 \lesssim h^{k+2} \sum_T s_T^2 = D. \quad (3.9)$$

Here $c_J = \int_{\mathbb{R}} (1 + s^2)^{-J} ds < \infty$. Indeed the cap projections have bounded overlap, and integration by parts in the original normalized frequency data gives summable Gram rows for the fine packets, using finitely many prescribed localization seminorms. This charges the original data before every transition mask. No division by χ_h is used. Choose an even Q sufficiently large compared with the original envelope order P and k , and put $j_K(x) = c_Q K^{-k} (1 + |x/K|^2)^{-Q/2}$.

Convolve in space with heat variance K^2 and with independent symmetric jumps having density j_r and intensity dr/r at scale $r \leq K$. The limiting small-jump distribution exists because $\int_0^K r^2 dr/r < \infty$. Denote the resulting kernel by J_K . It has

$$C^{-1} j_K \leq J_K \leq C j_K, \quad |\nabla J_K| \leq C K^{-1} j_K, \quad |\hat{J}_K(\xi)| \leq e^{-cK^2|\xi|^2}. \quad (3.10)$$

Here is a useful direct verification of the polynomial tails. For the lower bound request one jump in $[K/2, K]$ and all other displacement at most CK , an event of fixed positive probability by the second moment bound. The density of that jump is comparable to j_K after such a displacement. For the upper bound group jumps into $[2^{-j-1}K, 2^{-j}K]$, with Poisson counts m_j . If their total plus the Gaussian has magnitude comparable to $|x| \gg K$, either the Gaussian has such magnitude or one of the jumps in group j has magnitude at least $c2^{-j/2}|x|/(1+m_j)$. Conditioning on the other variables and summing candidates bounds its density by

$$C(1+m_j)^{Q+1}K^{-k}2^{jk-jQ/2}(K/|x|)^Q.$$

Poisson moments are bounded uniformly in j and the series converges for $Q > 2k$. The same argument after differentiation of a slightly broader Gaussian proves the gradient bound. The Fourier bound follows from the heat factor and the characteristic function of a probability measure.

In the coordinates moving with $2b$, write $\partial_{t,b} = \partial_t + 2b \cdot \nabla_x$, and also convolve in time by a Gaussian of width

$$T = T(K) = KH/P_K, \quad P_K = \min(H, K/H)^\theta, \quad 0 < \theta < 1/4,$$

for $H \leq K \leq h$, using any increasing continuation below H . Write $\rho_{K,b}, F_{K,b}$ for these convolutions. The spatial generator in logarithmic scale is, up to fixed heat conventions, $L_K = K^2\Delta_x + j_K * -I$.

Generator bounds. We claim that for some $c' > 0$

$$H^{-k} \sum_b \|L_K \rho_{K,b}\|_2^2 \lesssim S^C \|\rho\|_1 \min(H, K/H)^{-c'}. \quad (3.11)$$

Indeed (3.7) and the Gaussian identity for Fourier ball mass give

$$\int_{|\zeta| \lesssim 1/r} |\hat{\rho}(\zeta)|^2 d\zeta \lesssim S^C \|\rho\|_1 (1 + H^2/r), \quad r \geq H.$$

To see the normalization, the local mass is at most $S^C r^k (H^2 + r)$ and the smoothing volume is r^{k+1} . For $r < H$ the same bound loses only a fixed polynomial in H/r . On $|\xi| \asymp 1/r$ the squared spatial multiplier is bounded by $C_N (K/r)^2 (1 + K/r)^{-N}$. For $r \leq T$, the time Gaussian restricts $|\tau + 2b \cdot \xi| \lesssim T^{-1}$, with rapidly decreasing dyadic tails. Of the $O(H^k)$ cap centers, a fraction at most $C(r/T + H^{-1})$ obeys this restriction; this is the elementary count of an H^{-1} grid in a slab. The corresponding spacetime Fourier ball has radius $O(r^{-1})$. For $r \geq T$ use radius $O(T^{-1})$ and the trivial cap fraction one. At $r = K$ the resulting factor is

$$(P_K/H)(1 + H^2/K) \leq 2 \min(H, K/H)^{\theta-1}.$$

The dyadic sums for $r < K$, $K < r < T$, and $r \geq T$ converge, after choosing N large; enlargements in the Gaussian tails cost only fixed powers. This proves (3.11), also with the heat and jump multipliers treated separately.

The exact equation supplies a second estimate:

$$|\partial_{t,b} F_{K,b}| \lesssim (HK)^{-1} F_{K,b}. \quad (3.12)$$

After carrier removal U_b has frequency support $O(H^{-1})$, and its exact equation gives

$$\partial_{t,b} |U_b|^2 = -2\nabla_x \cdot \text{Im}(\overline{U_b}(\nabla_x - ib)U_b).$$

Move this divergence to J_K . The required weighted Bernstein bound is

$$\int j_K(x-y)|(\nabla_y - ib)U_b(y,t)|^2 dy \lesssim H^{-2} \int j_K(x-y)|U_b(y,t)|^2 dy.$$

To prove it, remove the carrier and reproduce with a smooth projection kernel $R_H = H^{-k}R(\cdot/H)$. Polynomial comparison of translates of j_K and weighted Young's inequality give operator norm at most

$$\int |\nabla R_H(u)| \langle u/K \rangle^{Q/2} du \lesssim H^{-1} \quad (K \geq H).$$

Together with (3.10) and Cauchy–Schwarz this bounds the current by $C(HK)^{-1}(J_K *_x F_b)$. The sole time-weight term satisfies $|w'_h| \leq C_J h^{-2} w_h$, and $h^{-2} \leq (HK)^{-1}$. Positive comoving time convolution preserves the inequality. The argument requires only compact-frequency L^2 data for the summed U_b ; smooth approximation in that fixed cap justifies the current identity and the limiting convolutions.

Propagation of the clipped bound. Put $\phi(u) = M_c(1 - e^{-u/M_c})$ and $I(K) = \sum_b \int \rho_{K,b} \phi(F_{K,b})$. For any symmetric diffusion or jump generator L , concavity gives $\phi'(F)LF \geq L\phi(F)$. Thus differentiating I gives a lower bound consisting of twice the generator pairing with ρ . For the spatial part Cauchy–Schwarz, (3.11), and $\phi(F)^2 \leq M_c F$ yield

$$|\text{spatial error}| \leq S^C \|\rho\|_1 \min(H, K/H)^{-c'/2},$$

where we used $D \leq S^C \|\rho\|_1$ from (3.6). Notice the cancellation of the factors $H^{k/2}$ and $M_c^{1/2}$. For the time part integrate by parts once. The Gaussian gives $\|T\partial_{t,b}\rho_{K,b}\|_1 \lesssim \|\rho\|_1$, while (3.12) and $u\phi'(u) \leq CM_c$ give cost $CM_c T/(HK)$ per cap. There are $O(H^k)$ caps, so the time error is at most $S^C \|\rho\|_1 / P_K$. Both errors integrate from K_0 to h to at most

$$S^C \|\rho\|_1 \{S^{-cL} + H^{-c} \log h\} = o(\|\rho\|_1)$$

when L is fixed sufficiently large. We used $H \geq h^{2c/k}$ and $S = h^{o(1)}$. Smooth regularization of ρ and piecewise smooth approximation at $K = H^2$ justify the differentiation; the positive convolution formulas allow passage to the limit.

At $K = h$, drop clipping. Convolution twice averages $w_h|U_b|^2$ on spatial width h and time width at most Hh along $2b$. The original transition masks remain outside this estimate. For a witness $z = (x_z, t_z)$, the Gram bound for the uncut original fine packets is

$$\sum_b \int j_h(x - x_z - 2b\tau) |U_b(x, t_z + \tau)|^2 dx \lesssim \langle \tau/(Hh) \rangle^{M_1} \sum_T s_T^2 w_T(z). \quad (3.13)$$

The relative displacement in width- h units is $O(|\tau|/(Hh))$, and elapsed native time is at most $C + |\tau|/h^2 \leq C + |\tau|/(Hh)$. The prescribed finite localization seminorms of the individual original packets grow polynomially in this variable. Spatial almost orthogonality and summable Gram rows therefore prove (3.13), retaining the original exponent on w_T by choosing the auxiliary kernel order Q larger. Since $w_h \leq 1$ and $T(h) \leq Hh$, the Gaussian moments give

$$I(h) \lesssim \int \rho(z) \sum_T s_T^2 w_T(z) \lesssim \|\rho\|_1.$$

At K_0 , kernel comparability in space and (3.12) in comoving time show that $F_{K_0,b}$ varies by bounded factors in its central smoothing box. Therefore the extra convolution against ρ dominates a constant

multiple of $\sum_{z \in Y} H^{k+2} \sum_b \min(F_{K_0,b}(z), M_c)$. We have proved that this last clipping sum has bounded mean on Y .

Recovery of packet envelopes. We now bound the actual square envelopes by the smoothed auxiliary density on the test interval. This also fixes the required finite orders. Partition the exact data of U_b into smooth caps of width K_0^{-1} . Form the actual packets by localizing $\chi_h P_v U_b$ in their scale- K_0 moving coordinates, and retain the original H masks externally. Since $K_0 \leq h$, fixed native derivatives of χ_h are bounded and its temporal Fourier thickness is at most $O(K_0^{-2})$. Use weighted suprema as packet sizes. Local constancy, $|\chi_h|^2 \leq C_J w_h$, and bounded overlap of the spatial projections give

$$K_0^{k+2} \sum_{v,\lambda} s_{b,v,\lambda}^2 \lesssim \sum_v \int |\chi_h P_v U_b|^2 \lesssim \int w_h |U_b|^2. \quad (3.14)$$

Thus (3.9) supplies the total integrated budget CD for this representation of the same actual cutoff field.

Here is an explicit fixed-order comparison. Work in native scale $K_0 = 1$ for a cap, write its uncut carrier-removed solution as $q_v(u, s)$, and take

$$N = Q + k + 2, \quad P_{\text{work}} = 3N + 3.$$

A smooth larger frequency projection has propagation kernel $|L_\sigma(r)| \leq C_N \langle \sigma \rangle^N \langle r \rangle^{-N}$. Consequently, for every fixed s_* ,

$$|q_v(u, s)|^2 \leq C \langle s - s_* \rangle^{2N} \int \langle u - y \rangle^{-N} |q_v(y, s_*)|^2 dy.$$

The bounded multiplier χ_h and its fixed native derivatives only change the constant when estimating the actual packet sizes from this uncut field. Choose the size-supremum exponent at least $\max(P'_{\text{work}}, 2N)$ for some fixed $P'_{\text{work}} > P_{\text{work}}$. For the packet centered at $(a, m) \in \mathbb{Z}^{k+1}$ its size then obeys

$$s_{v,a,m}^2 \leq C \langle m - s_* \rangle^{2N} \int \langle a - y \rangle^{-N} |q_v(y, s_*)|^2 dy.$$

Summing with the work packet weight of order P_{work} first in m leaves time decay $P_{\text{work}} - 2N - 1 = N + 2$, and then summing in a retains spatial decay at least N . Thus the packet envelope at z is bounded by a spatial j_{K_0} -weighted energy at any time $|t_* - z_t| \leq T(K_0)$. Here $T(K_0) \leq K_0^2$ and the relative velocity displacement is $O(T/H) \leq CK_0$.

For these cap energies use weighted Bessel:

$$\sum_v \int j_{K_0}(x - x_0) |P_v f(x)|^2 dx \lesssim \int j_{K_0}(x - x_0) |f(x)|^2 dx.$$

One proof avoids a nonsummable bound for each projection. With an even polynomial weight, differentiate the cap symbols and use their bounded overlap and Plancherel to bound synthesis $(g_v) \mapsto \sum_v P_v^* g_v$ in positive polynomially weighted $L^2(\ell^2) \rightarrow L^2$. Duality gives the displayed negative-weight analysis estimate. A partition into boxes and summable convolution extends it to translates of j_{K_0} . Apply it to U_b at the fixed time t_* . Average the preceding packet comparison over the central part of the time Gaussian. For z on the test slab and $|t_* - z_t| \leq T(K_0) \leq h^2$, the polynomial weight $w_h(t_*)$ is bounded below by a fixed constant. No lower bound for χ_h outside the test slab is needed. Hence $e_b^2(z) \lesssim F_{K_0,b}(z)$ on the test interval. Markov's inequality, with any slowly diverging subpower threshold, now proves (3.8).

When $K_0 \leq h$, these reconstructed packets are the leaves of the subpower tree: no original scale- h fine squares are inserted below them. Their rapid output decay can be fixed above P'_{work} ,

and the same work exponent is used in each later application of Theorems 2.6, 2.10 and 2.11. The original order P enters only the input square hypothesis and the original-packet Gram bound used at $K = h$; no comparison of a reconstructed fine array with the original square envelope is asserted. If $K_0 > h$, use the unchanged original scale- h representation as the leaf, with order P . Its square envelope is bounded at the witnesses, so clipping is automatic. All orders are chosen before the finite scale grids and the finite original derivative order in Theorem 3.2; they depend on k, c and the original envelope order, not on hierarchy depth. $\square$

Retain only original H masks whose centers obey $e_b^2 \leq M_c$. The loss is negligible. Indeed, within a sufficiently small fixed power enlargement, a removed bin has $e_b^2 \gtrsim M_c/S$ at the witness, by polynomial weight comparison. Formula (3.8) therefore bounds the number of such bins by $H^k S^{1-C_3+o(1)}$. Multiply this by the original amplitude $SH^{-k/2}$ and choose C_3 large. Outside the enlargement, use the rapid cutoff tails and the same absolute amplitude bound.

At surviving transition masks use weights $b_T = e_b(z_T)$. For each bin b , sampling only the retained mask centers gives

$$\sum_{T \text{ retained in } b} e_b^2(z_T) w_T(z) \lesssim \min\{e_b^2(z), M_c\}.$$

Convolution gives the bound by $Ce_b^2(z)$; the center restriction $e_b^2(z_T) \leq M_c$ and the bounded sum of the mask weights give the bound by CM_c . At strict ancestors we may therefore use the square sum of $\min(e_b^2, M_c)$ over descendant bins, with proportional clipping inside each bin. At finer scales, up to the chosen leaf K_0 or h , use its full descendant squares with the same envelope order. The displayed comparison verifies Theorem 2.10 at the crossing of the clipping threshold; ordinary convolution verifies the other transitions. Root and transition square sums are $S^{o(1)}$. Sorting actual transition amplitudes by their ratios r to b_T , Proposition 2.6 from the endpoint gives energy $S^{o(1)}r^{2-q}D$; the square bound of this class is $S^{o(1)}r$. The ratios are polynomially bounded in S by Cauchy, and negligible negative powers make no root contribution. Theorem 2.11, with its exponent slack sent to zero, therefore gives

$$|Y| \leq S^{o(1)} D_i H_i^{-k-2}, \quad H_i/H_g = S^{o(1)}, \quad |Y| H_i^{k+2}/D_i = S^{o(1)}. \quad (3.15)$$

The reverse comparisons follow from (3.4). Rounding H_i or the fine endpoint by fixed factors has no effect; if $H_i \gtrsim h$, use the original h label grid.

3.3 Finite scale trees and the limiting profiles

The logarithmic argument will always be realized on a fixed finite grid before limits are taken. In the power case use

$$\kappa_i(s) = h^s, \quad 0 < s < e_i = 1, \quad \Lambda = h.$$

In the subpower case use

$$\kappa_i(s) = H_i S^s, \quad -\infty < s < e_i, \quad e_i = \min\{L, \lim \log_S(h/H_i)\}, \quad \Lambda = S.$$

The actual fine endpoint and the original transition masks are included in every tree. A transition mask is a distinct linear layer even if its exponent coincides with the endpoint exponent. Physical lattices need not be nested. Frequency labels retain their assigned ancestors despite bounded overlaps of the enlarged supports.

We spell out the selection order because subsequent directional deletions change intermediate amplitudes. No assertion below assumes that a selected amplitude ratio survives such a deletion.

Lemma 3.4 (Finite-grid selection and discard). *On a fixed finite grid one can select ratios $r_{i,s}$, retain a subpower fraction of the high cubes, and retain packet masks with the following properties. Put*

$$M_{i,s} = \kappa_i(s)^{k+2} (r_{i,s}/A_i)^q. \quad (3.16)$$

At every selected scale the original selected amplitudes obey $a_T \asymp r_{i,s} b_T$, and

$$Q_{i,s} := \sum_T |T| a_T^q b_T^{2-q} = \Lambda^{o(1)} D_i, \quad (3.17)$$

$$E_{i,s} := \sum_T |T| a_T^2 = \Lambda^{o(1)} D_i r_{i,s}^{2-q}, \quad (3.18)$$

$$\sum_{T \text{ active}} b_T^2 \leq \Lambda^{o(1)} |Y| / M_{i,s}. \quad (3.19)$$

An active cell has at most $\Lambda^{o(1)} M_{i,s}$ eligible witnesses in its prescribed subpower enlargement. All witnesses are eligible in the power case and at strict ancestors of transition bins. Within a subpower transition bin b , eligibility means $e_b^2(z) \leq \Lambda^{o(1)} M_c$.

More generally, a collection of actual outputs in fixed masks whose weighted q budget is at most $\Lambda^{-\varepsilon} D_i$, for fixed $\varepsilon > 0$, has negligible root contribution. The same assertion holds for error summands in any later restricted tree.

Proof. Fix a small number $\beta > 0$ which bounds the exponents of all norm losses, fixed-grid constants, logarithmic sums, and initial balance errors. Work far enough along the sequence that these losses are at most $\Lambda^{C\beta}$. Choose the distance cutoff exponent next, the eligibility and enlargement exponents after that, and the occupancy cutoff exponent last. Each is larger than the fixed comparability multiples of its predecessors. They will all tend to zero, in this order, along a slow diagonal.

First prove the discard statement, without assuming selection or saturation. For an error U with actual amplitudes and budget Q_U , Propositions 2.6 and 2.10 through its masks give, in the power case,

$$\#\{z \in Y : |U(z)| > t A_i\} \leq \Lambda^{C\beta} t^{-q} A_i^{-q} Q_U. \quad (3.20)$$

One can jump directly from a current layer to the root: no not-yet assembled intermediate masks are needed, and any transition restriction already fixed remains imposed. In the subpower case propagate to the gap edge. Sort the *actual* output amplitudes there by their ratios to the propagated weights. Cauchy bounds the nonnegligible ratios by fixed powers of S , since only finitely many scales are involved and the clipped square is bounded at the transition. Theorem 2.11 across the gap gives (3.20) with a further $S^{C\eta}$, where $\eta > 0$ is its arbitrarily small exponent slack. Choose η after the desired discard saving. Very small ratios have negligible contribution by absolute Cauchy, and larger ratios have summably small weighted budget. Thus $Q_U \leq \Lambda^{-\varepsilon} D_i$ allows both a vanishing threshold fraction and a power-small exceptional set, by choosing β, η and the exponent in t sufficiently small relative to ε . This proof uses actual error amplitudes throughout.

Assemble from leaves toward the root, with the original transition masks fixed. At each stage Cauchy gives a polynomial upper bound for the amplitude-to-weight ratios. In the subpower case this includes the quotient $\kappa_i(s)^{k/2}/A_i$, which is a fixed power of S at each fixed scale. Ratios below a sufficiently small inverse power are harmless by Cauchy and the discard estimate. There remain $O(\log \Lambda)$ dyadic classes. Select a class for the sum actually reaching the root before packetizing a smaller inverse width. A finite number of selections costs $\Lambda^{o(1)}$ in thresholds and high set size.

Here is the occupancy deletion at this same stage. If n_T counts eligible witnesses in the prescribed enlarged cell, local weight comparison and (3.8) give

$$\sum_T b_T^2 n_T \leq \Lambda^{\beta'} |Y|,$$

where β' accounts for the already chosen enlargement and eligibility. This also holds with full squares in the power case. For a candidate ratio class r delete cells with $n_T > \Lambda^\gamma M_s$, choosing $\gamma > \beta' + C\beta$. Their weighted charge is

$$Q_{\text{bad}} \lesssim r^q \kappa_s^{k+2} \sum_{\text{bad}} b_T^2 \leq \Lambda^{\beta'-\gamma} |Y| A_i^q \leq \Lambda^{\beta'-\gamma+C\beta} D_i.$$

Use (3.2) or (3.15) in the last inequality. The discard estimate therefore permits this pruning before the final selection at that layer. Similarly, a size supremum realized only at distance Λ^γ from the packet center carries a negative power budget if its local amplitude is power-smaller: retain an extra fixed decay margin in the supremum and apply the same estimate.

The upper bound for $Q_{i,s}$ follows from the weighted norm estimate through the already assembled masks. The reverse follows by applying (3.20) to the entire selected field, which still has its root high set. Since $a_T \asymp r_{i,s} b_T$, the formulas for $E_{i,s}$ and $\sum b_T^2$ follow algebraically. Taking the diagonal of the specified slacks proves the displayed subpower assertions. $\square$

Lemma 3.5 (Profiles). *After taking subsequences and finer finite rational grids there are nonnegative, nonincreasing, $k/2$ -Lipschitz functions f_i such that*

$$r_{i,s} = \Lambda^{f_i(s)+o(1)}, \quad f_i(e_i) = 0.$$

In the power case $f_i(0) = \lim \log_h A_i$. In the subpower case

$$0 \leq d_i(s) := f_i(s) + ks/2 \leq ke_i/2 \quad (s \leq 0). \quad (3.21)$$

Proof. For $s < s'$, weighted Cauchy between the two layers gives

$$r_{i,s} \leq \Lambda^{o(1)} (\kappa_i(s')/\kappa_i(s))^{k/2} r_{i,s'}.$$

Conversely, integrated spatial L^2 orthogonality gives $E_{i,s} \leq C_{\text{grid}} E_{i,s'}$. To justify it with masks, first remove the outermost mask by its bounded sup norm; then separate its immediate children by their bounded Fourier overlap. Repeat in this order. One never commutes all outer masks past a final fine frequency decomposition. Polynomial spatial weights can be retained using translated band-limited cutoffs. Formula (3.18), and $q > 2$, now give $r_{i,s} \geq \Lambda^{-o(1)} r_{i,s'}$. At the fine endpoint the ratio is one by the choice of the input sizes.

In the power case root witness sampling and local constancy give $|Y| A_i^2 \lesssim \Lambda^{o(1)} E_{i,s}$. Together with saturation and $|Y| = \Lambda^{o(1)} D_i A_i^{-q}$ this gives $r_{i,s} \leq \Lambda^{o(1)} A_i$. The reverse endpoint comparison comes from root Cauchy, $A_i \leq \Lambda^{o(1)} \kappa_i(s)^{k/2} r_{i,s}$, as $s \downarrow 0$. The same root Cauchy in the subpower case gives $f_i(s) + ks/2 \geq 0$. The fine endpoint and the Lipschitz bound give the other inequality in (3.21).

On each finite grid the logarithms are uniformly bounded and have these two one-sided Lipschitz bounds up to vanishing errors. A diagonal subsequence on rational parameters gives the asserted continuous profiles. Every later test involves only a fixed finite grid: include its parameters first and take the subsequence far enough that its logarithms approximate the limiting values to the desired accuracy. Grid complexity can equivalently increase sufficiently slowly. No rate of convergence to a particular slope is assumed. $\square$

Freeze the selected masks together with the ratios and budgets recorded in Lemma 3.4. These records will charge later deletions and supply the broad witnesses. The original transition amplitudes in (3.3) remain separate records for the final incidence count. After a deletion the fields receive fresh actual amplitudes and fresh propagated positive weights; neither set of recorded amplitudes is asserted to survive that deletion.

3.4 Positive paths and the occupancy identities

We next turn the selected packet hierarchy into a law on directions through a high point. Its purpose is to compare two quantities: the square mass available in a frequency cap, and the number of high points in a physical cell directed by a sampled leaf. The normalization of the law must be recomputed when directions are removed, although its occupancy scale $M_{i,s}$ and its charging bounds come from the frozen selection.

Fix one field and one of its finite trees. Initially use the selected tree itself. Later we use the same construction after the particular angular omissions of Lemma 3.8. Those omissions keep the frozen masks and retain the root high threshold. Their fresh weighted norm bound is proved in that lemma using the packet estimates and the discard statement; the construction below uses that bound, not any assertion that the old amplitudes survive the omissions.

Expand the chosen positive square propagation through the retained cells, keeping their assigned cap ancestry. A surviving path $\pi = (T_0, \dots, T_\ell)$ ends at a terminal physical input $\gamma = T_\ell$. Denote its successive nonnegative propagation factors by $k_0(z; \pi), \dots, k_\ell(z; \pi)$. The first evaluates at the root point z ; an ordinary parent-to-child factor is the native sampling weight $w_{T'}(z_T)$ from (2.23), with its chosen fixed normalization and edge-decay reserve. Retain each factor's dependence on the assigned cap ancestry and its evaluation point. With the input square s_γ^2 , the weight of the path is

$$s_\gamma^2 \prod_{j=0}^{\ell} k_j(z; \pi).$$

Let $\omega(\gamma)$ denote its terminal direction label. Sum these weights over the surviving paths with $\omega(\gamma) = \omega$ to define $J_z(\omega)$, and put

$$W(z) = \sum_{\omega} J_z(\omega).$$

In particular, terminal physical inputs with the same direction label are summed. Wherever $W(z) > 0$, the provisional direction law is $p_z(\omega) = J_z(\omega)/W(z)$. The proof removes distant paths and normalizes once more; all conclusions below concern that resulting near-path law. Later angular omissions likewise require fresh path sums and a fresh normalization.

The reference cap weights $b_\tau^2(z)$ retain the available descendant squares of the selected tree. In the power case these are full squares. In the subpower case strict transition ancestors use proportionally clipped bins, and finer levels use full descendant squares only in eligible transition bins. This retains the transition-center restriction and the eligibility condition in Lemma 3.4. Indeed, in the subpower case,

$$\sum_{b \text{ eligible}} e_b^2(z) \leq \Lambda^{o(1)} \sum_b \min(e_b^2(z), M_c) \leq \Lambda^{o(1)},$$

because eligibility gives $e_b^2(z) \leq \Lambda^{o(1)} M_c$. Thus these level-specific reference squares remain controlled even where full squares are used. They are distinct from the surviving path mass J_z . Their comparison with the law will hold at sampled labels, not uniformly at every label after omissions.

For finite-valued labels A and B , $\Pr(A | B)$ denotes $p_{A|B}(a | b)$ evaluated at the sampled values $a = A$, $b = B$. Its logarithmic information in base $\Lambda > 1$ is $-\log_\Lambda \Pr(A | B)$. In $\Pr(T_s(Z, \Omega_i) | \Omega_i)$, conditioning on Ω_i fixes the tilted lattice, and $T_s(Z, \Omega_i)$ is its sampled cell label.

Lemma 3.6 (Restricted path law). *For the selected tree, and for each of the directional restrictions constructed below, one can draw a uniform root point Z from a subpower fraction of Y and then a path to a fine direction Ω_i in each field. The intermediate active cells lie within subpower normalized distance of Z . The fields' paths may be drawn independently conditional on Z .*

For the reference cap square weights just defined, at a typical sampled label

$$\Pr(\tau \mid Z) = \Lambda^{o(1)} b_\tau^2(Z). \quad (3.22)$$

Let $T_s(Z, \Omega_i)$ be the cell containing Z in the disjoint scale- $\kappa_i(s)$ lattice tilted by Ω_i . At a typical draw,

$$\frac{\Pr(T_s(Z, \Omega_i) \mid \Omega_i)}{\Pr(Z \mid \Omega_i)} = \Lambda^{o(1)} M_{i,s}. \quad (3.23)$$

Typical means outside probability tending to zero on each fixed finite test, with any prescribed positive exponent tolerance.

Proof. We first show that the available high points have enough surviving path mass to normalize the law. The weighted norm estimate applied afresh to this restricted tree gives a budget at most $\Lambda^{o(1)} D_i$, with its actual modified amplitudes. At the root its integrand is

$$|V(z)|^q W(z)^{1-q/2}.$$

The number of high points where $W \leq \Lambda^{-\alpha}$ is at most $\Lambda^{o(1)-\alpha(q-2)/2} D_i A_i^{-q}$. Across the subpower gap the ratio-class argument in the proof of Lemma 3.4 adds only $S^{C\eta}$; choose η smaller than $\alpha(q-2)/(4C)$. Thus low path mass is exceptional. Full-background convolution and the transition eligibility test give $W \leq \Lambda^{o(1)}$.

Use a fixed extra polynomial decay power on each path edge. A path with an intermediate center farther than Λ^α from z has a comparably large normalized edge displacement, since the inverse widths increase toward the leaves and the nested-cap shears are bounded in the appropriate coordinates. Dropping the extra power on this edge still leaves a convolution bounded by the full background; hence such paths have mass $o(W(z))$ after the fixed extra margin is chosen larger than the path-mass lower-bound exponent requires. This uses the same fixed margin on each edge, not an increasing endpoint weight exponent. Redefine J_z and W to count only the retained near paths, and normalize by this new W . Relabel the final retained root set as Y ; its cardinality differs by only a subpower factor from the preceding set. Every retained path is eligible at its transition bin: its surviving transition-mask center has $e_b^2 \leq M_c$, and proximity to that center and polynomial weight comparison give $e_b^2(z) \leq \Lambda^{o(1)} M_c$. On the near paths, telescoping local weight comparisons and the subpower neighboring-cell multiplicities give

$$p_z(\omega) := J_z(\omega)/W(z) \leq \Lambda^{o(1)} a_\omega(z),$$

where a_ω is the corresponding available leaf square weight. If a final direction label sums many final input cells, only its last localization layer and intermediate active cells need be near z ; the input coefficients are summed in a_ω .

Summing this upper bound over descendants proves the upper half of (3.22). For its lower half, at any fixed root point the total probability of sampled labels satisfying $p_z(\tau) < \Lambda^{-\varepsilon} b_\tau^2(z)$ is at most $\Lambda^{-\varepsilon} \sum_\tau b_\tau^2(z) = \Lambda^{-\varepsilon+o(1)}$. This is a statement at sampled labels, not a uniform lower bound for all labels.

Here are both sides of (3.23). For a fixed leaf direction ω and a tilted s cell C , put

$$\mu(\omega, C) = |Y|^{-1} \sum_{z \in Y \cap C} p_z(\omega).$$

At the realized pair (z, ω) the ratio in question is exactly

$$|Y| \mu(\omega, C) / p_z(\omega). \quad (3.24)$$

For the upper bound, all contributing active s cells lie in a subpower enlargement of C . The occupancy deletion counts at most $\Lambda^{o(1)}M_{i,s}$ eligible points there. The leaf background varies by a subpower factor within the tilted cell, and path domination and the typical leaf lower bound compare each summand to $p_z(\omega)$. This proves the upper bound.

For the lower bound, let $J_T^{\text{dom}}(\omega)$ be the unnormalized sum of positive descendant-path weights from an active scale- s mask T , evaluated at its center z_T on the selected tree before angular deletions. Retain the transition-center restriction and proportional clipping, and sum all terminal physical cells with direction label ω . Later macro-dependent omissions give subcollections of this one majorant, without creating extra copies. Positive convolution gives $\sum_{\omega} J_T^{\text{dom}}(\omega) \lesssim C_{\text{grid}} b_T^2$. At the transition this uses both the full-square convolution bound and the bound by $C_{\text{grid}} M_c$ from retained centers; take their minimum. At finer cells use the full descendant weights.

Let $\mathcal{N}_s(\omega, C)$ consist of active masks in the assigned s -cap whose Λ^η native enlargements meet the tilted s -cell C , and put

$$a_{\omega, C} = \sum_{T \in \mathcal{N}_s(\omega, C)} J_T^{\text{dom}}(\omega).$$

This reference charge is used only for domination; the actual law $p_z(\omega) = J_z(\omega)/W(z)$ is normalized using the fresh surviving mass of the restricted tree. The near-ancestor kernel sum and $W(z) \geq \Lambda^{-o(1)}$ give the following pointwise bound. For fixed (ω, T) , only $\Lambda^{o(1)}$ cells C occur, because the two scale- s frames differ by a bounded normalized shear. Consequently

$$p_z(\omega) \leq \Lambda^{o(1)} a_{\omega, C}, \quad \sum_{\omega, C} a_{\omega, C} \leq \Lambda^{o(1)} \sum_{T \text{ active}} b_T^2 \leq \Lambda^{o(1)} |Y|/M_{i,s}.$$

At a strict transition ancestor, eligible contributions are comparable at the charging center up to the prescribed subpower factor; unique label ancestry charges each clipped bin only once. The total probability of fibers satisfying

$$\mu(\omega, C) < |Y|^{-1} \Lambda^{-\varepsilon} M_{i,s} a_{\omega, C}$$

is at most $\Lambda^{-\varepsilon+o(1)}$. For every other sampled fiber, (3.24) proves the lower bound. Replacing boxes by balls or bounded-overlap coverings changes these sampled logarithms by $o(1)$: the total probability of a cell power-smaller than one of its boundedly many neighbors is power-small. The same argument handles subpower distortions. $\square$

Lemma 3.7 (Conditioning directions). *Inside a fixed cap τ_l , let C be a physical cell over which every descendant velocity has drift at most the endpoint width, and whose time range is at most the endpoint duration. Adding an independently redrawn descendant direction as conditioning changes the logarithmic information of position inside C by $o(1)$ at a typical draw. The cell may depend on a different conditionally independent drawn direction. The conclusion also holds for subpartitions and subpower distortions. In the power case it holds for a direction from another field whenever its full weights have the same bounded drift on C .*

Proof. Use as reference direction law the endpoint square weights at the center of C , normalized within τ_l . Clip whole transition bins proportionally if τ_l is a strict ancestor. The drift assumption makes these reference weights comparable throughout C . By (3.22) and path domination, the conditional density of the direction relative to this law is at most $\Lambda^{o(1)}$ at typical data. Its average conditional density is at least $\Lambda^{-o(1)}$ at a typical sampled direction: the exceptional mass is bounded by summing reference weights exactly as in the lower half of (3.22). Bayes' identity now bounds the likelihood ratio for position by $\Lambda^{o(1)}$.

For clarity, the remaining one-sided information comparison requires no independence assumption. If p and p' are the probabilities of a realized finite label before and after additional conditioning,

then $\mathbb{E}[p/p'] \leq 1$. Markov's inequality implies $p/p' \leq \Lambda^\varepsilon$ outside probability $\Lambda^{-\varepsilon}$. Apply this to each member of a finite chain of partitions. Combining it with the reference-law comparison proves equality of the relevant information increments to arbitrary logarithmic tolerance. $\square$

3.5 Angular pruning inside one profile block

Lemma 3.8 (Angular test and original broad witnesses). *Fix one field, a cap parameter l away from the endpoints and the transition seam, and a finite mesh of $0 < v \leq u$ with $l + Cu < e_i$. Suppose that*

$$f_i(l) - f_i(l + v) \geq xv$$

on this mesh, for some $x > 0$. For a number $\eta > 0$ depending only on x, k , one may restrict the frozen tree so that, at typical (Z, τ_l) , its new conditional direction law assigns mass at most $\Lambda^{-\eta v + o(1)}$ to every affine hyperplane strip of width Λ^{-v} in the native frequency coordinates of τ_l .

At one arbitrarily small mesh precision $v_0 > 0$, the sampled active l cells may also be required to have broad witnesses for their original pre-modification fields. Namely, there are $k + 1$ $(l + v_0)$ cap groups, transverse throughout their supports with determinant at least $\Lambda^{-C_k v_0}$, each having magnitude at least $\Lambda^{-C_k v_0} r_{i,l} b_{\tau_l}(Z)$ at a point within native distance $\Lambda^{C_k v_0}$ of the cell.

The high root set loses only a vanishing fraction, and Lemmas 3.6 and 3.7 apply to the modified tree with the same recorded occupancy sizes. This construction is made for one fixed block at a time; no simultaneous assertion over all blocks is needed.

Proof. Suppress the field index. In the native coordinates of τ_l , use macro boxes of side Λ^{3u} . The endpoint descendant weights, divided by the weight of τ_l , vary within bounded factors on these boxes: all affected cell widths and durations are at most $\Lambda^{2u + o(1)}$, and the endpoint is separated by Cu . If a transition ancestor occurs, clip each entire bin proportionally.

For each mesh value v , greedily remove all $(l + v)$ caps intersecting an enlarged hyperplane neighborhood whose reference measure is at least $\Lambda^{-2\eta v}$. There are at most $O(\Lambda^{2\eta v})$ choices, and the residue has the desired upper bound before normalization. Carry out the omissions from coarser to finer mesh values, so earlier omissions only restrict subsequent masks.

We verify that their contribution at level l is cheap. Let $q' = 2(k + 1)/(k - 1) > q$, the lower-dimensional exponent. The original selected budget at $l + v$ is bounded by

$$Q_{q', l+v} \leq \Lambda^{o(1)} r_{l+v}^{q' - q} D.$$

This follows from the recorded ratio class, before any modification. The plate-supported version of Proposition 2.6 at exponent q' propagates each omitted hyperplane contribution through *all* intervening masks. Plate support is inherited by each descendant in its own normalized coordinates. Summing the omitted planes costs at most $\Lambda^{2\eta q' v}$. On the active l masks, Hölder with measure $b_T^2 |T|$ and $\sum b_T^2 |T| \leq \Lambda^{o(1)} D r_l^{-q}$ gives

$$\begin{aligned} Q_{q, l}(\text{error}) &\leq (\Lambda^{2\eta q' v + o(1)} r_{l+v}^{q' - q} D)^{q/q'} (\Lambda^{o(1)} D r_l^{-q})^{1 - q/q'} \\ &\leq D \Lambda^{2\eta q v + o(1)} (r_{l+v}/r_l)^{q(q' - q)/q'}. \end{aligned}$$

Choose $2\eta < x(q' - q)/(2q')$. The profile assumption then makes this a negative fixed power budget, and Lemma 3.4 discards it at the root. Earlier restrictions are harmless: use actual modified amplitudes when propagating, and bill the initial restricted subfields to their original fine budgets.

We include the localization justification for the fact that the direction subsets can vary by macro box. Assign each l packet by its center to such a box and alter the field *before* its Schwartz packet

cutoff. No rough physical cutoff is inserted. At a single hierarchy step, packets centered away from a target macro box have an arbitrarily large fixed inverse polynomial box-distance gain, by their cutoff and supremum tails. The denominator weights may be left in the child coordinates, using Theorem 2.10; one need not compare them across an entire macro box. These gains form a summable convolution in macro offsets. An equivalent algebraic formulation is useful. For a summable polynomial macro weight Ψ and exponent $p > 2$, replace

$$b_T \quad \text{by} \quad b_T \Psi(z_T)^{-1/(p-2)}.$$

Then $a_T^p b_T^{2-p}$ acquires exactly the factor $\Psi(z_T)$. Polynomial comparability of Ψ requires only a fixed additional edge-decay margin, since all affected widths are below the macro scale. Use this for $p = q, q'$ at each step; the same margin suffices at every depth. Summing the target-box estimates bills each initial packet with bounded total weight. This proves the stated norm charge for varying direction restrictions, and also proves the fresh restricted-weight norm bound used in Lemma 3.6.

For broad witnesses, use separately macro boxes of side Λ^{3v_0} *before* modifying directions. In each box mark a group at level $l + v_0$ if its original contribution through the frozen intervening masks has supremum at least $\Lambda^{-C_k v_0} r_l b_{\tau_l}$ in a fixed enlargement. There are at most $\Lambda^{k v_0 + o(1)}$ groups, so for C_k sufficiently large the unmarked sum cannot supply a high witness defining an active l packet. The far-supremum deletion in Lemma 3.4 ensures such a witness occurs within the stated enlargement.

The elementary maximal-simplex test gives a dichotomy for the marked centers: either $k + 1$ of them have successive affine widths at least a constant times Λ^{-v_0} , or all marked centers lie in a constant enlargement of a hyperplane strip of that width. In the first case the determinant is at least a fixed power of Λ^{-v_0} ; enlarge the separation constants so it holds uniformly on the chosen supports. In the second case the same q' calculation shows that the marked contribution is cheap at level l . Thus the nonbroad masks themselves have negligible weighted q budget and may be deleted first.

Finally reconstruct the positive path law for the modified fields, as in Lemma 3.6. Low total path mass is exceptional by its fresh norm bound. Its conditional law is dominated by the surviving reference descendant weights divided by $b_{\tau_l}^2(Z)$; the typical reverse cap comparison loses only a subpower. The heavy strip cutoff $\Lambda^{-2\eta v}$ therefore implies the claimed $\Lambda^{-\eta v + o(1)}$ angular bound. Occupancy upper bounds still refer to the same frozen masks, and the charging proof of (3.23) is unchanged. The broad witnesses used later are the recorded original subfields, whose selected amplitudes remain available for norm and square estimates. $\square$

3.6 Ball inflation and the two allowed slopes

Lemma 3.9 (Ball inflation in a cap). *Work inside a cap τ_l of the preceding lemma, and suppose $f(l + v) - f(l) = -xv + O(\varepsilon u)$ on the required finite mesh in $[0, 2u]$. Put $d_0 = 1 - 2x/k$. For the isotropic cell Q_v of radius Λ^v in this cap's native coordinates, define the sampled effective occupancy*

$$N_v = \Pr(Q_v(Z) \mid \tau_l) / \Pr(Z \mid \tau_l).$$

For $u \leq v \leq 2u$ on that mesh,

$$\log_{\Lambda}(N_v/M_l) \leq (k + 1)d_0 v + O(\varepsilon u). \quad (3.25)$$

The error can be made arbitrarily small by taking the broad precision v_0/u small, refining the finite mesh, and then taking the sequence limit. All spatial tests stay on one side of the transition seam.

Proof. Every point under the cap conditioning is near an active l cell. Such a cell has at most $\Lambda^{o(1)}M_l$ eligible points. Path domination, the typical reverse cap comparison, and background variation bound their relative conditional weights by subpowers. It therefore suffices to count active l cells in an isotropic ball. They have the original broad witnesses from Lemma 3.8. Pigeonholing the separated tuple costs $\Lambda^{O_k(v_0)}$, since there are only a fixed power of Λ^{v_0} possible labels.

Write $\kappa_s = \Lambda^s$ in the l coordinates and, for the j th broad group, let $\mathcal{E}_{j,s}$ be the local square sum of the original selected packet amplitudes with their native weights at relative scale s . The witness inequalities, including their allowed offsets, imply at every counted unit cell

$$r_l b_{\tau_l}(Z) \lesssim \Lambda^{O_k(v_0)+o(1)} \prod_{j=1}^{k+1} \mathcal{E}_{j,v_0}^{1/(2(k+1))}.$$

Masks between l and $l + v_0$ have bounded absolute sums, and the native weights absorb the witness offsets. The background b_{τ_l} is comparable over the larger test ball.

For $s < t \leq 2s$, average the resulting product of squares with exponents $1/k$ on a κ_t ball. Proposition 2.2 gives

$$\frac{1}{|B_{\kappa_t}|} \int_{B_{\kappa_t}} \prod_{j=1}^{k+1} \mathcal{E}_{j,s}^{1/k} \lesssim \Lambda^{O_k(v_0)} \prod_{j=1}^{k+1} \left(\frac{1}{|CB_{\kappa_t}|} \int_{CB_{\kappa_t}} \mathcal{E}_{j,s} \right)^{1/k}. \quad (3.26)$$

Indeed these are nonnegative sums essentially constant on tubes of width κ_s and duration $\kappa_s^2 \geq \kappa_t$. On the ball replace them by columns crossing the ball, where each envelope is comparable. Such a column costs a cross section of width κ_s times length at least κ_t in the enlarged-ball average. The quantitative determinant factor is $\Lambda^{O_k(v_0)}$. Weighted offsets are summed with the fixed integrable envelope powers.

We verify that each last average is bounded by the corresponding $\mathcal{E}_{j,t}$, even when intermediate mask levels are skipped. This verification prevents an aspect-ratio loss. Put $S_0 = \kappa_s$ and $T_0 = \kappa_t$. Reproduction in the native s coordinates bounds the square of a weighted local supremum by a normalized weighted L^2 integral of the pre-cutoff field. After summing s cells, the displacements from an averaged point have the form

$$(S_0 U + 2v_s S_0^2 w, S_0^2 w), \quad \text{density} \leq C(1 + |U| + |w|)^{-P}.$$

For each fixed displacement and time, integrate in space over width T_0 , replacing the ball by a smooth polynomial weight. Remove the next mask by its bounded sup norm and separate its immediate child caps by spatial almost orthogonality; localization at width T_0 preserves bounded Fourier overlap. Iterate in the actual order of the masks. Translated band-limited cutoffs positive on boxes prove the same assertion for the polynomial weight. At the final layer, pointwise Cauchy bounds the fields by the t packet envelopes.

In the frame of a final t cap, the normalized shift just displayed has sizes

$$(S_0/T_0)(U + O(1)w), \quad (S_0/T_0)^2 w,$$

because its velocity differs from v_s by $O(S_0^{-1})$. Convolution with this contracted shift density preserves the order P of the final weight by the distance and volume comparison in Theorem 2.10. The auxiliary spatial ball weight has all the finitely many moments needed for this comparison. Thus no factor depending on T_0/S_0 is lost. One can use the same fixed P at each inflation step, with a fixed excess supremum margin.

At the terminal scale, the recorded ratio class and weight convolution give

$$\mathcal{E}_{j,v} \lesssim \Lambda^{o(1)} r_{l+v}^2 b_{\tau_l}^2(Z).$$

Iterate (3.26) along a finite chain from v_0 to v , each step satisfying $s < t \leq 2s$. There are $O(1 + \log(v/v_0))$ steps; covering larger balls by smaller balls at each step gives the cell-count bound

$$\Lambda^{(k+1)v + O_k(v_0 \log(v/v_0)) + o(1)} (r_{l+v}/r_l)^{p_k}.$$

Since $p_k x = (k+1)(1 - d_0)$, this proves (3.25). The finite mesh can include the whole inflation chain from the outset, and v_0/u can be as small as needed before taking logarithmic limits. $\square$

Proposition 3.10 (Binary slope rigidity). *Every profile satisfies*

$$f'_i(s) \in \{0, -k/2\} \quad \text{for almost every } s. \quad (3.27)$$

Proof. Suppose a differentiability point, away from the transition seam, has slope $-x$ with $0 < x < k/2$. Choose a short rational block based at l very near this point and a finite relative mesh so that $f(l+v) - f(l) = -xv + O(\varepsilon u)$ throughout $[0, 2u]$. Differentiability allows errors arbitrarily small relative to every mesh spacing. Put $d_0 = 1 - 2x/k \in (0, 1)$ and apply the angular test with its fixed positive exponent, together with the broad-witness test at a much smaller v_0 .

Condition on τ_l and Q_{2u} . Rescale this parent isotropically to bounded size and write $\Delta = \Lambda^{-u}$. The information of the fine isotropic cell is

$$\log_{\Lambda}(N_{2u}/N_u) + o(1).$$

Projecting parallel to a drawn direction gives information

$$\log_{\Lambda}(N_{2u}/M_{l+u}) + o(1).$$

Its inverse-image cells, clipped to the parent, are comparable to T_{l+u} : transverse width is Λ^u and length is Λ^{2u} . Thus Lemma 3.6 gives the second formula. Adding independent directions costs $o(1)$ information by Lemma 3.7, because all directions in τ_l drift by at most the endpoint width on this box.

Draw $k+1$ directions conditionally independently. The angular test shows, with probability tending to one, that their lifted vectors form a basis with inverse norm at most $\Lambda^{O(\varepsilon u)}$. Indeed expose them successively; proximity to the span of the previous ones is an affine hyperplane-strip event in the frequency coordinates, and use one mesh precision much smaller than u . In this basis each projection records all but one coordinate, up to the same precision loss. Order the coordinates once and use the chain rule for their information. The information of a coordinate conditioned on all its predecessors is no larger, up to the sampled $O(\varepsilon u)$ slack of Lemma 3.7, than its information when one predecessor has been omitted. Each coordinate is counted k times in the $k+1$ projections. This is the finite-partition Loomis–Whitney information inequality, and gives

$$\log_{\Lambda}(N_{2u}/M_{l+u}) \geq \frac{k}{k+1} \log_{\Lambda}(N_{2u}/N_u) - O(\varepsilon u).$$

The sample form follows from Markov's inequality for the finitely many conditional probability ratios, as in the proof of Lemma 3.7; no uniform pointwise entropy inequality is being asserted.

Let $g(v) = \log_{\Lambda}(N_v/M_l)$. The ratio definition gives

$$\log_{\Lambda}(M_{l+u}/M_l) = (k+2)u + q(f(l+u) - f(l)) = (k+2)d_0u + O(\varepsilon u).$$

The preceding inequality becomes

$$g(2u) + kg(u) \geq (k+1)(k+2)d_0u - O(\varepsilon u).$$

Lemma 3.9 bounds $g(u)$ and $g(2u)$ above by $(k+1)d_0u$ and $2(k+1)d_0u$, respectively. Their weighted sum is exactly the last right side. Hence both upper bounds are equalities up to $O(\varepsilon u)$.

We now apply the robust discretized projection theorem in Proposition 2.3, with ambient dimension $k+1$, projection dimension k , and $\alpha = (k+1)d_0 \in (0, k+1)$. We give the set extraction because exceptional points can depend on direction. Choose a cap and parent box with conditional success probability close to one for all the finite comparisons. Within this fixed cap and parent box, the probabilities of typical Δ cells, without conditioning on the drawn direction, lie between $\Delta^{\alpha+O(\varepsilon)}$ and $\Delta^{\alpha-O(\varepsilon)}$. Take their centers to form one set $\mathcal{A}$. It is Δ separated and

$$\Delta^{-\alpha+O(\varepsilon)} \leq |\mathcal{A}| \leq \Delta^{-\alpha-O(\varepsilon)}.$$

For coarser balls, the mesh instances of (3.25), divided by the saturated parent occupancy, give the probability upper bounds with exponent α . Passing from probability to number by the preceding individual-cell bounds gives precisely the nonconcentration inequalities for $\mathcal{A}$, with $O(\varepsilon)$ exponent tolerance. Intermediate radii follow by nesting between consecutive mesh values, whose ratio is chosen sufficiently fine on the logarithmic scale.

Restrict the joint position-direction law first to successful positions, then to directions for which conditional spatial and projected success have probability bounded below. Starting with success close to one makes both restrictions retain fixed positive mass. Mixing the conditional angular laws preserves their hyperplane nonconcentration, and these fixed-mass restrictions change it by only a constant. For every retained direction, its successful positions occupy at least $\Delta^{O(\varepsilon)}|\mathcal{A}|$ cells, because its conditional individual-cell probabilities have the same upper bound. The conditional projected information bound shows that these cells project to at most

$$\Delta^{-kd_0-O(\varepsilon)} = \Delta^{-k\alpha/(k+1)-O(\varepsilon)}$$

projection cells. The direction law obeys the required Schubert nonconcentration: for lifted vectors $(2\xi, 1)$, the relevant Schubert test is exactly an affine hyperplane-strip test on ξ , up to fixed constants. The theorem supplies a fixed positive projection gain for every sufficiently large subset for some retained directions, contradicting this upper bound once ε and the finite mesh tolerances are sufficiently small. All nonconcentration exponents were fixed using $x > 0$ before ε was sent to zero. This proves (3.27). $\square$

3.7 The location of the full-slope interval

Call a differentiability parameter a *full-slope point* if $f'_i = -k/2$, and a *zero-slope point* if $f'_i = 0$. The logarithmic derivative of $M_{i,s}$ is zero at the former and $k+2$ at the latter.

Lemma 3.11 (Midpoint closure). *For almost every pair $l < a$ of full-slope points in one field, $(l+a)/2$ is a full-slope point. Consequently each field's full-slope set is an interval modulo null sets. In the power case, if a is a full-slope point in one field, then $a/2$ is a full-slope point in every other field, for almost every such a .*

Proof. Use pairs for which all relevant parameters are differentiability points, avoiding the one possible transition seam. Fubini's theorem permits this for almost every pair. Set $b = (l+a)/2$ and

choose $0 < \eta \ll \sigma \ll a - l$. By the angular test in a short block above l , two independently drawn paths inside τ_l have velocity difference Δv satisfying

$$\kappa_i(l)^{-1} \Lambda^{-\eta} \lesssim |\Delta v| \lesssim \kappa_i(l)^{-1}$$

with probability tending to one. Work inside the second path's parent tube at $b + \sigma/2$. Adding the first direction costs no logarithmic information by Lemma 3.7; the maximal relative drift there is

$$\kappa_i(l)^{-1} \kappa_i(b + \sigma/2)^2 = \kappa_i(a + \sigma),$$

which is below the endpoint width when the block and σ are chosen in the interior. We used the exact scale identity $\kappa_i(b)^2 = \kappa_i(l) \kappa_i(a)$.

Suppose b were a zero-slope point. The occupancy identity between its scales $b - \sigma/2$ and $b + \sigma/2$ gives information increment $(k + 2)\sigma + o(\sigma)$. Spatial labels have at most $k\sigma + o(\sigma)$ information, so the time label has at least $2\sigma + o(\sigma)$.

On the other hand, a spatial intercept label for the first path at scale $a - \sigma$ almost determines this time label. The spatial uncertainty in the second parent is $\kappa_i(b + \sigma/2) \ll \kappa_i(a - \sigma)$. Once the first intercept is specified to width $\kappa_i(a - \sigma)$, the relative motion determines time to accuracy at most

$$C \frac{\kappa_i(a - \sigma)}{|\Delta v|} \leq C \kappa_i(l) \kappa_i(a - \sigma) \Lambda^\eta = C \kappa_i(b - \sigma/2)^2 \Lambda^\eta.$$

There are therefore at most $\Lambda^{\eta+o(1)}$ possible smaller time labels. The whole second parent meets only boundedly many first parents at $a + \sigma$, by the preceding drift identity and the smaller time extent. Since a is full slope, the first-path information increment from $a + \sigma$ to $a - \sigma$ is $o(\sigma)$. It remains an upper bound after conditioning on the second parent, and the extra first-parent label has negligible cost. The chain rule would bound the second time-label information by $\eta + o(\sigma)$, contradicting $2\sigma + o(\sigma)$. First let the sequence tend to its limit at fixed rational approximations, then let $\eta/\sigma \rightarrow 0$ and $\sigma \downarrow 0$.

In the power case repeat the same geometry with the second path in another field and with $l = 0$. Root transversality gives $|\Delta v| \geq \Lambda^{-o(1)}$, so the same time-recovery bound holds and the drift is $O(h^{a+\sigma})$. The conditioning rule applies to the other field's full weights. Full slope at a in the first field therefore forces full slope at $a/2$ in the second field.

We finish with the measure-theoretic assertion. Let E be a measurable set satisfying the midpoint rule for almost every pair of its points. For two density points x, y choose equally short intervals centered there on which E has relative measure greater than $9/10$. The convolution of their restricted indicators is strictly positive on a neighborhood of $x + y$: after a small translation the two high-density subsets still intersect in positive measure. Fubini and the midpoint rule imply that a neighborhood of $(x + y)/2$ belongs to E modulo a null set. Taking $x = y$ shows that the density points form an open set modulo null sets; taking arbitrary x, y shows its midpoint convexity. Repeated midpoints, followed by openness, fill the interval between any two such points. Thus E is an interval modulo null sets. $\square$

In the power case let $s_H = \lim \log_h H_i$, the same for every field by (3.2). It is positive, since $A_i \geq h^c$. The endpoint values of f_i and the two allowed slopes show that its full-slope interval has length s_H . If the left endpoints of these intervals had positive minimum, the cross-field halving rule would give a smaller left endpoint, a contradiction. An interval beginning at zero then forces every other interval to begin at zero by that rule. Therefore

$$f_i(s_H) = 0 \quad \text{for every } i. \tag{3.28}$$

The analogous subpower conclusion needs control at the unbounded left endpoint. We give the incidence calculation explicitly.

Lemma 3.12 (Cross-field one-step inflation). *Let $K \leq R \leq K^2$ be two inverse widths in a fixed finite packet tree. At scale K , let W_j , $1 \leq j \leq m = k+1$, be the original selected packet fields testing the root high set, with no outer smaller-inverse-width mask. Each W_j is assembled from exactly its selected scale- R descendants through the frozen bounded masks and scale- K localizers; no omitted branch contributes to W_j . The witnesses below test these same masked fields W_j . Suppose their frequency labels, including fixed enlargements, are δ -transverse. Write $\mathcal{E}_{j,K} = \sum_T a_{j,T}^2 w_T$ for their local square envelopes and $\mathcal{E}_{j,R}$ for the envelopes of their descendants at scale R in the same frozen tree. Let Y be a set of distinct unit cubes with separate field witnesses at thresholds A_j . Suppose that, on every K -ball meeting Y ,*

$$K^{k/2} \mathcal{E}_{j,K}(z)^{1/2} / A_j \leq U, \quad 1 \leq j \leq m,$$

at one, hence every, point z in that ball, where $U \geq 1$. Then for every fixed $\varepsilon > 0$ and every R -ball centered at a surviving root point z ,

$$\#(Y \cap B_R(z)) \lesssim_{\varepsilon, \mathcal{G}} \delta^{-C_\varepsilon} U^\varepsilon R^{k+1} \prod_{j=1}^m (\mathcal{E}_{j,R}(z) / A_j^2)^{1/k}. \quad (3.29)$$

The constant may depend on the fixed finite grid $\mathcal{G}$, but not on the aspect ratio R/K .

Proof. Cover $B_R(z)$ by boundedly overlapping K -balls D . On one meeting Y , the separate-unit-supremum form of Theorem 2.8, with $p = p_k + \varepsilon$, and comparability of packet envelopes on D give

$$\#(Y \cap D) \lesssim_\varepsilon \delta^{-C_\varepsilon} K^{kp/2} \prod_j (\mathcal{E}_{j,K}(z_D) / A_j^2)^{p/(2m)}.$$

Since $kp_k/2 = k+1$ and $p_k/(2m) = 1/k$, this is

$$C_\varepsilon \delta^{-C_\varepsilon} K^{k+1} \prod_j (\mathcal{E}_{j,K}(z_D) / A_j^2)^{1/k} \left[K^{k/2} \prod_j (\mathcal{E}_{j,K}(z_D) / A_j^2)^{1/(2m)} \right]^\varepsilon.$$

The bracket is at most U , up to a fixed factor. Summing over D and replacing the sum by an integral uses only the fixed variation of each envelope on a K -ball.

For these globally transverse groups, the local weighted cylinder consequence of Theorem 2.2 is

$$\frac{1}{|B_R|} \int_{B_R} \prod_j \mathcal{E}_{j,K}^{1/k} \lesssim \delta^{-C} \prod_j \left(\frac{1}{|CB_R|} \int_{CB_R} \mathcal{E}_{j,K} \right)^{1/k}.$$

To verify its normalization, expand the polynomial envelopes on B_R into summably weighted translated columns of radius K . Their time length is at least R because $R \leq K^2$. A column meeting the ball contributes a fraction comparable to $(K/R)^k$ of its coefficient to the average on a fixed enlargement. The radius- K multilinear Kakeya estimate contributes K^{k+1} ; the $m = k+1$ averages contribute $(R/K)^{k+1}$, giving exactly R^{k+1} . The fixed reserve in the packet envelope power sums the translated-column tails. This is the profile-free geometric calculation in (3.26), with the determinant factor now δ^{-C} .

For each field separately, the weighted spatial L^2 calculation following (3.26) gives

$$\frac{1}{|CB_R(z)|} \int_{CB_R(z)} \mathcal{E}_{j,K} \lesssim_{\mathcal{G}} \mathcal{E}_{j,R}(z).$$

Indeed reproduce the coarse weighted suprema, remove each intervening bounded mask before separating its immediate child caps by spatial almost orthogonality, and then use the final R -packet envelope. In the final cap frame the normalized shift has sizes $((K/R)(U_0 + O(1)w), (K/R)^2w)$, so convolution preserves the same fixed polynomial weight, as in the calculation preceding (3.26) and following it. No power of R/K is lost. Combining these estimates proves (3.29). $\square$

Lemma 3.13 (Vanishing left limit). *In the subpower case,*

$$\lim_{s \rightarrow -\infty} d_i(s) = 0 \quad \text{for every } i. \quad (3.30)$$

Consequently $f_i(0) = 0$, including when $e_i = 0$.

Proof. Use the path law without an extra angular test, retaining each path's attachment to an original surviving H_i mask. Fix a field, write $H = H_i$, $K = HS^s$, and let μ_z be the original local direction law at a typical root point z . The clipped cap bound and path domination give the slot estimate

$$\mu_z(B(\omega, r)) \lesssim S^{C+o(1)} r^k, \quad r \geq K^{-1}. \quad (3.31)$$

Indeed a slot at inverse width K contains at most its frequency volume times H^k , up to the appropriate bounded grid enlargement, transition bins, each with clipped weight at most $H^{-k}S^{C_3}$; path normalization loses only a subpower. The same bound summed over slots proves the displayed formula.

There is also a lower bound at a typical sampled direction, simultaneously at all relevant radii:

$$\mu_z(B(\omega, r)) \gtrsim S^{-1} r^k. \quad (3.32)$$

To prove this without a number-of-radii loss, remove maximal dyadic cubes on which the measure is less than cS^{-1} times volume. The maximal cubes are disjoint and lie in a bounded frequency region, so their total measure is $O(S^{-1})$. Use finitely many shifted dyadic grids to put a comparable cube containing ω inside any ball in question. Every violating ball is thereby charged to one of the removed cubes.

Let $y \in Y$ and $\tau = |t_y - t_z|$. The probability, under the typical-direction restriction, that y belongs to the sampled enlarged K tube is at most

$$S^{C+o(1)} (K/\tau)^k. \quad (3.33)$$

Only $\tau \lesssim K^2 S^{o(1)}$ matters. Hence the angular window K/τ is no smaller than the slot precision up to a subpower, and (3.31) proves the bound.

If this event is nonempty, choose a typical direction ω_0 realizing it. Every original path with direction within $c \min(1, H/\tau)$ of ω_0 has its attached enlarged H mask containing y . Those masks are already near z , the extra transverse displacement is $O(H)$, and their enlarged duration contains the relevant time interval. Thus (3.32) gives

$$\Pr_{\text{original}} \{y \text{ hits the attached enlarged } H \text{ mask}\} \gtrsim S^{-1-o(1)} \min(1, H/\tau)^k.$$

For $\tau \geq KS^{C_4}$, divide (3.33) by this last bound. For $\tau \leq H$ use $K/\tau \leq S^{-C_4}$; for $\tau \geq H$ use $K/H = S^s$. This gives the ratio bound

$$S^{C+1+o(1)} (S^{-kC_4} + S^{ks}).$$

Sum this comparison *once* over y . Every attached original mask has occupancy at most S^{C_1} by the deletion preceding (3.7). Therefore

$$\mathbb{E}[\text{occupancy at time distance} \geq KS^{C_4}] \lesssim S^{C+C_1+1+o(1)}(S^{-kC_4} + S^{ks}) = o(1)$$

for sufficiently large fixed C_4 and $s < -2C_4$. There is no iteration over a number of intermediate scales.

The occupancy identity (3.23), together with the typical leaf probability comparisons, turns its effective occupancy into an ordinary count up to $S^{o(1)}$. With probability tending to one there are no far points, so a ball of radius $O(KS^{C_4})$ contains at least

$$M_{i,s}S^{-o(1)} = S^{qd_i(s)-o(1)} \quad (3.34)$$

root points. The identity is exact on the logarithmic scale:

$$M_{i,s} = (H_i S^s)^{k+2} (S^{f_i(s)} / H_i^{k/2})^q = S^{qd_i(s)},$$

because $kq/2 = k + 2$.

For the upper bound use a common ball scale at $s' = s + C_4 + 1 < 0$. The inverse widths in the fields differ by only subpowers, by (3.15). Choose a common first width below all field-specific first widths and a common terminal width above all field-specific terminal widths, each within an $S^{o(1)}$ factor. Inserting these levels into the fixed grid preserves the tested field; the local polynomial-weight convolution and bounded-multiplicity comparison in Theorem 2.10 change its square-envelope bounds by at most $S^{o(1)}$. Begin at the first gap-edge parameter $s_{\min} \leq s$ of a fixed finite grid, where no earlier smaller-inverse-width masks intervene. Write $\mathcal{E}_{j,s}(z) = \sum_T a_{j,T}^2 w_T(z)$ for the local square envelope of the original selected outputs at that layer; this is distinct from the integrated energy $E_{j,s}$ in (3.18). On every first-edge ball carrying witnesses, the selected ratio class and clipped square give $\kappa(s_{\min})^{k/2} \mathcal{E}_{j,s_{\min}}^{1/2} / A_j \leq S^C$: the remaining exponent is $d_j(s_{\min}) \leq ke_j/2$. The transversality loss is $S^{o(1)}$. We may apply Theorem 3.12 once from this first edge to s' , because for every fixed finite range

$$\frac{\kappa(s_{\min})^2}{\kappa(s')} = H_i S^{2s_{\min}-s'} \longrightarrow \infty.$$

Here H_i grows faster than every fixed power of S . The single step keeps the determinant power and the ε cost independent of grid length. At the root-centered terminal ball, $\mathcal{E}_{j,s'} \leq S^{o(1)} r_{j,s'}^2$ by the clipped square bound. Substitution in (3.29) gives the count upper bound

$$\begin{aligned} S^{o(1)+C\varepsilon} \kappa(s')^{k+1} \prod_j (\mathcal{E}_{j,s'} / A_j^2)^{1/k} &\leq S^{(k+1)s' + (2/k) \sum_j f_j(s') + o(1) + C\varepsilon} \\ &= S^{\frac{p_k}{k+1} \sum_j d_j(s') + o(1) + C\varepsilon}. \end{aligned} \quad (3.35)$$

The H_j factors cancel since their ratios are subpowers. Take limits for this finite comparison and then send ε to zero. Equations (3.34) and (3.35) yield

$$qd_i(s) \leq \frac{p_k}{k+1} \sum_j d_j(s + C_4 + 1).$$

By (3.21), every d_i is bounded, nonnegative, and nondecreasing. Let $L_i = \lim_{s \rightarrow -\infty} d_i(s)$. Sending s to $-\infty$ and summing the preceding inequalities gives

$$(q - p_k) \sum_i L_i \leq 0.$$

Since $q > p_k$ and $L_i \geq 0$, all the limits vanish.

The full-slope interval must extend to the unbounded left endpoint: otherwise f_i is constant sufficiently far to the left and $d_i(s) = f_i(s) + ks/2$ becomes negative. On this interval $d'_i = 0$, so the just proved limit gives $d_i = 0$. After its right endpoint f_i is constant, and its fine endpoint value makes that constant zero. Continuity at the right endpoint therefore places it at $s = 0$. If the full-slope interval fills the domain, the endpoint value instead gives $e_i = 0$. In either case $f_i(0) = 0$. $\square$

The profiles forced by these arguments are shown in Figure 1.

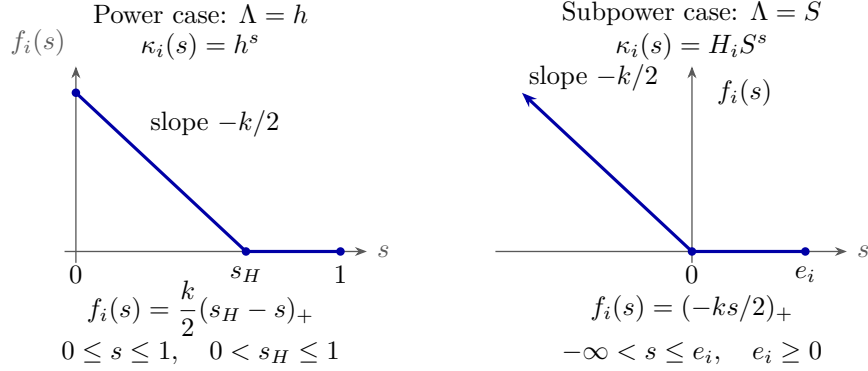


Figure 1: The forced amplitude-ratio profiles $r_{i,s} = \Lambda^{f_i(s)+o(1)}$. Their shapes follow from Theorems 3.5, 3.10 and 3.11, (3.28), and Theorem 3.13. The panels show generic cases; the terminal flat segment may collapse at $s_H = 1$ or $e_i = 0$, respectively. The subpower domain continues to the left without a finite endpoint. The recomputed transition ratio is $\Lambda^{o(1)}$ in both cases, as used in the final incidence contradiction.

3.8 The final incidence contradiction

We finish the proof of Theorem 3.1. In the power case (3.28) says that the recomputed transition ratio is $\Lambda^{o(1)}$; in the subpower case the same assertion follows from Lemma 3.13. The original transition masks have remained fixed throughout.

One final occupancy deletion makes the use of packet tails independent of any convergence rate for the profiles. Fix an arbitrarily small $\eta > 0$ and enlarge transition cells by Λ^η in their native coordinates. In the power case count all visits. In the subpower case count points with $e_b^2(z) \leq CM_c \Lambda^{C_0 \eta}$, where C_0 is sufficiently large for polynomial weight comparison. This includes every neighbor of a surviving original mask. The incident squares sum to at most $\Lambda^{C_2 \eta + o(1)} |Y|$. Since the transition ratio is $\Lambda^{o(1)}$ and $H_i^{k+2} A_i^{-q} = 1$, the weighted q charge of cells with more than $\Lambda^{C_1' \eta}$ visits is at most

$$\Lambda^{(C_2 - C_1')\eta + o(1)} D_i.$$

Choose $C_1' > C_2$ sufficiently large. Lemma 3.4 propagates this error to the root with a negative power saving. Do not reselect ratios after this deletion: all other transition fields remain the same. The fixed packet decay, with the square bound at surviving masks, makes tails outside the chosen radius negligible relative to the high threshold. Thus each remaining root point receives a substantial contribution from nearby transition masks, each visited at most $\Lambda^{C_1' \eta}$ times.

The field threshold is $A_i \Lambda^{-o(1)} = H_i^{k/2} \Lambda^{-o(1)}$. At a root point the square sum of the transition backgrounds is $\Lambda^{o(1)}$, and each transition amplitude is at most its background times $\Lambda^{o(1)}$. Weighted

Cauchy and bounded absolute sums of the ancestor cutoffs therefore require at least $H_i^k \Lambda^{-o(1)}$ nearby active masks. Letting the fixed η be arbitrarily small, double counting visits gives at least

$$|Y|H_i^k \Lambda^{-o(1)}$$

distinct original transition masks.

Choose a field with $S_i = S$. Each of these masks had original amplitude $a_T^{\text{orig}} \geq cSH_i^{-k/2}$ in (3.3), and their original integrated square budget is bounded by (3.5). Hence

$$D_i \gtrsim H_i^{k+2} (|Y|H_i^k \Lambda^{-o(1)}) (S^2 H_i^{-k}) = |Y|H_i^{k+2} S^2 \Lambda^{-o(1)}.$$

This contradicts (3.2) in the power case, because S is a fixed positive power of $\Lambda = h$, and contradicts (3.15) in the subpower case, because $S = \Lambda$. The exact-packet estimate follows. Lemma 3.2 transfers it to compact profiles and fixes a finite derivative order after all the fixed weight margins have been chosen. The countersequence construction proves the uniform quantitative form $C\delta^{-C}$ in (3.1), completing the proof.

4 Fractal norms, compact arrays, and norm transport

From this point the spatial dimension is $n \geq 3$. This section supplies a weighted fractal estimate with an occupancy gain and a power saving for frequency supports near affine $(n-2)$ -planes. In Section 5, these bounds will locate densely occupied boxes and then permit pruning of the incident tube families so that their direction laws do not concentrate near such planes. We use

$$\beta = \frac{2}{n+1}, \quad e = \frac{n^2}{n+1}, \quad \gamma_n = \frac{2n}{(n+1)(n+2)}. \quad (4.1)$$

The exponent β is the measure homogeneity in the fractal square norm. Keeping that homogeneity explicit will distinguish a genuine sparsity gain from a change of normalization.

4.1 Arrays and measures

Definition 4.1 (Normalized compact arrays and admissible measures). A compact array of scale $L \geq 1$ is a finite sum

$$U(x, t) = \sum_{\alpha} c_{\alpha} L^{-n/2} e^{i(x \cdot \omega_{\alpha} - t|\omega_{\alpha}|^2)} \phi_{\alpha} \left(\frac{x - y_{\alpha} - 2t\omega_{\alpha}}{L}, \frac{t}{L^2} \right). \quad (4.2)$$

The frequencies ω_{α} belong to a fixed bounded box. The native profiles ϕ_{α} are supported in a fixed box and have uniform bounds for derivatives through a prescribed sufficiently large finite order. The number of labels $(L\omega_{\alpha}, y_{\alpha}/L)$ in every unit phase-space ball is bounded by a fixed constant. Its coefficient energy is $E(U) = \sum_{\alpha} |c_{\alpha}|^2$. Subarrays retain their original profiles and labels.

A nonnegative Borel measure μ has n -dimensional growth with parameter m if

$$\mu(B(z, r)) \leq mr^n \quad (z \in \mathbb{R}^{n+1}, r \geq 1). \quad (4.3)$$

At array scale L its native time range is bounded by a fixed constant, so $\text{supp } \mu \subset \{|t| \leq CL^2\}$ after a common time translation. Fix a sufficiently large M . The tube moment condition with sparsity parameter D is

$$\int \left(1 + \frac{|x - y_{\alpha} - 2\omega_{\alpha}t|}{L} \right)^{-M} d\mu(x, t) \leq \frac{mL^{n+1}}{D} \quad \text{for every included label } \alpha. \quad (4.4)$$

We use half-open unit cubes for all measure sums, so their masses add without boundary ambiguity.

To compare these conventions with Theorem 2.1, set $k = n$ and $N = L$, and let C_{prof} be a fixed common bound for the required native-profile derivatives. The choice $s_\alpha = C_{\text{prof}} L^{-n/2} |c_\alpha|$ is admissible, so the packet energy parameter in (2.5) is $C_{\text{prof}}^2 L^2 E(U)$. This is a normalization of packet sizes, not a spacetime orthogonality assertion. The parameter D in (4.4) measures sparsity.

At any fixed bounded native time, transporting the intercepts along their velocities preserves phase-space multiplicity: in the normalized label variables it is a linear shear with bounded operator norm and bounded inverse. In particular,

$$\|U\|_\infty \leq CE(U)^{1/2}. \quad (4.5)$$

Indeed only $O(L^n)$ frequency/intercept labels can meet a point in the fixed profile support; Cauchy–Schwarz cancels their square-root count against $L^{-n/2}$. Condition (4.4) with $D = 1$ follows from (4.3), with fixed inflation of m . To see this, on the annulus of transverse radius $r = 2^a L$, the bounded-time tube is covered by $C(1 + L^2/r)$ balls of radius Cr . Its mass is at most $Cm(r^n + L^2 r^{n-1})$. Multiplication by 2^{-aM} and summation over $a \geq 0$ give CmL^{n+1} when M is sufficiently large.

For later use, if labels lie in a frequency cube of side H^{-1} centered at v , put

$$X_{H,v}(x, t) = \left(\frac{x - 2vt}{H}, \frac{t}{H^2} \right). \quad (4.6)$$

After removal of the common carrier, the field is $H^{-n/2} U'(X_{H,v}(x, t))$, where U' has scale L/H , frequency $\omega'_\alpha = H(\omega_\alpha - v)$, intercept $y'_\alpha = y_\alpha/H$, the same c_α , and exactly the same native profile. Thus native regularity and normalized phase-space multiplicity are unchanged.

4.2 The precise fractal input and its weighted form

Write

$$\mathcal{E}g(x, t) = (2\pi)^{-n} \int e^{ix \cdot \xi - it|\xi|^2} g(\xi) d\xi,$$

where $g \in L^2$ is supported in a fixed bounded frequency box. Fixed changes of that box alter constants only.

Theorem 4.2 (Du–Zhang input in squared form). *Let $\mathcal{Q}$ be a collection of distinct unit lattice cubes in a ball of radius $S \geq 1$. Suppose their counting function in every ball of radius $r \geq 1$ is at most Γr^n , and their number in any ball of radius $\sqrt{S}$ is at most λ_0 . Then for every $\varepsilon > 0$,*

$$\sum_{q \in \mathcal{Q}} \sup_q |\mathcal{E}g|^2 \leq C_\varepsilon S^\varepsilon \Gamma^{4/((n+1)(n+2))} \lambda_0^{\gamma_n} S^{\gamma_n} \|g\|_2^2. \quad (4.7)$$

All fixed enlargements and translations of the cube lattice are allowed, with fixed changes of the constant.

Proof. The exact external input is Du and Zhang [12, Theorem 1.6] with counting exponent n . On a union X of unit cubes for which every occupied $\sqrt{S}$ lattice parent has comparable occupancy λ , it states

$$\|\mathcal{E}g\|_{L^2(X)} \leq C_\varepsilon \Gamma^{2/((n+1)(n+2))} \lambda^{n/((n+1)(n+2))} S^{n/((n+1)(n+2))+\varepsilon} \|g\|_2.$$

The original unit-ball frequency hypothesis is changed to our fixed box by a fixed parabolic dilation and bounded coverings. The same holds for balls with arbitrary centers by translating the extension, which only modulates its input. Squaring and renaming ε gives the exponents in (4.7) for an integral over X . No comparable-field-norm assumption occurs in this theorem; that additional hypothesis belongs to Proposition 3.1 of the cited paper, which is not needed here.

Choose a comparable ambient scale with a compatible square-root grid. Group whole square-root parents according to their dyadic occupancy. A parent is covered by boundedly many radius- $\sqrt{S}$ balls, so its occupancy is at most $C\lambda_0$. Each group inherits the counting bound. Sum the squared theorem over these groups; the resulting series $\sum_{2^a \leq C\lambda_0} 2^{a\gamma_n}$ is bounded by $C\lambda_0^{\gamma_n}$. This removes the equal-occupancy hypothesis.

Finally, $\mathcal{E}g$ has spacetime Fourier support in a fixed compact paraboloid patch. Choose a Schwartz reproducing kernel equal to one on that patch in Fourier space. Cauchy–Schwarz and local boundedness of translates of this kernel imply, for every sufficiently large A ,

$$\sup_q |\mathcal{E}g|^2 \leq C_A \int_{\mathbb{R}^{n+1}} (1 + |h|)^{-A} \int_q |\mathcal{E}g(z + h)|^2 dz dh.$$

For each fixed h , the translated field is an exact extension of an input of the same norm and support. Apply the integral estimate on the unchanged cube collection, then integrate in h . This proves (4.7) without enlarging the ambient radius for distant translations. $\square$

Proposition 4.3 (Weighted fractal norm and occupancy gain). *Suppose μ is supported in a ball of radius CS , satisfies (4.3), and, in addition,*

$$\mu(B(z, \sqrt{S})) \leq d m S^{n/2}, \quad S^{-A_0} \leq d \leq 1. \quad (4.8)$$

Then

$$\sum_q \mu(q)^\beta \sup_q |\mathcal{E}g|^2 \leq C_{\varepsilon, A_0} m^\beta S^{n/(n+1)+\varepsilon} d^{\gamma_n} \|g\|_2^2. \quad (4.9)$$

With $d = 1$ no extra assumption beyond growth is needed. The estimate is homogeneous in μ and in the squared input norm.

Proof. Normalize $m = \|g\|_2 = 1$; zero cases are immediate. Sort unit cubes by $a < \mu(q) \leq 2a$, where a runs over a fixed multiple of the nonpositive dyadic powers. Cubes meeting a radius- r ball lie in a fixed enlargement, and their disjoint masses are at least a times their number. Thus this class has counting parameter at most Ca^{-1} and square-root occupancy at most $CdS^{n/2}a^{-1}$. Applying Theorem 4.2 and multiplying by $(2a)^\beta$ gives

$$\begin{aligned} \sum_{a < \mu(q) \leq 2a} \mu(q)^\beta \sup_q |\mathcal{E}g|^2 &\leq C_\varepsilon S^\varepsilon a^\beta (a^{-1})^{4/((n+1)(n+2))} (dS^{n/2}a^{-1})^{\gamma_n} S^{\gamma_n} \\ &= C_\varepsilon S^{n/(n+1)+\varepsilon} d^{\gamma_n}. \end{aligned}$$

The two cancellations are

$$\beta - \frac{4}{(n+1)(n+2)} - \gamma_n = 0, \quad (1 + n/2)\gamma_n = \frac{n}{n+1}.$$

This exact cancellation leaves no positive power of a to sum, so a small-mass truncation is necessary. Bounded frequency support gives $\|\mathcal{E}g\|_\infty \leq C$, and there are $O(S^{n+1})$ relevant unit cubes. Those with $\mu(q) \leq S^{-A}$ contribute at most $CS^{n+1-A\beta}$. Choose A so large, in terms of A_0 and a prescribed error power, that this is bounded by $S^{-B}d^{\gamma_n}$. There remain $O_A(1 + \log S)$ mass classes. Use half the requested loss in the preceding class estimate and absorb this count into the remaining S^ε . Restoring homogeneity proves (4.9). $\square$

4.3 Exact Fourier layers and spatial localization

The following reduction permits compact native profiles in estimates that tolerate an arbitrarily small power loss. Its common frequency shift is useful when the chosen subarray changes from one test to another.

Lemma 4.4 (Exact Fourier layers of compact profiles). *For every prescribed finite collection of derivative and native spatial decay orders, a sufficiently high finite profile regularity in (4.2) gives a representation*

$$U(x, t) = \sum_{k \in \mathbb{Z}^n} \int_{\mathbb{R}} \rho(k, u) e^{iut/L^2} \mathcal{E} g_{k,u}(x, t) du, \quad (4.10)$$

where $\rho(k, u) \leq C_A(1 + |k| + |u|)^{-A}$ for any prescribed A . For every subarray $\mathcal{A}$,

$$\|g_{k,u}^{\mathcal{A}}\|_2^2 \leq C \sum_{\alpha \in \mathcal{A}} |c_{\alpha}|^2. \quad (4.11)$$

Each label contributes an exact input supported within $O(L^{-1})$ of $\omega_{\alpha} + k/L$, with uniformly bounded native frequency seminorms. On a fixed native time range, its extension has all the prescribed spatial decay about the original trajectory after the factor ρ is extracted. The shift k/L and the time modulation are common to the entire layer and every subarray. For any fixed $\eta > 0$, the portion $|k| > L^{\eta}$ can be made an arbitrarily negative power error in the array supremum, and hence in tests involving polynomially many cubes and polynomially bounded mass parameters.

Proof. Fourier transform ϕ_{α} in its native variables, denoting the dual variables by (η, ζ) . Insert a smooth bounded-overlap partition $\sum_k \vartheta_k(\eta) = 1$ into bounded boxes about $k \in \mathbb{Z}^n$, and change variables $u = \zeta + |\eta|^2$. For fixed (k, u) define the input, in $\xi = \omega_{\alpha} + \eta/L$ coordinates, by

$$g_{\alpha,k,u}(\omega_{\alpha} + \eta/L) = C c_{\alpha} L^{n/2} e^{-iy_{\alpha} \cdot \eta/L} \vartheta_k(\eta) \hat{\phi}_{\alpha}(\eta, u - |\eta|^2).$$

Fourier inversion gives

$$U = \sum_k \int e^{iut/L^2} \mathcal{E} \left(\sum_{\alpha} g_{\alpha,k,u} \right) du.$$

Compact support and sufficiently many derivatives of ϕ_{α} give any prescribed polynomial decay in (η, ζ) , also after a fixed number of frequency derivatives. The change $\zeta = u - |\eta|^2$ costs only polynomial factors. Extract $\rho(k, u) = C_A(1 + |k| + |u|)^{-A}$ with C_A chosen to bound the needed seminorms, and put the remaining normalized sum into $g_{k,u}$. Increasing the finite initial derivative order supplies these bounds simultaneously.

The input norm estimate follows by a summable Gram-matrix bound. Two normalized input bumps can interact only if $|L(\omega_{\alpha} - \omega_{\alpha'})| \leq C$, since the shift k is common. For these pairs, integration by parts in the native frequency variable gives any prescribed decay in $|y_{\alpha} - y_{\alpha'}|/L$. The coefficient factors are $c_{\alpha} \overline{c_{\alpha'}}$ and all remaining seminorms are uniform. Phase-space counting therefore bounds the absolute row and column sums of the Gram matrix. Schur's inequality proves (4.11), without requiring the subarray to be a Fourier projection.

For spatial decay, the phase in native variables is $z \cdot \eta - \tau|\eta|^2$. On bounded τ and η near k , integration by parts gives rapid decay in z away from the original trajectory, at the cost of a fixed polynomial in k . This was included in the orders used to extract ρ ; derivatives of the extension are handled at the same time. The common shift preserves all relative frequency labels and phase-space counts. Finally, Cauchy–Schwarz over labels and the prescribed rapid spatial decay bound every normalized layer by a fixed polynomial in (k, u) times $E(U)^{1/2}$. Increase A , sum over $|k| > L^{\eta}$, and integrate in u to obtain $C_B L^{-B} E(U)^{1/2}$ for any required B . All orders here are fixed before the varying scale L . $\square$

Lemma 4.5 (Exact packet localization on a time slab). *Let $T \geq 1$ and let g have bounded frequency support. On a time interval of length $O(T)$, an exact extension can be localized to spatial boxes of side T with summed input energies $C\|g\|_2^2$. The localization errors have arbitrarily high negative powers of T , with summable local squared budgets. No factor equal to the number of spatial boxes is needed.*

Proof. Set $R = \sqrt{T}$ and absorb the initial time into the input by a unit-modulus multiplier. Partition frequency into R^{-1} caps and expand each rescaled cap density in a Fourier series on a fixed larger box. With smooth bumps equal to one on the cap supports, this gives exact packets

$$c_{\theta,m} R^{-n/2} e^{i(x \cdot \omega_\theta - t|\omega_\theta|^2)} H_\theta \left(\frac{x - \gamma R m - 2t\omega_\theta}{R}, \frac{t}{R^2} \right), \quad \sum_{\theta,m} |c_{\theta,m}|^2 \leq C\|g\|_2^2.$$

Here H_θ is the extension of a fixed smooth native frequency bump. It and any fixed derivatives decay arbitrarily rapidly in the first variable for bounded second variable. Parseval proves the coefficient bound; differentiating the input density is unnecessary. The same cap-overlap and integration-by-parts Gram bound used in Theorem 4.4 bounds the squared norm of any retained exact input by its coefficient energy.

For a spatial box B of side T , retain packets whose trajectories on the time slab come within CT of B . A bounded-velocity trajectory of duration $O(T)$ can be retained for only $O(1)$ such boxes. Therefore these retained energies sum to $C\|g\|_2^2$. For the omitted packets, the native distance from the box is at least $c\sqrt{T}$. Extract an arbitrary negative power of T from the rapid packet decay and keep a further weight in the distance from the trajectory to B , measured in units of T . Cauchy–Schwarz and phase-space counting bound the squared supremum of the omitted field on B by

$$C_A T^{-A} \sum_{\theta,m} |c_{\theta,m}|^2 \left(1 + \frac{\text{dist}(B, \text{trajectory}_{\theta,m})}{T} \right)^{-n-2}.$$

The right-hand budgets sum boundedly in B . This proves the asserted error control and also permits arbitrarily long spatial unions of such boxes. Translations modulate the exact inputs and preserve their norms, so the construction is uniform in the slab and box centers. $\square$

Corollary 4.6 (Power-loss norm for compact arrays). *Let U be a compact array of scale L , and let μ satisfy (4.3) in a spacetime box of side $O(L^2)$. For every $\varepsilon > 0$, with sufficiently many fixed profile derivatives,*

$$\sum_q \mu(q)^\beta \sup_q |U|^2 \leq C_\varepsilon m^\beta L^{2n/(n+1)+\varepsilon} E(U). \quad (4.12)$$

Proof. Apply Theorem 4.4, discarding $|k| > L^\eta$ with a sufficiently high negative power. For $0 < \eta < 1$, every retained layer has support in a fixed bounded frequency box. Apply Theorem 4.3 with $S = L^2$ and $d = 1$ to each exact layer. Minkowski’s inequality for the weighted ℓ^2 norm of unit-cube suprema, followed by (4.11), gives the result because $\sum_k \int \rho(k, u) du < \infty$. The common time modulation has absolute value one. The discarded error is negligible: each cube has mass at most Cm , there are $O(L^{2(n+1)})$ relevant cubes, and its supremum has any required negative power of L . Rename the arbitrarily small scale loss after replacing S by L^2 . $\square$

4.4 Coarse decoupling and exact transport powers

For dyadic $K \geq 2$, $0 < b < 1$, and $P = 2/(1 - b)$, define

$$\mathcal{T}_b(V, \nu) = \sum_{Q \in \mathcal{Q}_{K^2}} \nu(Q)^b \|V\|_{L^P(Q)}^2, \quad (4.13)$$

where $\mathcal{Q}_{K^2}$ is a fixed grid of half-open spacetime cubes of side K^2 .

Lemma 4.7 (Decoupling inside a strip). *Group a field $V = \sum_{\theta} V_{\theta}$ by frequency-label bins of side K^{-1} . Suppose the field supports lie in fixed enlargements of these bins and within vertical Fourier distance $O(K^{-2})$ of the paraboloid. If the labels lie within $O(K^{-1})$ of an affine j -plane, $1 \leq j \leq n$, and $2 \leq P \leq q_j = 2(j+2)/j$, then*

$$\|V\|_{L^P(Q)}^2 \leq C_{\varepsilon, A} K^{\varepsilon} \sum_{\theta} \sum_{a \in \mathbb{Z}^{n+1}} \langle a \rangle^{-A} \|V_{\theta}\|_{L^P(Q+K^2a)}^2. \quad (4.14)$$

The constituent fields may be arbitrary prescribed subfields with these supports. Without the strip hypothesis the same estimate holds with the factor CK^n in place of $C_{\varepsilon} K^{\varepsilon}$.

Proof. Rotate frequency into tangent and normal coordinates for the affine plane, remove its central normal carrier, and shear away the normal linear time phase. The remaining normal frequencies have size $O(K^{-1})$, so their quadratic contribution is $O(K^{-2})$. On each fixed normal physical slice, the tangential spacetime Fourier support is therefore within $O(K^{-2})$ of the positive j -dimensional paraboloid. Multiply first by a Schwartz cutoff bounded below on Q with Fourier width $O(K^{-2})$. It preserves this support statement with fixed enlargement.

Apply paraboloid decoupling [7] in the range $P \leq q_j$. After this localization, use smooth tangential cap projections; their spatial convolution kernels have bounded L^1 norm. Only boundedly many original enlarged grid bins in the strip can meet a projected bin, since both the normal thickness and tangential cap size are $O(K^{-1})$. Bounded overlap and Minkowski in the normal physical variables give the squared-norm estimate with a Schwartz weight outside Q . Decomposing that weight into translates of Q gives (4.14), with any fixed A . This proof uses only supports and not the way a subfield was selected. Without a strip, there are $O(K^n)$ bins, and Cauchy-Schwarz gives the stated alternative directly. $\square$

Lemma 4.8 (Coarse norm transport). *Let $F = X_{K,v}$ with bounded v , and suppose $V(z) = K^{-n/2} e^{i\varphi(z)} V'(Fz)$ with $|e^{i\varphi}| = 1$. For a fixed $a \in \mathbb{Z}^{n+1}$,*

$$\sum_Q \nu(Q)^b \|V\|_{L^P(Q+K^2a)}^2 \leq CK^{2-(n+2)b} \sum_{r,s \in \mathcal{R}} \mathcal{T}_b(V', \nu_{a,r,s}), \quad (4.15)$$

where $\mathcal{R}$ is a fixed finite set, independent of Q, K, a , and

$$\nu_{a,r,s} = (\tau_{F(K^2a)+K^2(s-r)})_{\#} F_{\#} \nu, \quad \tau_h(z) = z + h. \quad (4.16)$$

Their translations have size at most $CK^2(1+|a|)$. The same conclusion holds with any prescribed subset of parent cubes on the left. If ν has growth parameter m , each child measure has parameter CmK^{n+1} . Direct pushforward transports (4.4) from scale L to L/K with the same D .

Proof. The Jacobian of F is $K^{-(n+2)}$. Thus, with $A_Q = F(Q + K^2a)$,

$$\begin{aligned} \|V\|_{L^P(Q+K^2a)}^2 &= K^{-n+2(n+2)/P} \left(\int_{A_Q} |V'|^P \right)^{2/P} \\ &= K^{2-(n+2)b} \left(\int_{A_Q} |V'|^P \right)^{1-b}. \end{aligned}$$

No density factor accompanies the mass: the measure is the direct pushforward. Let $\nu_a = (\tau_{F(K^2a)})_{\#} F_{\#} \nu$. Then $\nu_a(A_Q) = \nu(Q)$, and the A_Q are disjoint. Group them according to the

child K^2 cube C containing their centers. Each A_Q has diameter $O(K)$, so all sets in a group are contained in $C^* = \bigcup_{r \in \mathcal{R}} (C + K^2 r)$ for a fixed finite $\mathcal{R}$. Hölder with exponents $1/b$ and $1/(1-b)$ gives

$$\sum_{Q \text{ in the group}} \nu(Q)^b \left(\int_{A_Q} |V'|^P \right)^{1-b} \leq \nu_a(C^*)^b \|V'\|_{L^P(C^*)}^2.$$

Subadditivity of the powers b and $1-b$ bounds this by

$$\sum_{r,s \in \mathcal{R}} \nu_a(C + K^2 r)^b \|V'\|_{L^P(C + K^2 s)}^2.$$

The two indices allow the mass and the field norm to occupy different neighbors. Reindex by $D = C + K^2 s$ to obtain precisely (4.16) and (4.15). None of the translations depends on which parent cube was selected.

The inverse image of a radius- r ball under F has n transverse widths $O(Kr)$ and longitudinal length $O(K^2 r)$. It is covered by $O(K)$ balls of radius CKr . Hence its mass is at most $CmK^{n+1}r^n$ for $r \geq 1$, and translations preserve this bound. For the tube moment, writing $L' = L/K$, $\omega' = K(\omega - v)$, and $y' = y/K$ gives the identity

$$\frac{x' - y' - 2\omega't'}{L'} = \frac{x - y - 2\omega t}{L}, \quad (mK^{n+1})(L/K)^{n+1} = mL^{n+1}.$$

This proves the direct-pushforward assertion. When the tube centers are kept fixed, translated measures preserve the moment up to a fixed factor only if their native displacements are bounded. Accordingly, applications requiring this moment truncate the Schwartz shifts while the child packet widths are much larger than the retained shifts. The freely chosen exponent A in (4.14) makes the omitted tail smaller than any prescribed power once the global suprema and occupied cube counts are polynomially bounded. For exact-field applications using only ball growth, arbitrary translations are harmless. $\square$

4.5 A power gain from codimension two

Proposition 4.9 (Codimension-two strip saving). *Fix $0 < \alpha < 1/8$. Suppose g has bounded frequency support contained in an $O(S^{-\alpha})$ neighborhood of an affine $(n-2)$ -plane, and $\mathcal{Q}$ is a collection of unit cubes in a ball of radius S with at most $m_0 r^n$ cubes in every radius- r ball, $r \geq 1$. Fix any b with $\beta < b < 2/n$. There is $c_0 > 0$, depending only on the fixed dimensional choices and independent of α , such that for every $\varepsilon > 0$,*

$$\sum_{q \in \mathcal{Q}} \sup_q |\mathcal{E}g|^2 \leq C_{\alpha,\varepsilon} m_0^b S^{n/(n+1)-c_0\alpha+\varepsilon} \|g\|_2^2. \quad (4.17)$$

In particular, for $m_0 = S^{o(1)}$ this is the saving $S^{n/(n+1)-c_0\alpha+o(1)}$. Finite disjoint subinputs are allowed, with their squared norms summed on the right.

Proof. Choose a sufficiently large integer J , fixed independently of S and α , and dyadic K so that $H = K^J \asymp S^\alpha$, with fixed rounding factors. Set $P = 2/(1-b)$. Since $b < 2/n$, one has $P < q_{n-2} = 2n/(n-2)$. Use the counting measure of the selected unit cubes, assigning unit mass to their centers. Local constancy and Hölder bound the initial sum of squared suprema by the coarse functional (4.13), with a fixed K^C endpoint cost allowed. Explicitly, in each coarse cube the sum of the local unit norms is bounded by the number of selected unit cubes to the power $1 - 2/P = b$,

times the squared L^P norm on a fixed enlargement. Schwartz reproduction supplies the required neighboring translates. Its measure parameter is Cm_0 .

Iterate Theorems 4.7 and 4.8 for J levels. After a rescalings the strip width is $O(K^a/H)$, which is at most $O(K^{-1})$ before the last step. Thus the codimension-two decoupling hypothesis persists. The product of transport factors is $H^{2-(n+2)b}$, up to constants depending on the fixed J , while the final measure parameter is

$$m' \leq C_J m_0 H^{n+1}.$$

Smooth frequency decompositions have bounded overlap, so final child input squared norms sum to $C_J \|g\|_2^2$. The J decoupling losses can be made S^ε by starting with a sufficiently small loss. Schwartz shifts may be truncated at a polynomial radius with negligible error, using bounded-frequency suprema and polynomial cube counts. Their weight sums cost fixed constants at each of the fixed number of levels.

The final time length is $T = S/H^2$, still a fixed positive power of S . The spatial region can have side S/H ; to avoid paying for this aspect ratio, cover it by spatial boxes of side T and use Theorem 4.5 on the child time slab. Retained exact inputs have squared norms summable to the child energy. Their omitted errors have arbitrarily high negative powers of T , with summable local budgets. Each resulting spacetime box has side $O(T)$.

Here is the terminal coarse-functional estimate on one such box:

$$\mathcal{T}_b(V', \nu') \leq C_\varepsilon K^{C_*} T^{n/(n+1)+\varepsilon} (m')^b \|\text{child input}\|_2^2. \quad (4.18)$$

Its exponent C_* is independent of J . Indeed, replace the field norm on each K^2 cube by its volume factor times a unit-cube witness for its supremum. Move the entire mass of that coarse cube to that witness. The resulting measure has ball parameter at most $CK^{2n}m'$, and each moved mass is at most $CK^{2n}m'$. Since $b \geq \beta$, its b th power is at most $(CK^{2n}m')^{b-\beta}$ times its β th power. Apply Theorem 4.3 at scale T to the moved masses. The volume factor, the moved growth parameter, and this last comparison produce only K^{C_*} , proving (4.18). A bounded number of witness-grid choices and enlargements have the same cost. The inequality holds separately for every translated child measure, and translations are absorbed into modulations of the exact inputs. Consequently no count of spatial boxes or translation labels is introduced.

Combining the three scale contributions, apart from $S^{n/(n+1)+\varepsilon}$ and the fixed endpoint factor K^{C_*} , gives

$$\begin{aligned} H^{2-(n+2)b} H^{(n+1)b} H^{-2n/(n+1)} &= H^{2-b-2n/(n+1)} \\ &= H^{\beta-b}. \end{aligned}$$

Enlarge C_* to include the initial endpoint cost, still independently of J . Choose J so large that $C_*/J < (b-\beta)/2$. Then $K^{C_*} H^{\beta-b} \leq H^{-(b-\beta)/2}$. Taking, for example, any $c_0 < (b-\beta)/2$ absorbs the fixed rounding and proves (4.17). Constants may depend on fixed α, J, ε , but the saving exponent c_0 does not depend on α . For disjoint subinputs the same construction is applied with their coefficient energies, whose sum is the original energy. $\square$

5 A transverse saving in every fixed dimension

Throughout this section the spatial dimension $n \geq 3$ is fixed, and

$$\beta = \frac{2}{n+1}, \quad e = \frac{n^2}{n+1}.$$

For a frequency ξ write $v(\xi) = (2\xi, 1)$; replacing these vectors by unit vectors changes determinants by bounded factors on a fixed bounded frequency region. We write $Ef = \mathcal{E}f$ for the exact extension operator of Section 4. The following statement concerns exact extension fields. Its use for smooth packet arrays will come after the exact Fourier-layer reduction in the next section.

Proposition 5.1 (Transverse saving). *Fix a bounded frequency region in $\mathbb{R}^n$. There are $c > 0$ and $L_0 < \infty$, depending only on this region and n , such that the following configuration is impossible for $L \geq L_0$. Let $\mathcal{Q}$ be distinct unit lattice cubes in a ball of radius L^2 in $\mathbb{R}^{n+1}$, with*

$$\#\mathcal{Q} \geq L^{2n-c}, \quad \#\{q \in \mathcal{Q} : q \cap B(z, r) \neq \emptyset\} \leq L^c r^n \quad (r \geq 1). \quad (5.1)$$

Suppose $f_1, \dots, f_{n+1}$ have L^2 norm at most one, frequency support in the fixed region, and

$$\sup_q |Ef_i| \geq L^{-\epsilon-c} \quad (q \in \mathcal{Q}, 1 \leq i \leq n+1).$$

It is impossible that every choice of one frequency from each support has direction determinant at least L^{-c} in absolute value.

We prove the equivalent assertion that no countersequence exists with $c \downarrow 0$ and $L \rightarrow \infty$. In this section $o(1)$ refers to such a sequence, after subsequences when necessary. Estimates available with every fixed positive power loss may be diagonalized to have $L^{o(1)}$ loss. The constants in each such estimate are fixed before diagonalization. The proof uses the limiting information calculus of OpenAI [24, Section 3, “Limiting information calculus”]. We state the usable facts below and prove the changes needed in dimension n . In particular, the planar transverse proposition in that source is not used as a higher-dimensional assertion.

5.1 Uniform multilinear estimates and incident weights

We first specify the dependence on transversality. This prevents a subpower determinant loss from concealing an uncontrolled constant.

Lemma 5.2 (Multilinear estimates with polynomial dependence). *If $n+1$ bounded-frequency exact inputs are transverse with determinant at least $\nu \in (0, 1]$, then, for each $\epsilon > 0$,*

$$\sum_{q \in B_S} \prod_{i=1}^{n+1} \left(\sup_q |Ef_i| \right)^{2/n} \leq C_\epsilon \nu^{-M_\epsilon} S^\epsilon \prod_{i=1}^{n+1} \|f_i\|_2^{2/n}. \quad (5.2)$$

The same bound holds with the sum replaced by the integral of the product. For transverse families of tubes of width δ in a fixed bounded region, with nonnegative weights $w_i(T)$, the corresponding Kakeya bound is

$$\int \prod_{i=1}^{n+1} \left(\sum_T w_i(T) \mathbf{1}_T \right)^{1/n} \leq C_\epsilon \nu^{-M_\epsilon} \delta^{n+1-\epsilon} \prod_i \left(\sum_T w_i(T) \right)^{1/n}. \quad (5.3)$$

Fixed enlargements are allowed. Subpower enlargements and $\nu = S^{-o(1)}$ give only subpower losses.

Proof. Subdivide direction supports into neighborhoods of radius a sufficiently small fixed power of ν . There are $\nu^{-O_n(1)}$ choices of $(n+1)$ neighborhoods. For each choice, send representative directions to the coordinate axes. The map, its inverse, and its determinant have size bounded by fixed powers of ν^{-1} . The near-coordinate-axes multilinear Kakeya theorem [2, Theorem 1.15 and

Remark 1.10], or its strengthened form [16], gives (5.3) after summing the choices. Weights follow by repetition and approximation.

Here is the finite induction giving the required uniform restriction constant. Let $A_B(S, \nu)$ be the best normalized integral constant for inputs in a fixed box of size B . Decompose each input into exact packets at radius S , and on each $\sqrt{S}$ -box retain the packets meeting its $S^\theta \sqrt{S}$ enlargement. Their input squared norms are bounded by incident coefficient energy, and the discarded fields are $O_N(S^{-N})$ relative to their parent input norms. The enlarged supports remain $\nu/2$ transverse if $S^{-1/2} \leq c_B \nu$. After rescaling by S , the padded tubes have width $\delta = S^{-1/2+\theta}$. Apply (5.3) with loss σ . Dividing its bound by the volume $S^{-(n+1)/2}$ of a rescaled $\sqrt{S}$ -box gives

$$A_B(S, \nu) \leq C \nu^{-M_\sigma} S^a A_{B+1}(C_B \sqrt{S}, \nu/2) + O_N(S^{-N}), \quad a = (n+1)\theta + \sigma/2.$$

Increasing N absorbs all errors into the same inequality for $1 + A_B$; when expanding products with exponent $2/n \leq 1$, use subadditivity to bound terms containing a discarded factor. For a prescribed ε , choose a finite J with $(n+1)2^{-J} < \varepsilon/2$, and then θ, σ with $2a < \varepsilon/2$. Iterating J times and using the terminal trivial bound $O(S_J^{n+1})$, where $S_J \asymp S^{2^{-J}}$, proves the integral estimate when $S \geq (C/\nu)^{2^J}$. In the complementary range the trivial bound is absorbed by increasing M_ε . Thus the dependence on ν^{-1} is a fixed power.

For the cube-supremum version the exponent $p = 2/n$ can be below one, so we include the submean argument. If F has Fourier support in the fixed compact region, choose a Schwartz reproducing kernel ψ and write $F = \psi * F$. For $A > 0$ let $M_A F(x) = \sup_y |F(x-y)|(1+|y|)^{-A}$, which is finite because these extensions are bounded. In the convolution insert $|F(w)| \leq |F(w)|^p M_A F(x)^{1-p} (1+|x-w|)^{A(1-p)}$. Peetre's elementary inequality $(1+|x-w|) \leq (1+|x-z|)(1+|z-w|)$ then gives, uniformly in z ,

$$\frac{|F(z)|}{(1+|x-z|)^A} \leq C_A M_A F(x)^{1-p} \int (1+|x-w|)^{-Ap} |F(w)|^p dw,$$

using the boundedness of $(1+|z-w|)^A |\psi(z-w)|$. Taking the supremum and dividing by $M_A F(x)^{1-p}$ proves the required Schwartz-average bound for $M_A F(x)^p$; the zero case is immediate. Choose Ap arbitrarily large. In particular this bounds $(\sup_q |E f_i|)^{2/n}$ by that average for each $x \in q$. Integrate the product over q and sum. For each independent list of shifts the translated fields are exact extensions with unchanged norms and supports. Apply the integral estimate and integrate the Schwartz weights. This proves (5.2). $\square$

Lemma 5.3 (Pruned incident-tube model). *A countersequence to Proposition 5.1 yields a set $\mathcal{P}$ of centers of distinct L^{-1} -scale cubes in a bounded region, satisfying*

$$\#\mathcal{P} = L^{n+o(1)}, \quad \#(\mathcal{P} \cap B(z, r)) \leq L^{n+o(1)} r^n \quad (L^{-1} \leq r \leq 1), \quad (5.4)$$

and $n+1$ transverse weighted tube families of width $L^{-1+o(1)}$. Their weights satisfy $\sum_T w_i(T) \lesssim 1$. For every $x \in \mathcal{P}$ there is an incident subfamily of retained weight $w_i^(x) = L^{-e+o(1)}$ whose normalized law satisfies*

$$\mathbb{P}\{\xi(T_i) \in N_{L^{-\alpha}}(H) \mid X = x\} \leq L^{-c_1 \alpha + o(1)} \quad (5.5)$$

for every affine $(n-2)$ -plane $H \subset \mathbb{R}^n$ and every fixed rational $0 < \alpha < 1/8$. Here $c_1 > 0$ is fixed independently of α , and the estimate is uniform in x, H for each fixed α .

Proof. Sort the original unit cubes by their occupancy in L -boxes. A dyadic class retains $L^{2n-o(1)}$ cubes, on which the squared supremum sum of each field is at least $L^{2n-2e-o(1)} = L^{2n/(n+1)-o(1)}$.

The refined fractal estimate, Proposition 4.3, at radius L^2 improves this upper bound by a fixed power if the L -box occupancy is at most $L^{n-\delta}$ for any fixed $\delta > 0$. The counting hypothesis supplies the upper bound $L^{n+o(1)}$. Consequently the retained occupancies are $L^{n+o(1)}$. Divide coordinates by L^2 and use the occupied box centers. Counting the original cubes in an enlarged ball, then dividing by the minimum occupancy, proves (5.4).

Exact packet localization at radius L^2 gives coefficient weights with bounded total sum. On each retained physical L -box, keep trajectories in a padded enlargement. The padding exponent can tend sufficiently slowly to zero; rapid tails make the discarded field negligible. The retained exact input has squared norm at most $Cw_i(x)$, where $w_i(x)$ is its incident weight. Apply Proposition 4.3 at radius L to its $L^{n+o(1)}$ selected unit cubes:

$$L^{n-2e-o(1)} \leq L^{n/(n+1)+o(1)} w_i(x), \quad w_i(x) \geq L^{-e-o(1)}.$$

The counting version of (5.3) gives $\sum_x \prod_i w_i(x)^{1/n} \leq L^{o(1)}$. If some weight exceeds $L^{-e+\delta}$, this and the other lower bounds allow only $L^{n-\delta/n+o(1)}$ such points. Diagonal deletion leaves $w_i(x) = L^{-e+o(1)}$ uniformly.

Let c_0 be the constant in the codimension-two strip gain, Proposition 4.9, and fix $0 < c_1 < c_0/2$. For a finite list of rational α , successively remove, at each x, i , strips of width $L^{-\alpha}$ carrying remaining weight greater than $L^{-c_1\alpha} w_i(x)$. At one scale there are at most $L^{c_1\alpha}$ removals. Assign a packet to its first removal, so the corresponding subarray energies sum to $O(w_i(x))$. On the physical L -box the exact support enlargement is $O(L^{-1})$, smaller than the strip width. Proposition 4.9, followed by Cauchy over the removed strips, bounds their combined squared supremum sum by

$$L^{n/(n+1)-(c_0-c_1)\alpha+o(1)} w_i(x).$$

This is negligible compared with the required local squared norm. The triangle inequality in that square-sum norm, followed by the unrestricted local fractal estimate, forces surviving weight $L^{-e+o(1)}$. On termination every affine strip has remaining weight at most the deletion threshold. Normalizing by the surviving weight costs $L^{o(1)}$. Diagonalize over finite rational lists to obtain (5.5). Constant or subpower enlargements of the strips are covered by applying the assertion at a slightly smaller rational exponent. The pruning is allowed to depend on x ; it defines the conditional laws used next. $\square$

5.2 Information profiles and the cost of direction bins

Draw X uniformly from $\mathcal{P}$. Conditional on X , draw T_i, T'_i independently from the retained law of family i , for $1 \leq i \leq n+1$. A tube label contains its central line and exact direction. Let X_s be its position label in nested cube grids of side comparable to L^{-s} , for rational $s \in [0, 1]$, with the trivial partition at zero. For a discrete variable Y and conditioning V , information at the sampled labels means

$$I_L(Y | V) = -\log_L \mathbb{P}(Y = Y(\omega) | V).$$

All spatial, coordinate, and direction partitions below have at most $L^{C+o(1)}$ labels. Exact tube labels may be more numerous; only their role as conditioning variables is used.

Lemma 5.4 (Limiting information calculus). *After passing to a subsequence, the countably many information variables at rational block and prefix scales have a common limiting joint law. Their conditional informations are uniformly integrable and bounded in the limit. The chain rule is exact, and conditioning lowers information almost surely in this law. If, on a specified event and given (Z, V) , there are at most $L^{a+o(1)}$ possible labels for Y , then the limiting information $I(Y | Z, V)$ is*

at most a on that event. Event indicators can be included in the joint law; finite systems of limiting inequalities of positive probability pass to finite scale with fixed positive slack.

Proof. This is the dimension-free lemma with this name in OpenAI [24, Section 3]. The short argument also explains its pointwise scope. If a variable has at most $L^{C+o(1)}$ labels, then $\mathbb{P}\{I_L(Y | V) > C+a\} \leq L^{-a+o(1)}$, uniformly in conditioning; integrating this bound gives uniform integrability. Writing $p = \mathbb{P}(Y | V)$ and $q = \mathbb{P}(Y | V, W)$ at the sampled labels, the conditional expectation of p/q is at most one. Thus

$$\mathbb{P}\{I_L(Y | V) - I_L(Y | V, W) < -a\} \leq L^{-a}.$$

This proves the limiting conditioning inequality, without asserting finite-scale pointwise monotonicity. The restricted determination rule follows by summing probabilities over the permitted small label set. Tightness and a diagonal subsequence give the joint law. Include all rational tests, their costs, and the restricted-event indicators; open-set implications with slack justify their passage to the limit. $\square$

The information profiles are

$$F(s) = I(X_s), \quad G_i(s) = I(X_s | T_i), \quad G'_i(s) = I(X_s | T'_i).$$

Thus F records the limiting exponent of the reciprocal probability of the sampled spatial cell, whereas G_i records that exponent after one incident tube is specified. We shall compare these profiles in coordinates adapted to the tube directions; their endpoint values will force equality in that comparison.

These profiles extend to nondecreasing Lipschitz functions on $[0, 1]$, with constants $n+1$ and 1 , respectively, and vanish at zero. Indeed a tube intersects at most $L^{t+o(1)}$ children inside a parent when the exponent increases by t . The determination rule gives the conditional Lipschitz constant; unrestricted grid counts give the other one. Moreover

$$F(s) \geq ns, \quad F(1) = n, \quad G_i(1), G'_i(1) \geq \frac{n}{n+1}. \quad (5.6)$$

The first two assertions follow from (5.4). For the third, the retained weights give

$$\mathbb{P}(X = x, T_i = T) \leq L^{-n+e+o(1)} w_i(T) = L^{-n/(n+1)+o(1)} w_i(T).$$

The probability of sampling a tube with marginal probability smaller than $L^{-a} w_i(T)$ is at most $L^{-a} \sum_T w_i(T) = O(L^{-a})$. Outside this event the conditional point mass is at most $L^{-n/(n+1)+a+o(1)}$. Let $a \downarrow 0$, also for the primed tubes.

To choose coordinates adapted to reference tubes, we reveal their coarse direction bins. Conditioning on these bins may lower the limiting information; the costs below measure this loss. Fix a rational block $[s, s+b]$, a rational $b_* \geq b$, and $C = X_s$. Let D contain direction bins of precision L^{-b_*} for a fixed sublist of the $2(n+1)$ tubes, including one reference tube per family. Normalize the parent to size one and use the basis given by deterministic representatives of the reference bins. Transversality bounds its norm and inverse by $L^{o(1)}$. For a prefix $0 < t \leq b$, let $Z = (Z_1, \dots, Z_{n+1})$ be the coordinate grid labels at resolution L^{-t} . Adding bins to D does not change this basis or these labels.

Lemma 5.5 (Block comparison, additivity, and coordinate subsets). *Put $J = F(s+t) - F(s)$ and $J_i = G_i(s+t) - G_i(s)$ for the selected references. The nonnegative costs*

$$\ell = I(X_{s+t} | C) - I(X_{s+t} | C, D), \quad \ell_i = I(X_{s+t} | C, T_i) - I(X_{s+t} | C, T_i, D)$$

satisfy

$$\sum_i (J_i - \ell_i) \leq \sum_i I(Z_i \mid Z_{\hat{i}}, C, D) \leq I(Z \mid C, D) = J - \ell. \quad (5.7)$$

For disjoint blocks or prefixes with the same bin list and the same exact tube conditioning, the sum of expectations of each cost is $O_n(b_*)$. Almost surely, for every choice of references,

$$F = \sum_{i=1}^{n+1} G_i, \quad G'_i = G_i, \quad G_i(1) = \frac{n}{n+1}. \quad (5.8)$$

For every coordinate subset $A \subset \{1, \dots, n+1\}$,

$$I((Z_i)_{i \in A} \mid C, D) = \sum_{i \in A} J_i + O_n\left(\sum_i \ell_i\right). \quad (5.9)$$

No unused coordinates are included in the conditioning in this formula.

Proof. A reference tube determines its off-axis coordinates to $L^{o(1)}$ choices: its relative width and its direction-bin error are both at most $L^{-t+o(1)}$. Given (C, D) , the block child label X_{s+t} and the coordinate label Z determine each other up to $L^{o(1)}$ choices: the bin-defined change of basis has inverse norm $L^{o(1)}$, and grid-boundary ambiguity has only subpower multiplicity. Thus

$$J_i - \ell_i = I(Z_i \mid C, D, T_i) = I(Z_i \mid Z_{\hat{i}}, C, D, T_i) \leq I(Z_i \mid Z_{\hat{i}}, C, D).$$

In any ordering of the coordinates each chain-rule term conditions on a subset of $Z_{\hat{i}}$, proving (5.7).

For a finite-valued label D , its Shannon entropy is $H(D) = \mathbb{E}[-\log p_D(D)]$, with the natural logarithm. At finite scale the expected cost is the conditional mutual information of child and D given the parent, divided by $\log L$, with the same fixed exact tube additionally conditioned on if present. These mutual informations telescope along the spatial filtration. Filling gaps between disjoint blocks only adds nonnegative terms. The total is at most $H(D)/\log L = O_n(b_*)$, since D has $L^{O_n(b_*)}$ labels. Uniform integrability passes this to the limiting law. This is exactly the hypothesis of the “Block comparison and telescoping costs” lemma in OpenAI [24, Section 3]; the number of reference directions only changes the dimensional constant.

Partition a rational interval into blocks with $b = b_* \downarrow 0$. The expected positive excess of the summed G_i increments over the F increment is $O_n(b)$, so in the limit every interval has $\sum_i \Delta G_i \leq \Delta F$. At $[0, 1]$, (5.6) forces equality. Nonnegative additive deficits must therefore vanish on every subinterval. Repeat this argument for all finitely many reference replacements and subtract the identities. This proves (5.8).

Finally choose any ordering that puts A first. Each chain-rule term is at least $J_i - \ell_i$, and their total surplus is $\sum_i \ell_i - \ell \leq \sum_i \ell_i$ by additivity. Summing just the terms for A proves (5.9). Equivalently, the lower bound for the complement subtracted from the total gives the upper bound for A . This is why a coordinate pair can be treated without fixing the other $n - 1$ coordinates. $\square$

5.3 Codimension two and a single common profile

The pruning in Lemma 5.3 was against affine $(n - 2)$ -planes, rather than just angular balls. The following linear algebra is the reason for that choice.

Lemma 5.6 (Restriction of two direction equations). *Let a basis of $\mathbb{R}^{n+1}$ consist of bounded vectors $v(\xi_i)$ with inverse norm $L^{o(1)}$. Normalize the i th coefficient of another bounded vector $v(\xi)$ to one, assuming it and its inverse are $L^{o(1)}$, and denote two of its other coefficient ratios by a_j, a_k . A ball*

of radius L^{-u} in these two ratios pulls back, on the bounded frequency region, into an $L^{-u+o(1)}$ -neighborhood of an affine $(n-2)$ -plane. The same conclusion holds when both specified ratios are L^{-u} -small.

Proof. Let r_ℓ be the dual basis rows. Equations for ratios centered at (a_j, a_k) are the homogeneous equations $(r_j - a_j r_i) \cdot v = 0$ and $(r_k - a_k r_i) \cdot v = 0$. Their row map has smallest singular value $L^{-o(1)}$: the coefficients of r_j, r_k in any linear combination are the coefficients of that combination itself, and the primal and dual basis norms are subpower. Choose the center at a realized ratio in the ball, enlarging its radius by a fixed factor. Both rows then vanish at a bounded vector $v_0 = (2\xi_0, 1)$. If a linear combination is (a, b) , this implies $b = -2a \cdot \xi_0$, so $\|(a, b)\| \lesssim \|a\|$. Consequently the spatial restrictions of the two rows still have smallest singular value $L^{-o(1)}$. Their time-one slice is an affine $(n-2)$ -plane, and approximate equations give distance $L^{-u+o(1)}$ from it. For two small ratios one can center the equations at zero; they vanish at the i th reference direction, giving the same argument. This verifies both rank and its quantitative conditioning after restriction to the time-one hyperplane. $\square$

Lemma 5.7 (One common profile). *Almost surely all $n+1$ conditional profiles coincide:*

$$G_i = G'_i = G, \quad F = (n+1)G, \quad 0 \leq \dot{G} \leq 1, \quad G(1) = \frac{n}{n+1}, \quad (n+1)G(s) \geq ns. \quad (5.10)$$

Proof. In the exact reference basis, normalize the i th coefficient of the direction of T'_i to one. Replacement transversality and Cramer's rule bound this normalization and all ratios by subpowers. Include in the joint limiting law the clipped inverse-size exponents

$$r_j = \lim \min\{1, \max\{0, -\log_L |a_j|\}\}, \quad j \neq i,$$

with value one when a ratio vanishes. At most one of these n exponents can be positive. Otherwise two coefficients are small at a fixed positive-power scale, placing T'_i in a codimension-two frequency strip by Lemma 5.6. Conditional on X and the exact references, the alternative still has its original law. Conditional independence and (5.5) make this event have probability tending to zero.

On $\{r_j = 0\}$ compare profile increments using the j th coordinate. For fixed rational $0 < a < t/2$, the finite event $r_{j,L} < a/2$ implies $|a_j| > L^{-a/2}$. The approximate basis fixed only by the reference bins changes this to a lower bound L^{-a} for large L . Given T'_i, Z_j, C, D , the child position then has at most $L^{O_n(a)+o(1)}$ possibilities, since that coordinate parametrizes the tube with inverse slope at most $L^{a+o(1)}$. This permitted set of child labels is determined by the displayed (T'_i, Z_j, C, D) , uniformly over hidden exact reference directions. Apply restricted determination before taking limits, and then let $a \downarrow 0$. On $\{r_j = 0\}$ it gives

$$G_i(s+t) - G_i(s) - \ell'_i \leq I(Z_j \mid C, D) \leq G_j(s+t) - G_j(s) + O_n\left(\sum_k \ell_k\right).$$

The comparison uses only the indicated bin-defined coordinates; exact reference directions have not been silently added to the conditioning.

On dyadic exponent partitions with block size $b = b_*$, distribute each cost uniformly over its block. Its expected integral is $O_n(b)$. Dyadic summability makes all these cost densities tend to zero for almost every path and exponent. At simultaneous differentiability points, $\dot{G}_i \leq \dot{G}_j$ whenever $r_j = 0$. These ratio events belong to the entire sampled path, independently of the current block. All endpoint integrals equal $n/(n+1)$, so each such derivative inequality implies equality of the two profiles. Each profile thus agrees with at least $n-1$ others. Every equivalence class has at least n members, and two disjoint classes would need $2n > n+1$ members. There is a single class, proving (5.10). $\square$

Use family one and two specified off-axis ratios a_2, a_3 of T'_1 . Conditional on the exact X and exact references, redraw T'_1 from its original law. Let $h_j(u)$ be the limiting negative base- L logarithm of the mass of the radius- L^{-u} ball centered at its sampled a_j . The functions are nonnegative and nondecreasing and satisfy

$$h_2(u) + h_3(u) \geq c_1 u \quad (0 < u < 1/8, u \in \mathbb{Q}). \quad (5.11)$$

Indeed the joint ball pulls back to a codimension-two strip by Lemma 5.6, and the original conditional law obeys (5.5). Joint grid information is at most the sum of marginal grid informations by the chain rule and limiting conditioning. Centered-ball and grid exponents agree: if p_z is a sampled grid-cell mass and q_z the sum over its boundedly many neighbors, then

$$\sum_{z: q_z > L^a p_z} p_z \leq L^{-a} \sum_z q_z \lesssim L^{-a}.$$

Apply this to joint and marginal grids, with a boundedly finer grid where necessary. This is the “Angular marginal information” argument of OpenAI [24, Section 3]; Lemma 5.6 supplies its new higher-dimensional hypothesis.

5.4 A planar projection test with no frozen extra coordinates

We record the precise external projection statement.

Lemma 5.8 (Robust planar projection). *Fix $0 < d < 1$ and $\rho > 0$. There are $\varepsilon_{\text{pr}}, \eta_{\text{pr}} > 0$ such that, for sufficiently small Δ , the following holds. Let $A \subset B(0, 1) \subset \mathbb{R}^2$, and write $\mathcal{N}_\Delta$ for covering number at radius Δ . Suppose*

$$\Delta^{-2d+\eta_{\text{pr}}} \leq \mathcal{N}_\Delta(A) \leq \Delta^{-2d-\eta_{\text{pr}}}, \quad \mathcal{N}_\Delta(A \cap B(z, r)) \leq \Delta^{-\eta_{\text{pr}}} r^\rho \mathcal{N}_\Delta(A)$$

for $\Delta \leq r \leq 1$. If a probability σ on projection lines obeys $\sigma(B(V, r)) \leq \Delta^{-\eta_{\text{pr}}} r^\rho$ on the same range, then some $V \in \text{supp } \sigma$ satisfies

$$\mathcal{N}_\Delta(\pi_V A') \geq \Delta^{-d-\varepsilon_{\text{pr}}} \quad \text{for every } A' \subset A \text{ with } \mathcal{N}_\Delta(A') \geq \Delta^{\eta_{\text{pr}}} \mathcal{N}_\Delta(A).$$

The tolerance and gain may be decreased.

Proof. This is He [21, Theorem 1] with ambient dimension 2, projection dimension 1, and set exponent $2d$. Its Grassmannian nonconcentration condition is angular-ball nonconcentration in this case. The theorem gives the conclusion for all large subsets simultaneously, outside a direction set of probability a positive power of Δ . Decrease the tolerance to obtain the stated version. $\square$

Lemma 5.9 (Projection block test). *Fix $0 < d < 1$, $d_0 > 0$, and $\kappa_0 > 0$. A sufficiently small $\zeta > 0$, depending on these parameters and fixed n , has the following property. Fix rational $[s, s+b] \subset [0, 1]$, $b < 1/8$, $b_* \geq b$, and $j \in \{2, 3\}$. Let D_0 be the reference direction bins at precision L^{-b_*} , and D_1 additionally contain the bin of T'_1 . The following cannot hold on an event of positive limiting probability:*

- (i) $|G(s+b) - G(s) - db| \leq \zeta b$, and $G(s+u) - G(s) \geq d_0 u - \zeta b$;
- (ii) every cost for D_0, D_1 , the reference tubes, and T'_1 , at the endpoint and the tested prefixes, is at most ζb ;
- (iii) $h_j(u) \geq \kappa_0 u - \zeta b$.

The prefix tests need only use a fixed finite rational grid in $0 < u \leq b$ containing b , of mesh a sufficiently small fixed multiple of ζb . All error constants and relative meshes are independent of s, b, b_* .

Proof. Set $K = L^b$, and use the same reference-bin basis for D_0, D_1 . Let $Y_u = (Z_1, Z_j)$ at prefix u . Lemmas 5.5 and 5.7 give, with either bin conditioning,

$$I(Y_b \mid C, D_a) = 2db + O_n(\zeta b), \quad I(Y_u \mid C, D_a) \geq 2d_0u - O_n(\zeta b), \quad a = 0, 1. \quad (5.12)$$

Let $\tilde{a}_j$ be the alternative ratio determined by D_1 , and let Π be the interval label at resolution K^{-1} of $z_j - \tilde{a}_j z_1$. Given T'_1, C, D_1 , it has $L^{o(1)}$ possible values: the central line is part of the tube label, the slope error is $L^{-b_*+o(1)}$, and the relative width is $L^{s-1+o(1)} \leq L^{-b+o(1)}$. Moreover coordinate one parametrizes the full child position along this tube to subpower ambiguity. Thus

$$I(Y_b \mid \Pi, C, D_1) \geq I(Y_b \mid \Pi, C, D_1, T'_1) = I(X_{s+b} \mid C, D_1, T'_1) \geq db - O_n(\zeta b).$$

Since Π is determined by Y_b, D_1 , the chain rule and (5.12) imply

$$I(\Pi \mid C, D_1) \leq db + O_n(\zeta b). \quad (5.13)$$

Every statement so far conditions on parent and direction bins, not on the unused coordinates.

Pass the finite test to finite L with slack. It succeeds with probability at least a fixed $p_* > 0$ along a subsequence. Fix parent and reference-bin labels whose conditional success probability $p \geq p_*$, and write μ for that conditional law. No lower bound on the probability of this individual fiber is needed. Absorb fixed slack into $\alpha = C_n \zeta + o(1)$. If A_{bin} denotes the alternative bin, success implies the following bounds for the displayed labels whenever they have a successful realization:

$$\begin{aligned} K^{-2d-\alpha} &\leq \mu(Y_b = y) \leq K^{-2d+\alpha}, \\ K^{-2d-\alpha} &\leq \mu(Y_b = y \mid A_{\text{bin}} = a) \leq K^{-2d+\alpha}, \\ \mu(Y_u = y_u) &\leq K^\alpha L^{-2d_0u}, \\ \mu(\Pi = \pi \mid A_{\text{bin}} = a) &\geq K^{-d-\alpha}. \end{aligned} \quad (5.14)$$

Let A be the centers of all endpoint pair cells with a successful realization in this fixed fiber. The first line gives

$$pK^{2d-\alpha} \leq |A| \leq K^{2d+\alpha}. \quad (5.15)$$

The mass upper bound for a prefix divided by the endpoint mass lower bound counts the successful descendants in that prefix. A ball is covered by boundedly many cells at a comparable coarser tested prefix; the finite mesh costs $K^{O_n(\zeta)}$. Hence

$$\frac{\mathcal{N}_{K^{-1}}(A \cap B(z, r))}{\mathcal{N}_{K^{-1}}(A)} \lesssim p^{-1} K^{O_n(\zeta)+o(1)} r^{d_0}, \quad K^{-1} \leq r \leq 1. \quad (5.16)$$

The direct exponent is $2d_0$; weakening it allows all grid and short prefix errors to be absorbed.

Retain alternative bins whose conditional success probability is at least $p/2$. Their successful mass is at least $p/2$. For each such bin, let $A_a \subset A$ be its successful endpoint pair cells. The second and fourth lines of (5.14) give

$$|A_a| \geq (p/2) K^{2d-\alpha}, \quad \mathcal{N}_{K^{-1}}(\pi_{\tilde{a}_j(a)} A_a) \leq K^{d+O_n(\zeta)+o(1)}. \quad (5.17)$$

Here initially $\pi_a(z_1, z_j) = z_j - az_1$; normalizing its projection vector can only decrease image distances.

We now construct one nonconcentrating direction measure without paying an inverse probability for an individual alternative bin. Restrict actual samples to success and the retained bins, normalize by their mass (at least $p/2$), and push them to the projection line spanned by $(-\tilde{a}_j, 1)$. To bound a slope ball, first condition on the exact point and exact references. They determine the fixed parent and reference-bin fiber, and the alternative law is still the original one given the point. If the slope ball meets success, choose a successful alternative in it. Its centered-mass test (iii), at a fixed enlarged radius, bounds the original probability of the whole ball. Rounding to a coarser tested radius costs $K^{O_n(\zeta)+o(1)}$. Restriction only decreases this probability. Integrating over exact points and references, and finally normalizing the entire successful mass, gives

$$\sigma(B(V, r)) \lesssim p^{-1} K^{O_n(\zeta)+o(1)} r^{\kappa_0}, \quad K^{-1} \leq r \leq 1. \quad (5.18)$$

Subpower slope bounds, condition numbers, and the sine-angle formula justify the passage from slopes to projective lines. The approximate ratios have error $K^{-1}L^{o(1)}$. Rescale the coordinate range of A , of size at most $L^{o(1)}$, into a unit ball; covering numbers and all displayed bounds change by subpowers only.

Choose $0 < \rho < \min\{d_0, \kappa_0\}$. Lemma 5.8 applies to (5.15), (5.16), and (5.18) when ζ is sufficiently small. For every direction in $\text{supp } \sigma$, however, (5.17) provides a sufficiently large subset of the same A with projection smaller than the guaranteed gain. This is a contradiction. These are exactly the pair-cell, projection-cell, cost, and angular hypotheses of the “Projection block test” in OpenAI [24, Section 3]; the proof above verifies their dimensional transfer explicitly. $\square$

5.5 From the block test to a full-to-empty transition

The remaining profile argument is one-dimensional in the scale parameter. We give its hypotheses and mechanism explicitly, so that the replacement of the endpoint fraction $2/3$ by $n/(n+1)$ is visible. The angular sum bound (5.11) does not select one marginal that is nonconcentrated at every radius. The following finite choice of comparable scales supplies the angular bounds needed by the block test.

Lemma 5.10 (Finite angular alternatives). *For fixed $d \in (0, 1)$ and $d_0 > 0$, there is a fixed finite list of positive rational ratios, bounded away from zero, with this property. For every sufficiently small rational b_0 , almost every path admits a length b' from that list times b_0 , and $j = 2$ or 3 , satisfying the angular hypothesis of Lemma 5.9, with spare slack, for all tested block lengths in $[b'/4, b']$. Each alternative has a fixed positive angular slope, tolerance, and finite relative mesh, independent of b_0 .*

Proof. Use the block-test tolerance ζ_2 for angular slope $c_1/4$. Choose a positive rational $a \ll c_1\zeta_2$, and then a sufficiently reduced tolerance ζ_1 for slope $a/2$. Test $h_2(u) \geq au$ on a fine fixed relative grid from $c\zeta_1 b_0$ to b_0 . If every test passes, monotonicity fills gaps and nonnegativity handles smaller radii with the allowed additive error; choose $j = 2$, $b' = b_0$. Otherwise a tested v obeys $h_2(v) < av$. By monotonicity and (5.11),

$$h_3(u) \geq c_1 u - h_2(u) > c_1 u - av \geq (c_1/2)u \quad (2av/c_1 \leq u \leq v).$$

Choose $j = 3$, $b' = v$. The choice $a \ll c_1\zeta_2$ makes the omitted smaller radii harmless for every block length in $[v/4, v]$. Reducing cutoffs and meshes leaves spare slack. The finitely many v/b_0 are fixed and bounded away from zero. This proves the “Finite angular alternatives” lemma of OpenAI [24, Section 3] from precisely its two monotone-marginal and positive-sum hypotheses. $\square$

Lemma 5.11 (Binary derivative and finite alternation). *Almost surely $\dot{G} \in \{0, 1\}$ almost everywhere, and $E = \{s : \dot{G}(s) = 1\}$ agrees modulo null sets with a finite union of intervals. There is an interior*

full-to-empty transition. Consequently, for every fixed $\eta > 0$, some rational block $[s, s + b]$ with $0 < b < 1/8$ satisfies, on an event of positive probability,

$$G(s + b) - G(s) = b/2 + O(\eta b), \quad G(s + b/2) - G(s) = b/2 + O(\eta b). \quad (5.19)$$

Proof. We use the arguments named “Binary derivative”, “Balanced subintervals inside a hill”, and “Finite alternation and a full-to-empty jump” in OpenAI [24, Section 3]. Their abstract hypotheses here are: G is 1-Lipschitz and nondecreasing; (5.11) holds; Lemma 5.9 holds with fixed positive tolerances for each d, d_0, κ_0 ; and costs on disjoint block lists total $O_n(b_*)$ in expectation. All have been proved. The following details show that neither the number of coordinates nor the planar endpoint value enters these arguments.

If interior derivative values occur on a set of positive path-times-exponent measure, choose $d \in (0, 1)$ and $0 < d_0 < d$ so that a small neighborhood of d captures positive measure. At dyadic $b_0 \downarrow 0$, form a rational mesh of candidate blocks for every alternative in Lemma 5.10. Endpoints have spacing qb_0 , with fixed rational q below all relative tolerances; lengths range between fixed positive multiples of b_0 . Include each finite prefix mesh and use direction precision $b_* = b_0$. Each family has bounded overlap, so can be colored into finitely many disjoint lists. If $\mathcal{C}_I$ is the sum of all costs for candidate I , then

$$\sum_I \mathbb{E} \mathcal{C}_I \lesssim_n b_0. \quad (5.20)$$

For $R_{b_0}(\omega, s) = \sum_{I \ni s} \mathcal{C}_I(\omega)/b_0$, integration over s gives $\mathbb{E} \int R_{b_0} \lesssim_n b_0$. Dyadic summability forces $R_{b_0} \rightarrow 0$ almost everywhere. At differentiability points, all nearby increments and tested prefixes have their linear approximation with error $o(b_0)$. The angular alternative and the vanishing costs then give a successful block test. The rational meshes and test parameters form a countable family. For each fixed test, Lemma 5.9 makes its all-success event null; the preceding positive-measure interior-derivative event would force one of those null events to have positive probability. A countable cover of interior derivative values proves that the derivative is binary.

We also need a deterministic fact about this binary derivative. If $a < b$ are respectively density points of E and E^c , then for every sufficiently small $r > 0$ there is $[x, y] \subset (a, b)$ with

$$r \leq y - x \leq 2r, \quad G(y) - G(x) = (y - x)/2, \quad G(x + u) - G(x) \geq u/2 \quad (0 \leq u \leq y - x). \quad (5.21)$$

To see this, set $H(s) = G(s) - s/2$. The density assumptions give a closed hill strictly inside (a, b) , with equal endpoint heights and larger interior values. Among levels admitting a closed subinterval of length at least r with values above the level inside, choose the highest, by compactness. No component strictly above this level has length greater than r , since it would admit a higher such interval. The level set has no gap longer than r . Starting at the left endpoint, its first point at distance at least r lies at distance between r and $2r$, and gives (5.21). Finitely many disjoint ordered density pairs give disjoint balanced intervals at every sufficiently small common r .

If a positive-probability set of paths had arbitrarily many disjoint ordered full/empty density pairs, apply the angular alternatives with $d = 1/2$, $d_0 = 1/4$. For a successful angular size $b' \asymp b_0$, use (5.21) at a size yielding lengths in $[b'/3, 2b'/3]$, and round endpoints to the fixed fine relative mesh. Any prescribed number of disjoint hills gives that many spatially and angularly successful candidates for all sufficiently small b_0 . Each must fail a cost test, by Lemma 5.9; such a failure costs a fixed positive multiple of b_0 . Equation (5.20) bounds the expected number of failures uniformly in b_0 . Fatou’s lemma contradicts its tending to infinity on a positive-probability set. Thus almost every path has finitely many alternations among density points. Their ordered types, together with the Lebesgue density theorem, make E a finite union of intervals modulo null sets.

Finally (5.10) gives $|E| = n/(n+1) < 1$ and $(n+1)|E \cap [0, s]| \geq ns$. The latter prevents an initial empty interval and the former forces a later empty interval. There is therefore an interior full-to-empty jump. Choose a small block with midpoint within ηb of the jump, round its endpoints rationally, and keep its halves inside the adjacent full and empty intervals except for this error. The countable rational choices cover almost every path, so one has positive probability. This gives (5.19) with $b < 1/8$. $\square$

5.6 The growing-degree partitioning contradiction

Lemma 5.12 (No full-to-empty block). *For sufficiently small fixed $\eta > 0$, the positive-probability event in (5.19) is impossible.*

Proof. Fix the rational block and put $K = L^b$. Endpoint and midpoint increments for F are both $(n+1)b/2 + O_n(\eta b)$; the conditional endpoint increment given $T = T_1$ is $b/2 + O(\eta b)$. Pass to finite scale with slack and fix a parent $C = X_s = c$ having conditional success probability at least a fixed $p_* > 0$. Write

$$p_c(z) = \mathbb{P}(X_{s+b} = z \mid C = c), \quad q_T(z) = \mathbb{P}(X_{s+b} = z \mid T, C = c).$$

Call (T, z) good when it has some successful realization. Let $\mathcal{Z}$ be the good endpoint labels, represented by their centers in normalized parent coordinates. Since the bounds depend only on the displayed labels, goodness by existence preserves

$$K^{-(n+1)/2 - O_n(\eta)} \leq p_c(z) \leq K^{-(n+1)/2 + O_n(\eta)}, \quad q_T(z) \leq K^{-1/2 + O_n(\eta)} \quad \text{for good } (T, z). \quad (5.22)$$

The midpoint mass upper bound divided by the endpoint mass lower bound gives

$$|\mathcal{Z}| \leq K^{(n+1)/2 + O_n(\eta)}, \quad \#(\mathcal{Z} \cap B(z, CK^{-1/2})) \leq K^{O_n(\eta)}. \quad (5.23)$$

Good incidences have central-line error $\delta = K^{-1}L^{o(1)}$ in this bounded parent.

Choose small fixed $\varepsilon > 0$ and $P = K^{1/2 - \varepsilon}$. Polynomial partitioning in $\mathbb{R}^{n+1}$ gives a nonzero polynomial of degree $O_n(P)$ whose open complementary cells each contain at most

$$O_n(|\mathcal{Z}|/P^{n+1}) = K^{(n+1)\varepsilon + O_n(\eta)} \quad (5.24)$$

points; points on the zero set are assigned to the wall. This is the finite-point polynomial partition theorem [19, Theorem 4.1], whose polynomial ham-sandwich construction works in every fixed ambient dimension.

Take a wall of thickness a sufficiently large constant times δ . At midpoint resolution $r = K^{-1/2}$, Wongkew's hypersurface neighborhood-volume bound [33, Main Theorem], in a fixed ball slightly larger than the parent, gives the number of boxes meeting this wall as

$$O_n \left(\sum_{a=1}^{n+1} P^a r^{a-(n+1)} \right) = O_n(PK^{n/2}). \quad (5.25)$$

Indeed $\delta = o(r)$, so thicken to $O(r)$, use the volume estimate, and divide by r^{n+1} . The equality follows from $Pr = K^{-\varepsilon}$. The formula includes singular zero sets and all degree powers; the degree is allowed to grow here. Equations (5.22) and (5.23) bound the discarded good mass by

$$O_n(PK^{n/2})K^{O_n(\eta)}K^{-(n+1)/2 + O_n(\eta)} = K^{-\varepsilon + O_n(\eta)} = o(1), \quad (5.26)$$

once $\eta \ll_n \varepsilon$.

The wall contains the incidence error, so the nearby central-line point of each surviving endpoint belongs to the same open cell. A line not contained in the zero set crosses at most $O_n(P)$ cells; a line in it has no off-wall incidence. Draw T with its original marginal given $C = c$, and then z, z' independently with law q_T . If $g(T)$ is the q_T mass of good off-wall endpoints, then $\mathbb{E}g(T) \geq p_*/2$. Cauchy over the cells visited by T , followed by Jensen, gives

$$\mathbb{P}\{z, z' \text{ good off-wall in the same cell} \mid C = c\} \gtrsim_n P^{-1} = K^{-1/2+\varepsilon}. \quad (5.27)$$

For $|z - z'| \lesssim K^{-1/2}$ there are at most $K^{O_n(\eta)}$ possible good second endpoints by (5.23); each costs $K^{-1/2+O_n(\eta)}$ under q_T . Near pairs therefore contribute at most $K^{-1/2+O_n(\eta)}$.

For a farther pair, a line incident to both endpoints to error δ has direction in one of two caps of radius $K^{-1/2+o(1)}$, hence in caps of radius $K^{-1/3}$ for large L . Under the double sampling, the (T, z) marginal is the original joint law. Thus

$$\mathbb{P}(T \in \mathcal{A} \mid z, C = c) = \sum_{x: X_{s+b}(x)=z} \mathbb{P}(X = x \mid z, C = c) \mathbb{P}(T \in \mathcal{A} \mid X = x). \quad (5.28)$$

An angular cap is contained, on bounded frequency sets, in a codimension-two strip of comparable width. Use (5.5) at exponent $b/3 < 1/8$ in each term of this mixture. Its cap probability is at most $K^{-c_1/3+o(1)}$. For each first endpoint, the same-cell list has at most $K^{(n+1)\varepsilon+O_n(\eta)}$ second endpoints by (5.24); this list is fixed before drawing T . On a good second incidence, bound $q_T(z')$ by (5.22), then apply the cap bound to the remaining indicator and sum the list. The far-pair contribution is at most

$$K^{-1/2-c_1/3+(n+1)\varepsilon+O_n(\eta)+o(1)}.$$

No angular estimate conditional on the success event has been used. After division by (5.27), the near and far bounds are

$$K^{-\varepsilon+O_n(\eta)}, \quad K^{-c_1/3+n\varepsilon+O_n(\eta)+o(1)}.$$

Choose $\varepsilon \ll_n c_1$, then $\eta \ll_n \varepsilon$. Both tend to zero, contradicting (5.27). $\square$

Proof of Proposition 5.1. A countersequence yields Lemma 5.3. Its information profiles coincide by Lemma 5.7; Lemma 5.11 supplies a full-to-empty block with any prescribed small fixed tolerance. Lemma 5.12 forbids it. Hence no countersequence exists, which supplies a fixed $c > 0$ and a large-scale threshold as asserted. $\square$

6 Sparse packet arrays

The transverse saving gives a strict improvement when each individual packet encounters little measure. The broad–narrow organization follows Bourgain–Guth [8, Section 2] and its fractal Schrödinger implementation in Du–Zhang [12, Section 3]. Here the iteration must also preserve the packet moment condition under parabolic rescaling. Broad leaves gain from Proposition 5.1; at the stopping scale the packet moment condition improves the square-root-scale density in the refined fractal estimate.

Proposition 6.1 (Sparse estimate). *There is $c_s > 0$, depending only on the fixed dimension and packet boxes, with the following property. Fix $\tau > 0$ and $L^\tau \leq D \leq L$. Let U be a full-dimensional*

compact-profile array of scale L , with coefficient energy $\mathcal{E} = \sum_{\alpha} |c_{\alpha}|^2$. Let μ be a nonnegative measure on the fixed native time range, obeying

$$\mu(B(z, r)) \leq mr^n \quad (r \geq 1), \quad \int \left(1 + \frac{|x - y_{\alpha} - 2t\omega_{\alpha}|}{L}\right)^{-M} d\mu(x, t) \leq \frac{mL^{n+1}}{D}$$

for every nonzero packet label, as in Definition 4.1. For sufficiently many bounded profile derivatives,

$$\sum_{q \text{ unit}} \mu(q)^{\beta} \sup_q |U|^2 \leq C_{\tau} m^{\beta} L^{2n/(n+1)} D^{-c_s} \mathcal{E}, \quad \beta = \frac{2}{n+1}. \quad (6.1)$$

The same estimate holds for any fixed finite set of field derivatives if the original profiles have sufficiently many more derivatives. The exponent c_s is independent of τ ; the constant and the required finite differentiability may depend on τ . Restrictions to submeasures and fixed enlargements of the native time range are allowed.

6.1 The local functional and exact layers

Choose a dyadic $K \asymp L^v$, with $v > 0$ chosen at the end, and put

$$P = \frac{2}{1-\beta} = \frac{2(n+1)}{n-1} = q_{n-1}, \quad \mathcal{T}_{\beta}(V, \nu) = \sum_{Q \text{ a } K^2\text{-cube}} \nu(Q)^{\beta} \|V\|_{L^P(Q)}^2.$$

This is the functional from Section 4; its square root is a seminorm. For a fixed finite set of multiindices a , unit-cube Sobolev followed by Hölder gives

$$\begin{aligned} \sum_{q \subset Q} \mu(q)^{\beta} \sup_q |U|^2 &\lesssim \sum_a \sum_{q \subset Q} \mu(q)^{\beta} \|\partial^a U\|_{L^P(q)}^2 \\ &\leq \sum_a \mu(Q)^{\beta} \|\partial^a U\|_{L^P(Q)}^2. \end{aligned} \quad (6.2)$$

Here $2/P = 1 - \beta$, and the grids may be chosen nested. The cube Sobolev inequality controls closures, so boundary suprema cause no problem. Differentiating a compact atom produces finitely many atoms with the same labels and comparable energy: carrier frequencies are bounded, and native derivatives carry only bounded powers of L^{-1} . It suffices to estimate $\mathcal{T}_{\beta}$ for these arrays.

First restrict space to boxes of side L^2 . On the stipulated time range a compact packet meets only boundedly many fixed enlargements of these boxes, because its velocity is bounded. The retained coefficient energies therefore sum with bounded overlap.

Apply Lemma 4.4 to each localized compact array. Its exact Fourier layers are indexed by $(k, u) \in \mathbb{Z}^n \times \mathbb{R}$, with an integrable common weight $w(k, u)$ and exterior phase e^{iut/L^2} . After division by that weight, every subarray A has exact input squared norm at most $C \sum_{\alpha \in A} |c_{\alpha}|^2$. Its frequency support is within $O(L^{-1})$ of $\omega_{\alpha} + k/L$, and on a fixed slightly enlarged native time range its packets have any prescribed finite spatial decay order about the original trajectories and bounded native frequency seminorms. These properties follow from the native Fourier identity $u = \zeta + |\eta|^2$ and the summable phase-space Gram rows; they apply to every subarray with the same constant.

Minkowski for $\mathcal{T}_{\beta}^{1/2}$ returns a bound for the original array after integrating the layer weights. The exterior phase has modulus one and is removed before the iteration. For the derivative version, differentiate compact atoms before taking layers; no derivative of this exterior phase enters the iteration. Choose a small fixed $\gamma > 0$ later. Layers with $|k| > L^{\gamma}$ have arbitrarily small negative-power contribution by choosing the weight decay last. Indeed, after normalizing $m = \mathcal{E} = 1$ on one original spatial box, the relevant masses, cell counts, and field suprema are polynomial in L . The layer input bound and bounded frequency volume give the uniform supremum bound needed for this tail estimate. Henceforth one retained normalized layer is fixed.

6.2 A fixed gain on broad cubes

At packet scale l , group labels into $1/K$ -frequency bins and write $V = \sum_{\theta} V_{\theta}$. Every subfield is exact and has support within $O(l^{-1})$ of its label bin plus the common shift k/l . Our choices will guarantee $l \gg K^2$ and $|k|/l = o(1)$. Call a bin active on Q when

$$\|V_{\theta}\|_{L^P(Q)} \geq K^{-A} \|V\|_{L^P(Q)}, \quad (6.3)$$

where $A > n + 1$ is fixed sufficiently large. The $O(K^n)$ inactive bins have total norm at most $CK^{n-A} \|V\|_{L^P(Q)}$, which can be absorbed.

Lemma 6.2 (The geometric alternative). *For a fixed sufficiently large C_0 , either all active centers lie within C_0/K of an affine hyperplane, or $n + 1$ active bins have direction determinant at least cK^{-n} throughout their exact supports. A common frequency shift does not affect this alternative.*

Proof. Choose one center ξ_0 , and then greedily choose a center of largest distance from the affine span of the previous choices. If a maximum distance is at most C_0/K , all centers lie in that neighborhood of the current affine span, hence of a hyperplane. Otherwise the n Gram–Schmidt heights $d_1 \geq \dots \geq d_n$ all exceed C_0/K . In the resulting orthonormal basis, the column matrix of $\xi_j - \xi_0$ is upper triangular with diagonal d_j and every entry in row j bounded by d_j , by the greedy choice. Factoring out these row heights leaves a unit upper-triangular matrix with bounded entries and bounded inverse, with constants depending only on n . Thus the inverse original matrix has norm $O_n(d_n^{-1})$. Perturbing the vertices by $O(K^{-1})$ changes its determinant by a relative $O_n((Kd_n)^{-1})$, which is small for sufficiently large C_0 . Its determinant is at least $\prod_j d_j \gtrsim K^{-n}$. The exact support enlargements are smaller than these bin perturbations. The determinant of the spacetime directions $(2\xi_j, 1)$ is a fixed multiple of this simplex determinant. Common translation preserves it; unit normalization changes it only by bounded factors because the frequencies remain bounded. $\square$

Call the first case narrow and the second broad.

Lemma 6.3 (Broad gain). *There exists $c_b > 0$, depending only on the fixed frequency box and n , such that if $K \leq l^{c_b}$, then in every box of side $O(l^2)$,*

$$\sum_{Q \text{ broad}} \nu(Q)^{\beta} \|V\|_{L^P(Q)}^2 \lesssim m_0^{\beta} l^{2n/(n+1)-c_b} \mathcal{E}_0. \quad (6.4)$$

Here $\nu(B(z, r)) \leq m_0 r^n$ for $r \geq 1$, and the input squared norms of the total field and every subfield are at most $C\mathcal{E}_0$. The fixed C may affect the implicit constant, but the gain is independent of packet differentiability and decay orders.

Proof. If no positive c_b works, normalize $m_0 = \mathcal{E}_0 = 1$ and the fixed input norm bound to obtain a sequence with $l \rightarrow \infty$, $K = l^{o(1)}$, and broad contribution at least $l^{2n/(n+1)-o(1)}$. Polynomially small mass and norm levels are negligible by the polynomial number of cubes and uniform exact-field supremum bound. Pigeonhole a common active transverse tuple and dyadic levels $\nu(Q) \asymp \chi$, $\|V\|_{L^P(Q)} \asymp A_0$ on N cubes. Tuple choices cost $K^{O_n(1)}$ and level choices cost logarithms, all subpower here. We have

$$\chi^{\beta} A_0^2 N \geq l^{2n/(n+1)-o(1)}, \quad \chi \leq l^{o(1)}, \quad \chi N \lesssim l^{2n}. \quad (6.5)$$

Each chosen active field has supremum at least $K^{-A-(n-1)} A_0$ on some unit cube in Q , since $|Q|^{1/P} = K^{n-1}$. Pigeonholing the $n + 1$ relative integer offsets of these witnesses costs $K^{O_n(1)}$. Translate the fields separately by these fixed offsets so that their witnesses lie on a common unit

cube associated to each retained Q . These cubes are distinct; translations modulate exact inputs and preserve supports and norms. Lemma 5.2 applies with determinant $K^{-n} = l^{-o(1)}$, and gives

$$A_0^{2(n+1)/n} N \leq l^{o(1)}. \quad (6.6)$$

In particular the inverse-transversality constant is controlled; no unspecified dependence is used.

For completeness, take limiting base- l logarithms (x, y, z) of (χ, A_0, N) along a subsequence. The retained levels give polynomial bounds, so these subsequences exist. Equations (6.5) and (6.6) imply

$$x \leq 0, \quad x + z \leq 2n, \quad \frac{2(n+1)}{n}y + z \leq 0, \quad \beta x + 2y + z \geq \frac{2n}{n+1}.$$

They force equality throughout

$$\frac{2n}{n+1} \leq \beta x + 2y + z \leq \frac{2x+z}{n+1} \leq \frac{2n+x}{n+1} \leq \frac{2n}{n+1}.$$

Thus

$$\chi = l^{o(1)}, \quad N = l^{2n+o(1)}, \quad A_0 = l^{-e+o(1)}. \quad (6.7)$$

The attached witness cubes have $l^{o(1)}r^n$ counts for $r \geq 1$: their full disjoint K^2 -cubes, each of mass at least $l^{-o(1)}$, lie in a ball of radius $r + CK^2$ whenever their witnesses lie in a radius- r ball. This gives at most $l^{o(1)}(r + CK^2)^n \leq l^{o(1)}r^n$ witnesses. The translated fields and (6.7) now contradict Proposition 5.1. This proves a fixed positive gain; decrease it if necessary so the same c_b occurs in both the hypothesis and conclusion of the lemma. $\square$

6.3 Narrow transport and the measure on each branch

On a narrow cube apply the codimension-one decoupling inequality in Lemma 4.7 to the full sum of active bins. Absorb the inactive sum using (6.3), and then extend the right side to all bins, with their complete child fields. With any prescribed large N_1 , the squared-norm bound is

$$\|V\|_{L^P(Q)}^2 \lesssim_\varepsilon K^\varepsilon \sum_\theta \sum_{a \in \mathbb{Z}^{n+1}} (1 + |a|)^{-N_1} \|V_\theta\|_{L^P(Q+K^2a)}^2. \quad (6.8)$$

The decoupling hypotheses hold because the active labels are in an $O(K^{-1})$ hyperplane strip, the exact support enlargement is $O(l^{-1}) \ll K^{-1}$, and after removing the normal linear phase the normal quadratic contribution has vertical thickness $O(K^{-2})$. The exponent is exactly $P = q_{n-1}$. Smooth cutoff localization and bounded overlap of projected tangential caps, as in Lemma 4.7, allow arbitrary prescribed active subfields, including overlapping enlarged supports.

For a child bin centered at v_0 , use

$$\Phi_{v_0}(x, t) = \left(\frac{x - 2v_0t}{K}, \frac{t}{K^2} \right), \quad l' = l/K, \quad y'_\alpha = y_\alpha/K, \quad \omega'_\alpha = K(\omega_\alpha - v_0). \quad (6.9)$$

After removing the carrier, $V_\theta = K^{-n/2}V'_\theta \circ \Phi_{v_0}$. Native variables, coefficient energies, and phase-space counting remain unchanged. For the direct pushforward $\nu' = (\Phi_{v_0})_\# \nu$,

$$\nu'(B(z, r)) \leq CK^{n+1}m_0r^n, \quad \int \left(1 + \frac{|x' - y'_\alpha - 2t'\omega'_\alpha|}{l'} \right)^{-M} d\nu' \leq \frac{m_0l^{n+1}}{D}. \quad (6.10)$$

Indeed the inverse ball is covered by $O(K)$ balls of radius $O(Kr)$. The normalized transverse distance in the moment integrand is unchanged pointwise, and $K^{n+1}(l')^{n+1} = l^{n+1}$.

Lemma 4.8 gives, for each fixed shift a , a finite sum of child functionals against translates of this direct pushforward, with total factor

$$CK^{2-(n+2)\beta} = CK^{-\beta}. \quad (6.11)$$

To see the exponent, the Jacobian is K^{n+2} , the normalized field contributes K^{-n} in squared norm, and $2/P = 1 - \beta$; the product is $K^{(n+2)(1-\beta)-n}$. Group disjoint shifted parent images by the child K^2 -cube containing the image of their centers. Hölder within the group bounds the sum by total pushed-forward mass to power β times the norm on a bounded enlargement. Subdividing the enlargement introduces only finitely many neighboring cube translates. There is no factor for the number of parent cubes in a group.

We keep each shift and each finite neighbor choice as a separate weighted branch of the iteration. The shift coefficients in (6.8) are summable, uniformly in their truncation; the finite neighbor choices cost a fixed constant. Thus the total weighted coefficient energy of the children is at most a fixed constant times the parent energy. The bins themselves partition the packet labels. This bookkeeping permits each branch to retain a measure with parameter $CK^{n+1}m_0$, without aggregating all translated measures into one.

Truncate $|a| \leq L^\eta$ for a sufficiently small fixed $\eta > 0$. The omitted tail is negligible by choosing N_1 last: exact fields have polynomial global supremum bounds, and normalized relevant masses, cells, and depths are polynomially bounded. Each remaining child translate has displacement d with $|d| \lesssim K^2 L^\eta$. Its effect on the child's transverse displacement is $d_x - 2\omega'_\alpha d_t$. The child label velocities are bounded, and we impose

$$K^2 L^\eta \ll l' \quad (6.12)$$

at every existing child scale. The moment weight therefore changes by at most a fixed factor; the same fixed inflation of $m_1 = CK^{n+1}m_0$ bounds both balls and moments. Time shifts are $o(l')$, so a fixed slightly enlarged native time range is preserved through the bounded depth.

In particular, a depth- j branch with $h = K^j$, $l = L/h$, has

$$m_j \leq C^j h^{n+1} m, \quad (6.13)$$

and the narrow prefactor, before the arbitrarily small decoupling losses, is $C^j h^{-\beta}$. The main factors cancel exactly:

$$K^{2-(n+2)\beta} K^{(n+1)\beta} K^{-2n/(n+1)} = 1, \quad h^{-\beta} m_j^\beta l^{2n/(n+1)} \leq C^j m^\beta L^{2n/(n+1)}. \quad (6.14)$$

6.4 Stopping and the terminal square-root-scale gain

Stop just before a further division by K would take the packet scale below $D^{1/2}$. Every existing scale has $l \geq D^{1/2}$, and terminal scales satisfy

$$D^{1/2} \leq l < KD^{1/2}. \quad (6.15)$$

The depth $J = O(v^{-1})$ is fixed once v is chosen.

At any node work spatial box by spatial box at side $O(l^2)$. Retain the exact packet labels whose original label trajectories meet a fixed enlargement of the box. Their energies have bounded overlap over the boxes on the node time range, also for every child subarray. The exact layers have arbitrarily prescribed decay about these trajectories, uniformly after normalization. Discarded fields are therefore smaller than any chosen negative power of L , by phase-space counting and Cauchy, since $l \geq L^\tau$. Use the same retained labels for the total field and all its subfields. Ignore

polynomially tiny total-field levels before the activity tests, and choose localization errors still smaller; this preserves the broad lower bounds. Equivalently apply the tests anew to the localized exact array. Its total and subfield input norms are bounded by retained coefficient energy through Lemma 4.4. These observations justify summing Lemma 6.3 over spatial boxes and summing energies over each depth of the tree.

At a terminal node, keep only K^2 -cubes within transverse distance lL^η of at least one label trajectory; the others have negligible contribution by the same rapid decay. Restrict the node measure to these cubes. If a radius- l ball meets this retained support, choose one nearby trajectory from a cube it meets. Since $K^2 \ll l$ and velocities are bounded, every point of the ball is at normalized transverse distance $O(L^\eta)$ from that one trajectory. Its moment bound gives

$$\nu(B(z, l)) \leq CL^{O(M\eta)} \frac{m_j l^{n+1}}{D}. \quad (6.16)$$

The assertion also holds for fixed ball enlargements. One trajectory is used for each ball; there is no summation over nearby packets.

Normalize the node energy to one and sort retained cubes by $\nu(Q) \asymp m_j \chi$, where $\chi \lesssim K^{2n}$. Polynomially tiny levels are negligible by the polynomial cell count and exact-input supremum bound, leaving only $O(\log L)$ levels. In each Q choose a unit witness q_Q of largest field supremum. Since $|Q|^{2/P} = K^{2(n-1)}$,

$$\sum_{Q \text{ in the class}} \nu(Q)^\beta \|V\|_{L^P(Q)}^2 \lesssim K^{2(n-1)} (m_j \chi)^\beta \sum_Q \sup_{q_Q} |V|^2. \quad (6.17)$$

These witnesses need not carry measure. Their counts are controlled by their disjoint full cubes, each of mass comparable to $m_j \chi$. For $r \geq 1$ the count is at most $C\chi^{-1}(r + CK^2)^n \lesssim K^{2n}\chi^{-1}r^n$. At radius l , (6.16) gives the improved count $CL^{O(M\eta)}\chi^{-1}l^{n+1}/D$.

Apply Theorem 4.2 at $S \asymp l^2$ with parameters

$$\gamma_0 \lesssim K^{2n}\chi^{-1}, \quad \lambda_0 \lesssim L^{O(M\eta)}\chi^{-1}l^{n+1}/D.$$

Writing

$$a_n = \frac{4}{(n+1)(n+2)}, \quad \theta_n = \frac{2n}{(n+1)(n+2)},$$

its bound is $O_\varepsilon(S^\varepsilon)\gamma_0^{a_n}\lambda_0^{\theta_n}S^{\theta_n}$. The normalized weight cancels exactly because $\beta - a_n - \theta_n = 0$. Combining with (6.17) and restoring energy proves

$$\mathcal{T}_\beta(V, \nu) \lesssim_\varepsilon K^{C_n} L^{O(M\eta)+\varepsilon} m_j^\beta l^{2n/(n+1)} (l/D)^{\theta_n} \mathcal{E}_{\text{node}}, \quad (6.18)$$

where one may take $C_n = 2(n-1) + 8n/((n+1)(n+2))$, increasing it for fixed grid enlargements. Indeed the unsimplified l power is $(n+3)\theta_n = 2n/(n+1) + \theta_n$. The logarithmic level count has been absorbed in the freely chosen small power loss. The input localization above handles spatial supports larger than one l^2 -box.

6.5 Order of parameters and summation of the iteration

Proof of Proposition 6.1. It remains to make the positive gains dominate every loss uniformly in D . Bounded L is absorbed in the constant, and $\tau > 1$ has no unbounded admissible scales, so assume $0 < \tau \leq 1$. Fix first

$$a_* = \min\{c_b/2, \theta_n/2\}, \quad c_s = a_*/8. \quad (6.19)$$

A broad leaf gains $l^{-c_b} \leq D^{-c_b/2}$, and a terminal leaf gains, by (6.15),

$$(l/D)^{\theta_n} \leq K^{\theta_n} D^{-\theta_n/2}.$$

Thus every leaf gains D^{-a_*} before its small losses and the displayed terminal K factors. In particular this preliminary gain is independent of τ .

Choose $v > 0$ sufficiently small, with fixed strict margins, that

$$v < \frac{\tau c_b}{4}, \quad 2v < \frac{\tau}{4}, \quad (C_n + \theta_n)v < \frac{\tau a_*}{16}. \quad (6.20)$$

These imply $K \leq l^{c_b}$ at all broad nodes, and permit the terminal K factors to consume at most $D^{a_*/8}$ after fixed rounding constants are absorbed. The depth $J = O(v^{-1})$ is now fixed. Choose the decoupling and refined-fractal losses so that their total product along a path is at most $D^{a_*/8}$. Choose $\eta > 0$ next, depending on J, M as well, so that all padding losses, including (6.16), cost at most $D^{a_*/8}$, and

$$2v + \eta < \tau/2.$$

This proves (6.12) at every existing child scale, because $l' \geq D^{1/2} \geq L^{\tau/2}$. Choose $\gamma < \tau/8$, so that the retained layer shift satisfies $|k|/l = o(1)$ at every node. Finally choose the frequency-tail powers, spatial decay orders, and profile differentiability large enough to make all discarded errors negligible. Those last choices do not change c_b or c_s .

On a depth- j branch, (6.14) makes the transported quantity $m_j^\beta l^{2n/(n+1)}$ equal, up to C^j , to the root quantity after multiplication by the narrow prefactor. At a fixed depth, angular bins partition energies, separate shifts have a uniformly summable total weight, finite neighbor choices cost a fixed constant, and spatial localizations have bounded energy overlap. Thus the total weighted leaf energy at that depth is at most C^j times the original energy. Summing over the bounded number of depths costs only a constant depending on τ . The allocations above leave more than the saving D^{-c_s} .

For clarity, ignored errors are also uniform after the whole tree is summed. On each original spatial box normalize mass parameter and energy; there are polynomially many labels after localization, cells, shift choices after truncation, and branches at bounded depth. Their parameters are polynomially bounded. The tail orders were chosen last and therefore beat this entire fixed polynomial, relative to $m^\beta L^{2n/(n+1)} D^{-c_s} \mathcal{E}$. Minkowski over the integrable exact-layer weights, the Sobolev reduction (6.2), and the initial spatial energy sum now give (6.1), also for the prescribed derivatives. $\square$

7 A lossless weak estimate

The power saving of Theorem 6.1 is available when the sparsity parameter is a positive power of the packet scale. We now reach every $D \geq 1$ by a second induction. Its geometric step places most of a hypothetical large level set near affine plates. The frame estimate can then be applied in one fewer spatial dimension. The additional unit density parameter in the next theorem makes the small-density part of this induction contract.

Theorem 7.1 (Weak estimate with density and sparsity). *Fix $n \geq 3$, and use the compact arrays, measure conditions and fixed time range of Theorem 4.1. There are constants*

$$p > 2, \quad \sigma > 0, \quad 0 < 2a < 1 - \beta, \quad \beta = \frac{2}{n+1}, \quad e = \frac{n^2}{n+1},$$

depending only on the dimension and the fixed packet conventions, for which the following holds. Suppose that the array has coefficient energy E , that $1 \leq D \leq L$, and that the measure has ball and

packet-moment parameter m as in Theorem 4.1. Suppose also that

$$\mu(B(z, 1)) \leq m\chi, \quad 0 < \chi \leq 1,$$

and that its support is in a box of side $O(L^2)$ within the fixed time range. For every $g > 0$,

$$\mu\{z : |U(z)| \geq \chi^a g L^{-e}\} \leq C_{\text{ind}} m L^{2n} g^{-p} D^{-\sigma} E^{p/2}. \quad (7.1)$$

The constants are independent of L, D, m, χ, g and the particular array. The assertion also holds for a subarray with its original native profiles.

We prove the theorem with $E = 1$; multiplying all coefficients by $E^{1/2}$ gives the general statement. A zero energy or a zero measure parameter is harmless. Every recursive application will use original compact profiles, possibly after the unrotated cap rescaling already specified in Theorem 4.8. In particular, derivatives introduced in a nonrecursive localization will never become extra hypotheses at the next induction level.

The induction is on dyadic ranges of L . For a small exponent $\kappa > 0$ to be chosen below, the inner parameter at a fixed range of L is the number of doublings of D needed to reach L^κ . Thus an application at smaller L is an outer induction step, while an application at the same L and at least twice the value of D is an inner step. Angular rescaling will use the outer hypothesis; removing subarrays with smaller packet moments will use the inner hypothesis. Spatial support constants remain fixed: partition a larger box into a bounded number of boxes of the stipulated size, keep the compact packets meeting each, and sum their $p/2$ -powers of energy. The energy overlap is bounded. The time-range constant is fixed to contain the packet supports.

7.1 Order of choices and the easy ranges

Choose

$$\beta < b < \frac{2}{n}, \quad 2a = 1 - b + \delta, \quad \zeta = 1 - \beta - 2a = b - \beta - \delta > 0, \quad (7.2)$$

with $\delta > 0$ small. We will use

$$X = BD, \quad R = K^J \asymp X^u,$$

where K is dyadic, J is a fixed large integer, $u > 0$ is small, and B is a fixed large constant. Dyadic rounding changes only constants depending on J . We list the order of choices because both inductions use strict gains.

- (i) Fix b , then small δ and a range $2 < p \leq p_{\text{max}}$. Choose J large enough for Theorem 7.2. That lemma gives a fixed gain R^{-c_2} , with $c_2 > 0$ uniform as an exponent in this range of p . Its numerical constants need not be uniform as $p \downarrow 2$.
- (ii) Fix a small unit-density cutoff exponent $v_0 > 0$. The positive dimensional margin

$$n - \frac{1}{2} - e = \frac{n-1}{2(n+1)} > 0 \quad (7.3)$$

allows the power cutoff in the eventual frame application to be fixed at this stage. For this fixed cutoff, fix the normalized profile bounds and the envelope and derivative orders for Theorem 3.1, including fixed reserves for its nonrecursive localization. The inverse-transversality, coarse-tuple and witness costs are then R^{C_f} , with C_f independent of all geometric powers chosen below. Take u so small that $uC_f < c_s/4$, reducing it for any other fixed frame cost. Then choose $\sigma > 0$ much smaller than uc_2 and c_s .

- (iii) Choose the finite powers of X required for the inner sparsity parameter, the wall tests, the plate count and width, the final ratio L/h , and the tiny-density cutoff. The exponent in the eventual polynomial bound on g is also determined by these choices. Use $p_{\max}$ in any upper estimate involving the still unfixed p .
- (iv) Choose $\kappa > 0$ small enough that, in the remaining range $D < L^\kappa$, L dominates every specified power of X at large scales for fixed B . Also impose $\kappa\sigma \ll v_0\zeta$. Choose $p - 2 > 0$ only afterwards, sufficiently small relative to κ and to the eventual level exponent. Then choose the small power losses and the finite derivative orders needed to make absolute errors negligible. Finally choose B , the base scale, and C_{ind} , in that order.

The sparse estimate may require more derivatives of the original profiles after the geometric powers have been chosen. This later regularity reserve does not change the fixed frame inputs in step (ii), and hence does not change C_f . Its multiplicative constants are absorbed by the final base scale and induction constant.

By coefficient Cauchy–Schwarz and phase-space counting, $\|U\|_\infty \lesssim 1$. Hence a nonempty level set has

$$g \lesssim L^e \chi^{-a}. \quad (7.4)$$

For $D \geq L^\kappa$, Chebyshev and Theorem 6.1 give

$$\mu\{|U| \geq \chi^a g L^{-e}\} \leq C_\kappa m L^{2n} D^{-c_s} \chi^\zeta g^{-2}. \quad (7.5)$$

The ratio of this bound to the desired right side, apart from its constant, is at most

$$D^{-(c_s-\sigma)} \chi^\zeta g^{p-2} \lesssim L^{e(p-2)-\kappa(c_s-\sigma)} \chi^{\zeta-a(p-2)}.$$

It is bounded if $p - 2$ is small enough. In the range $D < L^\kappa$ and $\chi \leq L^{-v_0}$, Theorem 4.6 gives the same comparison with

$$L^{\epsilon+\kappa\sigma+\epsilon(p-2)} \chi^{\zeta-a(p-2)}.$$

Choose ϵ small after $\kappa, p - 2$; the positive density power makes this bounded as well. These estimates also handle bounded scales uniformly as $\chi \downarrow 0$ by enlarging the final constant. If $g < D^{-1}$, the total mass bound $\mu(\mathbb{R}^{n+1}) \lesssim m L^{2n}$ is enough, since $g^{-p} D^{-\sigma} \geq D^{p-\sigma} \geq 1$.

It remains to consider

$$\chi > L^{-v_0}, \quad D^{-1} \leq g \lesssim L^{e+av_0}, \quad D < L^\kappa. \quad (7.6)$$

In this range the threshold $\lambda = \chi^a g L^{-e}$ is bounded below by a fixed negative power of L . Every absolute approximation error below can consequently be assigned a sufficiently large fixed negative power of L .

7.2 An angular induction with curved narrow sets

Besides a neighborhood of an affine codimension-two plane, we must exclude a small quadratic sublevel set inside a hyperplane. The latter condition will detect the second fundamental form of a polynomial wall. All frequency bins in the following lemma are bins of the original unrotated grid. Selecting a bin means selecting its entire field.

Lemma 7.2 (Angular induction). *Assume the outer induction hypothesis in Theorem 7.1 and the active range (7.6). Fix a subarray of energy $E_0 \leq 1$. In every cube of a fixed spacetime grid of side R^4 , prescribe some of its frequency bins of side $1/R$. Suppose their centers are covered by $O((1 + \log X)^{C_0})$ sets, each of one of the following types:*

- (a) a strip of width $O(R^{-1})$ about an affine $(n-2)$ -plane;
- (b) the $O(R^{-1})$ -neighborhood of

$$\{v \in H \cap B : |Q_H(v)| \leq R^{-4}\},$$

where H is an affine hyperplane, B is a fixed bounded region, and Q_H has coefficient norm one in orthonormal coordinates on H with bounded origin.

The cover and the prescribed bins may depend on the macro cube. Let U_{sel} denote the resulting prescribed-bin sum on that cube. For any fixed $c > 0$,

$$\mu\{|U_{\text{sel}}| \geq c\lambda\} \leq CC_{\text{ind}}R^{-c_2}mL^{2n}g^{-p}E_0^{p/2} + C_NmL^{-N}. \quad (7.7)$$

Here N can be any prescribed fixed exponent, given sufficiently many initial profile derivatives. The exponent $c_2 > 0$ depends on the dimensional choices in (7.2), not on u or C_0 .

Proof. We give the geometric iteration, the measure transport, and the exponent calculation separately.

The narrow geometry. Assign every selected bin to one covering set. In a set of type (b), choose one approximating point for each assigned bin and sort the sizes of $|\nabla Q_H|$ there into dyadic classes m_1 , down to $1/R$; put all smaller gradients in the last class. There are $O(\log R)$ classes. Each assignment concerns the whole bin, not a subset of its packet labels.

Consider one class through the J iterations of ratio K . If m_1 is a sufficiently small constant, the normalization of Q_H , its bounded coefficients, and its small value at an approximating point force its quadratic part to have norm bounded below. Some component of ∇Q_H then has a linear part bounded below. The condition $|\nabla Q_H| \lesssim m_1$ puts this class in a strip of width $O(m_1 + R^{-1})$ within the $O(R^{-1})$ -neighborhood of H . For parent inverse widths

$$s \leq \frac{c}{Km_1},$$

this is a codimension-two strip at the next child precision.

For a dyadic class with a positive lower gradient bound, the other regime is

$$\frac{CK}{m_1} \leq s \leq \frac{R}{K}.$$

Choose an approximating point in the parent. Taylor expansion around it places all assigned descendants within

$$C(m_1^{-1}(s^{-2} + R^{-4}) + R^{-1}) \lesssim (Ks)^{-1} \quad (7.8)$$

of the tangent affine $(n-2)$ -plane to its level set in H . The last, small-gradient class never needs this second regime. If the gradient is bounded below by a fixed constant, only a bounded number of initial steps are outside the second regime. In general the gap between the two regimes is a factor $O(K^2)$, so at most a bounded number of ratio- K steps need a crude estimate.

Use the functional $\mathcal{T}_b$ of Theorem 4.8, with $P = 2/(1-b)$. Since $b < 2/n$, we have $P \leq q_{n-2}$. At every good step Theorem 4.7 applies after rescaling the width in (7.8) to $O(K^{-1})$. The exceptional steps use Cauchy–Schwarz. Consequently the whole angular iteration, excluding the transport factors, costs

$$C_\epsilon K^C R^\epsilon, \quad (7.9)$$

where the exponent C is independent of J . Constants depending on the fixed J are allowed. This distinction is essential: good-step losses multiply to R^ϵ , rather than K^{CJ} .

For this nonrecursive calculation one may smoothly truncate compact profiles in native spacetime Fourier variables at radius $L^{1/4}$. With sufficiently many fixed derivatives, every relevant subfield and rescaling has error smaller than any prescribed negative power of L . There are only polynomially many labels of packets meeting the domain. The choice of κ makes the fixed hierarchy $R = K^J$ small enough relative to L that the truncated supports satisfy the thickness and overlap hypotheses of the packet estimates at each step. Supremum estimates for these approximations use ordinary convolution. At the terminal step we return to the original compact child fields, so none of this truncation changes the induction's profile class. The absolute error estimate is justified below, including when the tested mass is small.

Initial measure functional and recombination. Let ν be the original measure restricted to the set on the left of (7.7), and write $M_0 = \nu(\mathbb{R}^{n+1})$. For each cover/class index, pad the enumeration where needed and restrict ν separately to the macro cubes. On a unit cube q , the density bound gives $\mu(q)^{1-b} \lesssim (m\chi)^{1-b}$. Hölder on the constituent unit cubes, followed by the weighted supremum estimate for the bandlimited approximation, bounds $\lambda^2 M_0$ by the sum of the initial $\mathcal{T}_b$ functionals, multiplied by

$$C(m\chi)^{1-b}(1 + \log X)^C.$$

The sum of unit-cube convolution weights inside a K^2 -cube is bounded by the enlarged Schwartz weight used for that cube. Thus the weighted versions of the transport inequality apply without a factor counting unit cubes. During each calculation the angular subset prescribed on the original macro is kept fixed, including in every shifted norm.

After J steps, the transport factors and (7.9) give

$$C_\epsilon K^C R^{\epsilon+2-(n+2)b}. \quad (7.10)$$

We explain why the macro restrictions can now be recombined. Write

$$A_{K,v}(x, t) = ((x - 2vt)/K, t/K^2).$$

For a fixed angular chain, class index, and complete grid-shift sequence, every step is a translated pushforward under A_{K,v_j} . The grids on the right of the transport inequalities are global and unrotated; rotations used to prove a strip inequality do not change these lists. Hence the composition is

$$\tau_{z_{\text{sh}}} \circ A_{R,V}, \quad V = v_1 + v_2/K + \cdots + v_J/K^{J-1}, \quad |V| \leq C_n, \quad (7.11)$$

with a translation z_{sh} independent of the original macro cube. The inverse image of a terminal K^2 -cube has diameter at most $C_n R^2 K^2 \leq C_n R^4$, regardless of the size of z_{sh} . It therefore meets only boundedly many original macros. For their transported measures this gives, on every terminal cube Q ,

$$\sum_{\mathcal{M}} \nu_{\mathcal{M}}(Q)^b \leq C_n \left(\sum_{\mathcal{M}} \nu_{\mathcal{M}}(Q) \right)^b.$$

The combined measure is dominated by one translated pushforward of ν , has mass at most M_0 , and has ball parameter

$$m' \asymp C m R^{n+1}. \quad (7.12)$$

Infinite shift sequences are summed with their summable weights; neither the mass, the ball constant, nor the number of occupied cubes depends on their absolute translation. Since assignments were by whole original $1/R$ -bins, the terminal field is the same full compact bin field, or zero, on every contributing macro. Thus the outer induction will charge each original bin energy once.

The terminal induction. Put $l = L/R$. In each terminal K^2 -cube with nonzero original child supremum, move its mass to a point arbitrarily close to that supremum and bound its P -norm by the supremum. This changes ball bounds by at most K^C . Sort the cube masses as $\nu_{\text{child}}(Q) \asymp m'd$. Unit covering of the inverse image, whose volume scales as R^{n+2} while (7.12) scales as R^{n+1} , gives

$$d \leq CK^C R\chi. \quad (7.13)$$

The moved measure has ball parameter $CK^C m'$, relative unit density at most $\min(1, Cd)$, and, after fixed constant inflation, admissible moment parameter $D = 1$. Its relevant time range is that of the original compact child field. Spatial boxes can be partitioned as in the theorem.

Apply the outer induction at l and integrate the weak L^p bound against the trivial mass bound M_0 . Since $p > 2$, the resulting L^2 integral is bounded by

$$CK^C C_{\text{ind}}^{2/p} (m')^{2/p} l^{4n/p-2e} d^{2a} M_0^{1-2/p} E_{\text{bin}}, \quad (7.14)$$

where E_{bin} is that bin's original coefficient energy. Indeed, integrating a distribution estimate $\nu\{|V| > t\} \leq \min(M_0, At^{-p})$ gives $\int |V|^2 d\nu \leq C_p A^{2/p} M_0^{1-2/p}$. Multiplication by $(m'd)^{b-1}$ returns to the mass-power functional; the fixed K -volume factor costs another K^C . The dyadic d sum converges at its lower end because

$$b - 1 + 2a = \delta > 0.$$

By (7.13), its total cost is $C(K^C R\chi)^\delta$.

To justify returning from the Fourier approximations even on cubes with zero original supremum, normalize $m = 1$. There are polynomially many occupied cubes, and concavity gives

$$\sum_Q \nu(Q)^b \leq (\#Q)^{1-b} M_0^b.$$

Choosing the fixed truncation order large enough makes the entire error, with all prefactors restored, at most $L^{-N'} M_0^b$ for any prescribed N' . Either this absorbs into $\lambda^2 M_0$, or, using $\lambda \geq L^{-C}$ from (7.6),

$$\lambda^2 M_0 \lesssim L^{-N'} M_0^b \implies M_0 \lesssim L^{-(N'-2C)/(1-b)}.$$

The latter is the asserted absolute negligible mass after choosing N' large. Restoring homogeneity supplies the factor m .

The gain. Sum the disjoint bin energies. The factors of m and χ before solving for M_0 are exactly $m^{2/p} \chi^{2a}$. Apart from $K^C R^\epsilon$ and fixed logarithmic powers, the exponent of R is

$$\Gamma(p, \delta) = 2 - (n+2)b + (n+1)(b-1+2/p) - 4n/p + 2e + \delta. \quad (7.15)$$

At $p = 2$, $\delta = 0$, direct simplification gives $\Gamma = \beta - b < 0$. Solving for M_0 multiplies this exponent by $p/2 > 0$. First choose δ and $p_{\text{max}} - 2$ small, then J large so that $K^C = R^{C/J}$ is paid by the strict margin, and finally choose the decoupling loss small. The remaining logarithmic factors are absorbed by increasing B for the fixed C_0 . This proves (7.7) with a fixed $c_2 > 0$. $\square$

7.3 Heavy bins and a polynomial wall

Suppose, towards a contradiction to Theorem 7.1, that

$$Y = \{|U| \geq \lambda\}, \quad M_Y = \mu(Y) > C_{\text{ind}} m L^{2n} D^{-\sigma} g^{-p}. \quad (7.16)$$

All constants used in the geometric construction below are independent of C_{ind} . Translations of the spatial box are immaterial. For a macro cube with center $z = (x, t)$, call a $1/R$ -bin heavy if

$$\sum_{\alpha \text{ in the bin}} |c_\alpha|^2 \left(1 + \frac{|x - y_\alpha - 2\omega_\alpha t|}{L}\right)^{-M} \geq W, \quad W = X^{-P_0} \frac{mL^{n+1}}{M_Y}. \quad (7.17)$$

The power P_0 will be chosen after an inner sparsity exponent P_1 .

Lemma 7.3 (Reduction to broad heavy incidences). *There is a subset $Y' \subset Y$ with $\mu(Y') \geq cM_Y$ such that, at each point of Y' :*

- (i) *the heavy-bin field has modulus at least $\lambda/2$;*
- (ii) *the heavy-bin centers are not contained in a strip of any fixed large constant times R^{-1} about an affine $(n-2)$ -plane, nor in a single quadratic neighborhood of the form in Theorem 7.2;*
- (iii) *every heavy bin supplies at least $W/2$ of its weighted coefficient square mass from trajectories within distance $X^{C_w}L$.*

In particular some n heavy bins have direction vectors $(2\omega, 1)$ whose n -volume is at least R^{-C} throughout the bins.

Proof. For each bin restrict $\mu|_Y$ to the macros where that bin is light. Since $R^4 \ll L$, the weight in (7.17) is comparable at the macro center and at each point of the macro. Therefore the sum of its coefficient energies times their packet moments on this restricted measure is at most CWM_Y .

Put $D_1 = X^{P_1}$. Split off the labels whose moment exceeds mL^{n+1}/D_1 . In each bin their total energy is at most $CD_1X^{-P_0}$; summing over the $O(R^n)$ bins only contributes a fixed power of R . The other labels satisfy the inner induction hypothesis at D_1 , on the restricted measure, with the unchanged original profiles. Apply it at threshold $cR^{-n}\lambda$. Disjoint bin energies and $p/2 > 1$ give total exceptional mass at most

$$CC_{\text{ind}}mL^{2n}g^{-p}R^{np}D_1^{-\sigma}. \quad (7.18)$$

Choose P_1 so that $2D \leq D_1$ and $unp_{\text{max}} - \sigma P_1 + \sigma < 0$ with a fixed spare margin. The later choice of κ ensures $D_1 \leq L$. If D_1 has already reached L^κ , the large- D case is available; otherwise this is a strict inner induction step.

For each split-off bin, use unrotated cap scaling and the outer induction at L/R . The ball parameter is CmR^{n+1} , the relative unit density is at most $\min(1, C\chi R)$, and the scaled level parameter is at least gR^{-C} . Consequently all split-off fields together contribute at most

$$CC_{\text{ind}}mL^{2n}g^{-p}R^C(D_1X^{-P_0})^{p/2}. \quad (7.19)$$

The fixed power R^C includes the bin count and threshold splitting; its exponent can use p_{max} . Choose P_0 sufficiently larger than P_1 so that this expression, and (7.18), are small fractions of the lower bound in (7.16) once B is large. The thresholds of the $O(R^n)$ light-bin fields sum to at most $\lambda/2$ when c is small. This proves (i).

On every adverse macro for (ii), the heavy-bin sum is exactly a prescribed-bin field of the type in Theorem 7.2. Its mass is at most the target bound times $CR^{-c_2}D^\sigma$, plus an absolute negligible error. Since $\sigma < uc_2$, increasing B makes this a small fraction. The construction may choose the adverse hyperplane and quadratic separately on each macro, as allowed by that lemma.

For (iii), the measure ball bound and the fixed time range give, by covering tube annuli,

$$\int_{|x-y_\alpha-2\omega_\alpha t|>X^{C_w}L} \left(1 + \frac{|x - y_\alpha - 2\omega_\alpha t|}{L}\right)^{-M} d\mu \leq CmL^{n+1}X^{-A_w},$$

where A_w can exceed P_0 by choosing C_w large (the fixed exponent M was chosen sufficiently large first). Sum with $|c_\alpha|^2$ and compare with $WM_Y = X^{-P_0}mL^{n+1}$. Markov's inequality removes only a small fraction of Y and ensures the stated contribution in every heavy bin. Finally, successive maximal affine heights among the remaining centers give n robustly independent directions; failure would place all centers in a forbidden codimension-two strip. Choosing the strip constant large makes the volume bound stable throughout the bins. $\square$

Pass to tube coordinates $(x, t)/L$ and replace the measure by $\nu = \mu/(mL^n)$ in those coordinates. The relevant domain has diameter $O(L)$ and

$$\nu(B(z, r)) \leq Cr^n \quad (r \geq 1).$$

Write $N_Y = \nu(Y)$. Formula (7.17) becomes

$$W = X^{-P_0}L/N_Y. \quad (7.20)$$

The incidence tubes have width X^{C_w} in these coordinates.

Lemma 7.4 (Incidence mass and a wall). *The preceding level set satisfies*

$$N_Y W^{n/(n-1)} \leq X^C, \quad N_Y \geq X^{-C}L^n. \quad (7.21)$$

Moreover, a subset of Y' of mass comparable to N_Y lies within distance X^C of the zero set of a polynomial of degree at most $X^{C_{\text{part}}}$ in tube coordinates.

Proof. For a robust n -tuple, project its directions onto an n -dimensional space where their determinant is at least R^{-C} . Such a space exists by their n -volume lower bound. A further polynomially fine angular subdivision makes the projection common to the tuple's families. Apply endpoint n -linear Kakeya to the projected tubes.

To relate its cube sum to ν , divide into columns of side comparable to the tube width. Each tube meets only polynomially many width boxes over a projected column, uniformly in its label; its projected direction is bounded away from zero by the determinant condition. Each box has ν -mass at most X^C . If $a_{i,j} \geq 0$ denotes the incident weight of family i in the j th vertical box, Hölder with n factors followed by the monotonicity of finite sequence norms gives

$$\sum_j \prod_{i=1}^n a_{i,j}^{1/(n-1)} \leq \prod_{i=1}^n \left(\sum_j a_{i,j} \right)^{1/(n-1)}.$$

The total coefficient weight in each family is at most one. The projected Kakeya bound, its polynomial transversality cost and the polynomial number of tuples therefore bound the mass seeing weight W from every family by $X^C W^{-n/(n-1)}$. This proves the first inequality in (7.21). Substitute (7.20) to obtain the second. We also have $N_Y \lesssim L^n$ from the ball bound.

The separation into cells and a polynomial wall follows the Fourier-restriction partitioning framework of Guth [17, Sections 3.2–3.4]. For completeness, use polynomial partitioning with parameter $P_2 = X^{C'}$ on $\nu|_Y$. It gives a polynomial of degree $O(P_2)$, $O(P_2^{n+1})$ sign regions, and mass at most CN_Y/P_2^{n+1} in each open region. The finite-set partitioning theorem follows from iterative polynomial bisection [19, Theorem 4.1]; its integrable-density form is Guth [17, Theorem 1.4]. For the present finite Borel measure, smooth it first and use that bisection argument. Normalize the coefficient vector of each of the finitely many factors separately and take a convergent subsequence. A compact subset off the limiting zero set has stable signs; its limiting mass obeys the same bound.

Exhausting an open sign region by such compact subsets proves the assertion for the original measure. Mass on the zero set is already part of the wall. Connectedness of sign regions is not required.

A central line not contained in the wall meets only $O(P_2)$ open sign regions. Let $E_{\mathcal{O}}$ be the total coefficient energy of labels whose line visits $\mathcal{O}$. Then $\sum_{\mathcal{O}} E_{\mathcal{O}} \lesssim P_2$. Regions with $E_{\mathcal{O}} > C_* P_2^{-n}$ have total mass at most $C N_Y / C_*$ by the region mass bound. Choose C_* large. Away from a width-enlarged wall, a point and its nearby incident line point are in the same region. In every remaining region the preceding projected Takeya argument with local energies bounds the off-wall mass of Y' by

$$X^C W^{-n/(n-1)} P_2^{-n^2/(n-1)} \leq X^{C''} N_Y P_2^{-n^2/(n-1)}.$$

The last inequality uses (7.20) and $N_Y \lesssim L^n$. Since

$$\frac{n^2}{n-1} - (n+1) = \frac{1}{n-1} > 0,$$

choosing C' large pays the number of regions. A fixed positive fraction of the mass must therefore be near the polynomial wall. $\square$

7.4 Why the wall is flat

Testing the second fundamental form on incident line directions is related to the flat-point argument of Guth–Katz [18, Lemma 3.3 and Corollary 3.4]. Here the incidences are approximate and the ambient dimension is higher; the quadratic narrow estimate supplies the additional directional exclusion needed to bound the full Hessian. We prove the quantitative reduction to affine plates below.

We now work in macroscopic coordinates, obtained by dividing tube coordinates by L . The domain is a fixed bounded box, and the wall and incidence errors are at most

$$\Delta \leq X^{C_4}/L. \tag{7.22}$$

The exponent C_4 is already fixed. All further enlargements in this subsection are fixed powers of X and will be small on the macroscopic scale by the later choice of κ .

We use the following two elementary consequences of fixed-dimensional semialgebraic complexity. Here a description of complexity X^C means that the number and degrees of its defining polynomials are bounded by such a power; their coefficients are unrestricted.

- (a) A bounded set of dimension at most $n-1$ and complexity X^C is covered by $X^{C'} \rho^{-(n-1)}$ boxes of diameter $O(\rho)$. To see this, use a generic shifted mesh. A box meeting a component that does not meet any face contains that component, so the component bound pays such boxes. Intersections with grid faces generically lower dimension, have the same type of complexity bound, and are counted by induction on dimension, with the zero-dimensional component bound as base case.
- (b) A bounded set of complexity X^C has a partition into $X^{C'}$ cells within which any two points can be joined by a path of length $X^{C'}$. Use a compatible cylindrical decomposition, including the enclosing box. Recursively lift paths in the base to a graph or to the continuous middle section of a sector, adding vertical segments inside a sector. The paths have polynomial description complexity in fixed dimension. Each bounded coordinate has polynomially many monotonicity pieces, so the length is bounded by their total variation. The paths may be taken continuous and piecewise smooth.

The required cylindrical decomposition, projection and bounded-variable quantifier-elimination bounds are standard; see Basu [1]. These bounds are polynomial in description size when dimension is fixed and do not depend on coefficient magnitudes. They control the number of pieces and path lengths, not graph derivatives. The latter are obtained next.

Lemma 7.5 (Regular graph pieces and derivative bounds). *Let Z be the square-free polynomial wall. It has a semialgebraic exceptional subset of dimension at most $n - 1$ and complexity X^C such that, outside its $C\rho$ -neighborhood, every relevant point of Z has a local graph coordinate with bounded slope and second and third graph derivatives bounded by a fixed power of ρ^{-1} . The estimates are independent of the polynomial coefficients. The mass of the discarded neighborhood in tube coordinates is at most $X^C \rho L^n$.*

Proof. Square-freeness implies that the singular set has dimension at most $n - 1$. Decompose the bounded regular part into polynomially many pieces with a specified omitted coordinate j , on which the implicit graph $x_j = \phi(\bar{x})$ has slope bounded by a dimensional constant. Choose a finite set of base directions sufficient to polarize tensors through order three. Refine the pieces by the signs, including zero, of the corresponding fourth directional derivatives of ϕ . Implicit differentiation makes these tests semialgebraic with polynomial complexity. Put all singular points, lower-dimensional pieces and frontiers into the exceptional set; use a strictly larger box so that its artificial boundary is harmless. The frontier dimension property gives the stated dimension bound.

At distance $C\rho$ from this set, a straight base segment of length $c\rho$ lifts in the same graph/sign piece. Indeed its lift has bounded speed. If maximal continuation stopped earlier, its Cauchy endpoint would be a frontier or a singular point within distance ρ , unless the implicit graph could be continued. Both alternatives contradict the stopping assumption. Global injectivity of the base projection is unnecessary.

On such a two-sided segment let $g(t) = \phi(u + te)$, so $|g'| \leq C$. The fixed sign of g''' makes g''' monotone. If $|g'''(0)| = A$, on one half-interval of length comparable to ρ its magnitude is at least A , with constant sign. The midpoint second difference of g' on that half-interval has magnitude at least $cA\rho^2$ and at most $4C$. Thus $A \lesssim \rho^{-2}$. Apply the same argument on shorter interior intervals, then a difference quotient for g' , to get $|g''| \lesssim \rho^{-1}$. Polarization bounds the full second and third tensors.

By property (a), the exceptional set is covered by $X^C \rho^{-(n-1)}$ boxes. Each becomes a ball of radius $O(\rho L)$ in tube coordinates and has ν -mass at most $C(\rho L)^n$. Their total mass is $X^C \rho L^n$. Choose $\rho = X^{-C_5}$ with C_5 large and require $\Delta \ll \rho$. By (7.21), this discards an arbitrarily small fraction of N_Y . $\square$

Every remaining point of the level set near the wall can now be attached to a regular $p \in Z$ at distance $O(\Delta)$, with the preceding slope and derivative bounds. No regularity of this attachment is needed; all bad events below can be described by existence of an eligible nearby p .

Lemma 7.6 (Line tests). *For any prescribed large A , choose a fixed C_6 sufficiently large. For every label, outside polynomially many macroscopic time intervals of total length at most X^{-A} , an incidence with any eligible p satisfies*

$$|v_j - \nabla\phi(p) \cdot \bar{v}| \leq X^{C_6}/L, \quad |\nabla^2\phi(p)[\bar{v}, \bar{v}]| \leq X^{C_6}/L, \quad v = (2\omega, 1). \quad (7.23)$$

The exponent bounding the number of intervals is independent of the sizes of A, C_6 .

Proof. Lift the projection of the nearby line position onto the local graph. The lift is $O(\Delta)$ from p , and its first and second derivatives differ from those at p by $O(\Delta X^C)$ by Theorem 7.5. For the macroscopic line $\ell(s)$ define

$$H(s) = \ell_j(s) - \phi(\bar{\ell}(s)).$$

At a relevant incidence, $|H| \lesssim \Delta$. Failure of (7.23) forces $|H'|$ or $|H''|$ to be at least cX^{C_6}/L , after increasing C_6 beyond the fixed derivative error powers.

Order the regular roots over the projected line parameter by cylindrical decomposition and refine by these inequalities and signs. This gives polynomially many smooth branch intervals. They are graph sections, not sectors, and their implicit derivatives are the actual derivatives of H . On an interval where $|H| \lesssim \Delta$ and a first derivative of fixed sign has magnitude at least q , the length is $O(\Delta/q)$. For a second derivative of fixed sign and magnitude at least q , a midpoint second-difference estimate bounds the length by $O((\Delta/q)^{1/2})$. Set $q = cX^{C_6}/L$ and use (7.22). Increasing C_6 makes the sum of lengths at most X^{-A} . Thresholds change coefficients, not degrees or the number of quantified variables, so the interval-count exponent does not increase with A, C_6 . Quantifying the existence of any eligible nearby point preserves this complexity bound and proves the last assertion as well. $\square$

The bad parameters of one label contribute at most

$$X^C(LX^{-A} + X^C) \quad (7.24)$$

to its enlarged tube's ν -mass: cover the bad intervals in tube coordinates by width boxes and use the ball bound. The prefactor and count exponents are fixed independently of A, C_6 ; the initial Δ can be enlarged to include the incidence width. Choose A large, then C_6 , and finally let L dominate the fixed powers. Sum (7.24) with $|c_\alpha|^2$ and compare with $WN_Y = LX^{-P_0}$. Markov removes only a small fraction of Y' , and ensures that each heavy bin supplies at least one label passing both tests. In fact it supplies positive weighted mass of such labels. All assertions are simultaneous at each retained wall point.

Proposition 7.7 (Affine plate covering). *A subset $Y_0 \subset Y'$ with $\mu(Y_0) \gtrsim M_Y$ is contained in a union of at most X^C affine hyperplane plates, each of thickness X^C in tube coordinates.*

Proof. At a retained regular point p , let $\mathbf{n} = (\mathbf{n}_x, \mathbf{n}_t)$ be a unit normal. The first line test gives

$$|\mathbf{n}_x \cdot 2\omega + \mathbf{n}_t| \lesssim X^{C_6}/L$$

for a supplied bounded frequency, so $|\mathbf{n}_x| \gtrsim 1$. The exact tangent-frequency hyperplane is consequently well conditioned.

Suppose the graph Hessian has norm greater than X^{C_7}/L . Normalize its quadratic form and restrict it via $\omega \mapsto (2\omega, 1)$ to that exact hyperplane. The resulting polynomial on the hyperplane has coefficient norm comparable to one. To verify this, choose coordinates on the tangent space in which its time-one slice has bounded offset. Its direction space and one bounded point span the tangent space with bounded condition number because $|\mathbf{n}_x| \gtrsim 1$. A homogeneous quadratic has the form

$$Q(y, t) = y^T C y + 2tb \cdot y + at^2;$$

its restriction to $t = 1$ records all of C, b, a . The bounded graph slope makes base projection another bounded isomorphism. This proves the coefficient comparison. Divide the restricted polynomial by its coefficient norm, so that it has norm exactly one as required in Theorem 7.2; this changes the following bounds only by fixed factors.

Move each supplied frequency to the exact tangent hyperplane. Its distance is $O(X^{C_6}/L)$, and by the second line test its normalized quadratic value there is at most

$$CX^{C_6-C_7} + CX^{C_6}/L.$$

Choose C_7 large relative to C_6, u and then make L dominant. This quantity is less than R^{-4} . Every heavy-bin center is within $O(R^{-1})$ of these frequencies, so all heavy bins are trapped in a forbidden quadratic neighborhood from Theorem 7.3. The contradiction proves

$$\|\nabla^2 \phi(p)\| \leq X^{C_7}/L. \quad (7.25)$$

This is where the quadratic narrow estimate is essential: tangent directions on a nonflat ruled quadric may span an affine hyperplane, but their quadratic values still detect its curvature.

For each fixed graph coordinate j , partition the exact semialgebraic subset defined by regularity, the bounded slope condition and (7.25) into cells with property (b) above. Use paths inside these cells, not paths inside the larger regular wall. Even if the base projection has several sheets, the implicit gradient

$$g_j(p) = -\partial_{\text{base}} P(p) / \partial_j P(p)$$

is a single-valued function of the wall point. Along a piecewise smooth path p in this subset, with base coordinate u ,

$$dg_j = \nabla^2 \phi_j \, du, \quad dp_j = g_j \cdot du.$$

For path length $D_0 \leq X^C$, these identities bound the gradient change by $X^{C_7} D_0 / L$ and the height error from the starting tangent plane by $X^{C_7} D_0^2 / L$. A projected loop reaching a different sheet obeys the same identities. Thus every cell lies within X^C / L of one affine plane in macroscopic coordinates. Add the $O(\Delta)$ attachment error and return to tube coordinates. The number of cells and the plate widths are both fixed powers of X , as claimed. $\square$

7.5 Assigning packets to plates

The plate covering is used only to organize the original compact packets. It does not change their profiles. Assign a label to a plate if its central line comes within distance X^{C_8} of that plate somewhere in the enlarged tube-coordinate domain and its normal velocity deviation for the same plate is at most X^{C_9} / L . If there are several choices, choose one. Thus the assigned coefficient energies are disjoint. Increase each assigned plate's width by a fixed power of X so that it contains the compact packet support in the domain: the normal displacement over time $O(L)$ is at most X^{C_9} , in addition to the initial distance and the fixed packet width.

Every unassigned label has moment on $\mu|_{Y_0}$ at most mL^{n+1}/D_1 , after choosing the powers in order. Indeed choose C_8 larger than the plate-width power and truncate its moment at distance $X^{C_8}/2$, paying the tail by the same annular estimate as in Theorem 7.3. If its line is ever this close to a plate but is unassigned, its normal velocity exceeds X^{C_9}/L . The time spent within that distance is at most

$$CL \frac{X^{C_8} + X^{C_{\text{plate}}}}{X^{C_9}}$$

for each plate, in tube coordinates. Cover those intervals by width boxes and sum over the polynomial number of plates. Choosing C_9 large makes the resulting moment smaller than the asserted bound. Inner induction at D_1 therefore removes the unassigned field on only a small fraction of Y_0 . The assigned sum is at least a fixed multiple of λ on a set of mass comparable to M_Y .

Choose a dyadic inverse cap width h , nested with R , such that

$$b_1 = L/h \asymp X^{C_{10}}. \quad (7.26)$$

The exponent C_{10} is larger than the preceding plate-count and width powers and the loss in (7.21). Intersections of two enlarged plates whose projective normals differ by at least $1/h$ have small

total mass. In tube coordinates such an intersection fits in a box with $n - 1$ lengths $O(L)$ and two remaining widths at most $X^C h$. Covering by balls of the latter radius gives mass at most $X^C L^{n-1} h$. The same bound, with a larger fixed power, pays all pairs of plates. By (7.21) and (7.26), this is negligible relative to N_Y .

Group the plates by a bounded-overlap grid of projective normals at resolution $1/h$, choosing labels disjointly. At a retained point only boundedly many clusters contribute, so one cluster field has modulus $\gtrsim \lambda$. Cluster energies partition a subenergy of the original array. Within each cluster all frequencies lie within C/h of a single affine frequency hyperplane. To see this, the assignment condition gives an almost tangent direction of bounded velocity for each plate, so the spatial part of its normal is bounded below; changing the normal by $O(1/h)$ changes its tangent-frequency hyperplane by the same order. A bounded spatial rotation, shear and modulation now put this hyperplane in tangential frequency coordinates of dimension $k = n - 1$. These transformations are used only in the following nonrecursive estimates.

For each cluster, subdivide labels in the original frequency axes into $1/R$ -bins and then $1/h$ -bins. On a macro cube where the cluster is high, call a coarse bin significant if its field has a witness there of modulus at least $cR^{-n}\lambda$. The nonsignificant fields sum to at most a small multiple of λ at a tested point.

Except for a small total mass, the tangential projections of the significant centers cannot be covered by $O(C_{11} \log X)$ strips of width C_*/R about affine $(k - 1)$ -planes. Here C_* is any sufficiently large fixed dimensional constant, and C_{11} is the later fixed band cutoff exponent. Such a cover, together with the cluster's C/h normal width, would put the original bins in codimension-two strips in full frequency space. Apply Theorem 7.2 to the cluster subarray on each adverse macro. Summing disjoint energies and using $p/2 > 1$ preserves the bound. Increasing B absorbs the fixed logarithmic power. All ensuing coarse-bin witnesses are allowed to be anywhere within $O(R^4)$ of their tested points.

7.6 Flattened packets and their two energy budgets

For a fine bin θ centered at v , use the original-axis map $F_\theta = A_{h,v}$. After removing its carrier, the fine-bin field is

$$h^{-n/2} U'_\theta(F_\theta z),$$

where U'_θ is a child array of size b_1 . Partition its native spacetime into smooth compact unit cells T , with a fixed enlargement T^* containing each cutoff support. The resulting packets have carrier v , width h and duration h^2 . Their sizes and associated densities can be taken as

$$s_{\theta,T} = Ch^{-n/2} \max_{|\alpha| \leq N_*} \sup_{T^*} |\partial^\alpha U'_\theta|, \quad (7.27)$$

$$d_{\theta,T} = (mh^{n+1})^{-1} \int (1 + \text{dist}(F_\theta z, T))^{-P_*} d\mu(z). \quad (7.28)$$

Here P_*, N_* are the fixed orders committed to in step (ii) of Section 7.1. The envelope order $P = P_*$ was chosen strictly above both $M + n + 2$ and the lower envelope order in Theorem 3.1. Use it both for the full square-envelope weight w_T and for the density kernel. Thus

$$\int w_T(z) d\mu(z) = mh^{n+1} d_{\theta,T},$$

before replacing densities by their dyadic band values. The order N_* includes the frame derivative order and the fixed reserves for the normal Fourier expansion and cell Sobolev estimates. Dividing each localized profile by $s_{\theta,T}$ then gives the same fixed low-order profile bound in every frame

application. Extra original regularity needed by the sparse estimate does not redefine these sizes or frame inputs.

Keep the cells needed on the physical domain and its macro enlargements. The pushed measure has ball parameter Cmh^{n+1} , so $d_{\theta,T} \lesssim 1$.

Lemma 7.8 (Energy budgets after flattening). *For each physical time bin of length h^2 , including its fixed enlargement,*

$$\sum_{\text{bin}} s_{\theta,T}^2 \lesssim h^{-n}. \quad (7.29)$$

There is also the global bound, summed over all clusters, fine bins, cells and time bins with bounded overlap,

$$\sum_{\theta,T} d_{\theta,T}^\beta s_{\theta,T}^2 \lesssim_B h^{-n} b_1^{2n/(n+1)} D^{-c_s}. \quad (7.30)$$

The first assertion is only a per-time-bin statement.

Proof. A physical time interval of length h^2 has bounded length in the child time variable, even though a full child packet has duration b_1^2 . At each such time, the Gram matrix of child atoms, also after the fixed derivatives in (7.27), has summable rows. Integration by parts gives decay in the scaled carrier separation $b_1(\omega - \omega')$, while compact spatial profiles localize intercepts. Transported phase-space counting bounds the number of labels at each scaled distance. All child frequencies are in a fixed bounded box, so differentiation introduces no positive power of b_1 . Consequently

$$\sum_{\theta} \|\partial^\alpha U'_\theta(\cdot, t)\|_2^2 \lesssim \sum_{\theta} E_\theta \leq 1$$

for the required derivatives. Unit-cell Sobolev, summed in space and over the bounded child-time interval, proves (7.29).

For the weighted bound apply Theorem 6.1, including its fixed derivative versions, to every full child array. The pushforward of μ has ball parameter Cmh^{n+1} and its packet moment at scale b_1 retains the same D , since $h^{n+1}b_1^{n+1} = L^{n+1}$. The choice $b_1 \asymp (BD)^{C_{10}}$ ensures that D is at least a fixed positive power of b_1 once D is larger than a constant depending on B ; the remaining bounded b_1, D cases cost only a constant depending on B, C_{10} . Also $D \leq b_1$ by the choice of C_{10} .

Convolve the pushed measure by the isotropic polynomial kernel used in (7.28), and restrict to the suitably enlarged time range. Its ball and packet-moment bounds are preserved: translation of a packet weight costs at most its fixed polynomial order, and $P_* > M + n + 2$ makes that cost integrable. The convolved measure has mass at least a constant times $mh^{n+1}d_{\theta,T}$ on the cells where the suprema are taken. The sparse estimate therefore gives (7.30), after division by the normalization in (7.27). Cluster energies are disjoint and enlarged physical time bins have bounded overlap, so this is a global sum. $\square$

7.7 Density bands and the frame estimate

The two budgets will serve different parts of the summation. Very small densities will pay the polynomial number of clusters and time bins, so their energies can be bounded one time bin at a time. For the remaining densities we will choose transverse coarse bins from one common density band. The corresponding density weight can then remain attached to each packet energy and be summed with the global weighted budget. We now construct these two kinds of tuples and verify the hypotheses of the frame estimate.

Write the densities in dyadic bands $d_\ell = C_d 2^{-\ell}$, $\ell \geq 0$, with fixed upper constant C_d . Only $O(\log h)$ indices occur on the needed cells. Indeed their distances from the measure support are

bounded by a fixed power of h , while (7.21) gives a polynomial lower bound on total mass; the strictly positive kernel in (7.28) then gives $d_{\theta,T} \geq h^{-C}$.

At a witness for each significant coarse bin, at least one band field has modulus at least

$$A_\ell = c' R^{-n} \lambda (1 + \ell)^{-2}. \quad (7.31)$$

This follows by threshold splitting, since $\sum_{\ell \geq 0} (1 + \ell)^{-2} < \infty$. Choose one such band for each significant bin. There are two possibilities:

- (i) A chosen band is tiny, $d_\ell < X^{-C_{11}}$. Starting from its center, complete it to a robust $n = k + 1$ tuple of significant coarse bins in tangential frequency space.
- (ii) All chosen bands are non-tiny. There are only $O(C_{11} \log X)$ colors. Some single color supplies a robust n -tuple.

Both statements follow from the strip-union exclusion: successive maximal affine heights can all be chosen at least C_*/R . If a color failed to supply such a simplex, it would lie in one strip. The determinant lower bound for a successful tuple is cR^{-k} throughout the bins when C_* is large enough.

Fix a cluster, a time bin of length h^2 , an ordered coarse tuple and its band indices ℓ_i , $1 \leq i \leq n$. Let M_0 be the mass of tested points assigned to this tuple, and let $E_i = \sum s_{\theta,T}^2$ over its i th part, with the fixed time enlargement. A tuple of mass at most mh^{2n-c_4} can be discarded, where $c_4 > 0$ is a small fixed dimensional constant. The total number of tuples is $X^C (\log h)^C$, so their relative discarded mass, by (7.21), is at most

$$X^C (\log h)^C (h/L)^{2n} h^{-c_4} = o(1).$$

This is one of the requirements imposed on the later κ .

For every remaining tuple, Markov and (7.28) retain at least half its mass where all full square envelopes are bounded by

$$B_i = C \left(\frac{mh^{n+1} d_{\ell_i} E_i}{M_0} \right)^{1/2}. \quad (7.32)$$

They remain comparable at every point of the macro containing the tested point, since $R^4 \ll h$. Thus they also bound the envelopes at the separate coarse-bin witnesses. Zero-energy cases need no consideration. By (7.29) and the mass cutoff,

$$B_i \lesssim h^{(1-2n+c_4)/2}, \quad A_{\ell_i}/B_i \geq h^{c_3} \quad (7.33)$$

for a fixed $c_3 > 0$. For the latter assertion, use $g \geq D^{-1}$, $\chi > L^{-v_0}$ and the margin (7.3). Choose c_4, v_0, c_3 as sufficiently small dimensional fractions of that margin; the later choice of κ pays every fixed X power in L/h , R and the threshold, and logarithmic factors are smaller than any fixed positive power of h . This verifies the frame cutoff without allowing a geometric power to determine u retroactively.

Lemma 7.9 (A tuple bound). *For each retained tuple,*

$$\lambda^2 M_0 \leq CR^C (1 + \sum_i \ell_i)^C m \chi^{1-\beta} h^{n+2n/(n+1)} (\prod_{i=1}^n d_{\ell_i} E_i)^{\beta/n} (\sum_{i=1}^n E_i)^{1-\beta}. \quad (7.34)$$

The R exponent is independent of the number of plate clusters and time bins and of C_{10}, C_{11} .

Proof. Partition the sheared normal spatial coordinate into slabs of width h , with bounded enlargements to accommodate witnesses. The projected measure in tangential spacetime has unit-cube mass at most $Cm\chi h$ on each slab, by unit covering in the normal coordinate.

Only nearby normal and time labels of compact packets contribute in a fixed slab and time bin. Recenter at the time-bin origin. The residual normal frequency is $O(1/h)$, so its drift during h^2 is $O(h)$. Projected tangential packets consequently have bounded phase-space multiplicity. Rotating the lattice changes only bounded multiplicities and comparable polynomial weights. The omitted normal and time indices have bounded normalized displacements, so the projected envelopes are bounded by a constant times the full envelopes.

In the normal coordinate divided by h , expand each cutoff profile in a Fourier series. The residual normal carrier and time phase have bounded normalized derivatives and can be included in that profile. The exterior Fourier phase has modulus one. For mode $m \in \mathbb{Z}$, the tangential packet amplitude and envelope gain $\rho_m \lesssim (1 + |m|)^{-A_*}$, with any required fixed A_* supplied by the original derivative order. Split thresholds by $\gamma_m = c(1 + |m|)^{-2}$. The frame cutoff improves rather than deteriorates under the factor γ_m/ρ_m .

Witnesses need not be on the tested point's projected unit cube. Translate the tangential spacetime packets by an integer offset, separately for each family, so that its witness lies in that cube. Each offset is $O(R^4)$, so summing all choices costs R^C . Counting, transversality and envelope comparability are preserved.

Apply Theorem 3.1, normalizing family i by B_i , and multiply the resulting cube count by $Cm\chi h$. Before summing modes, the right side contains the decay factors

$$\prod_{i=1}^n (\rho_{m_i}/\gamma_{m_i})^{4/(kn)} \quad \text{and, in its } j\text{th summand, } (\rho_{m_j}/\gamma_{m_j})^2.$$

Every family has a positive decay exponent, even when it does not index the energy summand. All independent mode sums therefore converge for fixed large A_* . Sum slabs at this stage, keeping the same global B_i from (7.32) in every slab and putting only its nearby packet energies in the linear sum. Each packet occurs in boundedly many enlarged slabs. We obtain

$$M_0 \leq CR^C m\chi h \prod_{i=1}^n (B_i/A_{\ell_i})^{4/(kn)} \sum_{i=1}^n \frac{h^{k+2} E_i}{A_{\ell_i}^2}. \quad (7.35)$$

Summing slabs after solving a nonlinear mass inequality instead would lose a multiplicity factor; (7.35) avoids it.

Substitute (7.32) and (7.31). Up to the displayed R and band-index factors, the result is

$$M_0^{1+2/k} \leq CR^C (1 + \sum_i \ell_i)^C m^{1+2/k} \chi h^{n+2+2(n+1)/k} \lambda^{-2-4/k} (\prod_i d_{\ell_i} E_i)^{2/(kn)} \sum_i E_i.$$

Raise both sides to $k/(k+2) = 1 - \beta$. The exponent of h becomes $n + 2n/(n+1)$ and the product exponent becomes β/n ; the λ exponent is exactly -2 . This is (7.34). $\square$

7.8 Summation and completion of the induction

Completion of the proof of Theorem 7.1. First sum the tuples with a tiny band. The per-bin bound $E_i \lesssim h^{-n}$ gives

$$\left(\prod_i E_i \right)^{\beta/n} \left(\sum_i E_i \right)^{1-\beta} \lesssim h^{-n}.$$

Here the polynomial number of clusters and the $O((L/h)^2) = X^{2C_{10}}$ time bins may be counted explicitly. Their total power of X is fixed before C_{11} . To sum the density bands, extend the range to all nonnegative indices and use

$$\sum_{\min_i d_{\ell_i} < X^{-C_{11}}} (1 + \sum_i \ell_i)^C \prod_i d_{\ell_i}^{\beta/n} \leq CX^{-\beta C_{11}/(2n)}. \quad (7.36)$$

Indeed assign a tiny index, bound the polynomial by a product of polynomials, and absorb each into half its exponential decay. The remaining series are geometric. There is no factor $\log h$ in (7.36). Choose C_{11} large enough to pay all the previous cluster, time-bin and tuple powers and the desired negative power of D .

For a common non-tiny band, the energy expression in (7.34) is bounded by

$$Cd_\ell^\beta \sum_i E_i.$$

Each packet has one density index, and can occur in only R^C coarse tuple and witness-offset choices. Normal slabs and enlarged time bins have bounded overlap. Cluster labels are disjoint coefficient subarrays. Hence this sum is charged directly to the global weighted bound (7.30), with no factor counting clusters or time bins. Also $\ell \leq C + C_{11} \log_2 X$, so its band factor is only $C(C_{11})(1 + \log X)^C$. The scale factors cancel as

$$h^{n+2n/(n+1)} h^{-n} b_1^{2n/(n+1)} = L^{2n/(n+1)}.$$

The choices of u and then B absorb R^{C_f} and the fixed logarithmic powers into part of the sparse gain. Combining both types of tuples, and recalling that a fixed fraction of M_Y remained, gives

$$\lambda^2 M_Y \leq C_B m \chi^{1-\beta} L^{2n/(n+1)} D^{-c_s/2}. \quad (7.37)$$

We now verify the promised polynomial level bound, before choosing $p - 2$. From (7.21), $M_Y \geq cm L^{2n} X^{-C_{\text{geo}}}$. Substitute this and $\lambda = \chi^a g L^{-e}$ in (7.37). Since $2n - 2e = 2n/(n+1)$, we obtain

$$g^2 \leq C_B \chi^\zeta X^{C_{\text{geo}}} D^{-c_s/2} \leq C_B X^{C_{\text{geo}}}.$$

Thus $g \leq C_B X^{C_g}$ with C_g fixed by geometry, independently of $\kappa, p - 2, C_{\text{ind}}$. This confirms the noncircular order of choices in Section 7.1.

Finally (7.37) and $\chi^\zeta \leq 1$ imply

$$M_Y \leq C_B m L^{2n} g^{-2} D^{-c_s/2} \leq C'_B m L^{2n} g^{-p} D^{-\sigma},$$

provided

$$C_g(p - 2) + \sigma < c_s/2.$$

This requirement is compatible with the easy-range inequalities, since $p - 2$ is chosen after all geometric exponents and after κ . All uses of induction removed small fractions of the hypothetical mass: their ratios were negative powers of X , or $R^{-c_2} D^\sigma$, fixed before choosing C_{ind} . Every remaining constant C'_B is independent of that induction constant. Taking $C_{\text{ind}} > C'_B$ contradicts (7.16). Both inductions close, proving Theorem 7.1. $\square$

8 Frequency summation and the simultaneous Gaussian trace

Put $s = n/(2(n+1))$. We first turn Theorem 7.1 into a short-time estimate for exact extensions. We then sum the frequencies on a single dyadic time tree. This order matters: maximizing separately at each frequency would give a stronger frequency norm than H^s . Throughout this section

$$S(t)f(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \xi - it|\xi|^2} \widehat{f}(\xi) d\xi$$

when $\widehat{f}$ has bounded support. For general $f \in L^2$, $S(t)f$ initially denotes its L^2 equivalence class.

8.1 Exact data and compact arrays

Lemma 8.1 (Compact layers for bounded-frequency data). *Let $g \in L^2(\mathbb{R}^n)$ have support in a fixed bounded frequency box, and let $L \geq 1$. On $|T| \leq CL^2$, its extension is a sum of compact arrays of scale L ,*

$$\mathcal{E}g(X, T) = \sum_{q \in \mathbb{Z}^n} U_q(X, T).$$

For any prescribed finite profile derivative order J and any $A > 0$, the arrays can be chosen with uniformly bounded C^J native profiles and coefficient energies

$$E_q \leq C_{A,J} \langle q \rangle^{-2A} \|g\|_2^2. \quad (8.1)$$

Each fixed q has uniformly bounded phase-space multiplicity. Finite coefficient truncations converge uniformly on the stipulated time range, and the sum in q also converges uniformly. Consequently any homogeneous weak L^p estimate for these compact arrays passes to $\mathcal{E}g$ without a power of L or a count of layers.

Proof. Here $\mathcal{E}g$ uses the same Fourier normalization as $S(t)$; fixed normalizing constants below are harmless. Choose a smooth partition $\sum_\theta \chi_\theta = 1$ on the frequency box, with bounded overlap and caps of side L^{-1} centered at ω_θ . In a fixed larger box $B \subset \mathbb{R}^n$ form

$$G_\theta(\zeta) = L^{-n/2} (\chi_\theta g)(\omega_\theta + \zeta/L),$$

extended by zero. Its Fourier-series coefficients $c_{\theta,m}$ satisfy

$$\sum_{\theta,m} |c_{\theta,m}|^2 \leq C \|g\|_2^2 \quad (8.2)$$

by Parseval and bounded overlap. If ψ_θ is a smooth bump equal to one on the support of this rescaled cap and supported inside B , then

$$(\chi_\theta g)(\omega_\theta + \zeta/L) = L^{n/2} \psi_\theta(\zeta) \sum_m c_{\theta,m} e^{-i\gamma m \cdot \zeta}$$

as an L^2 identity, where the fixed lattice constant $\gamma > 0$ depends only on B . In particular, no derivative of g has been taken. Writing

$$H_\theta(u, \tau) = (2\pi)^{-n} \int e^{iu \cdot \zeta - i\tau |\zeta|^2} \psi_\theta(\zeta) d\zeta,$$

the corresponding extension terms are

$$c_{\theta,m} L^{-n/2} e^{i(X \cdot \omega_\theta - T |\omega_\theta|^2)} H_\theta \left(\frac{X - \gamma L m - 2T \omega_\theta}{L}, \frac{T}{L^2} \right). \quad (8.3)$$

For bounded τ , integration by parts in ζ gives arbitrary spatial decay for H_θ and every fixed number of its derivatives, uniformly in θ and L .

Take $\rho \in C_c^\infty(\mathbb{R}^n)$ with $\sum_{q \in \mathbb{Z}^n} \rho(u - q) = 1$, and a fixed time cutoff equal to one on the required native time range. In the q th summand replace the intercept $\gamma L m$ by $\gamma L m + L q$. With the new native spatial variable u , the profile is

$$\rho(u) \chi(\tau) H_\theta(u + q, \tau).$$

Its C^J norm is at most $C_{A,J}\langle q \rangle^{-A}$. Extract this factor into the coefficient. The divided profile is compact and uniformly bounded in C^J , and (8.2) gives (8.1). A fixed q translates the entire intercept lattice by the same vector; hence phase-space multiplicity is unchanged.

For completeness, convergence holds in the pointwise form needed here. At any point of a fixed compact layer only $O(1)$ intercepts for each cap can contribute, and there are $O(L^n)$ caps. Cauchy–Schwarz therefore bounds the contribution of a coefficient tail by

$$C\langle q \rangle^{-A} L^{-n/2} L^{n/2} \left(\sum_{(\theta,m) \text{ in the tail}} |c_{\theta,m}|^2 \right)^{1/2}.$$

This tends to zero uniformly. Taking $A > n$ also gives uniform summation in q . Independently, the cap Fourier-series partial sums converge to g in L^2 on a bounded frequency set; Cauchy–Schwarz on that set makes their exact extensions converge uniformly. The two limits coincide.

Finally choose positive b_q with $\sum_q b_q = 1$ and $b_q \asymp \langle q \rangle^{-n-1}$. The elementary implication

$$\left| \sum_q U_q \right| > \lambda \implies |U_q| > b_q \lambda \text{ for some } q$$

passes a weak p estimate to the sum, with cost $\sum_q b_q^{-p} E_q^{p/2}$. This is bounded by $C\|g\|_2^p$ after choosing the fixed A sufficiently large. $\square$

Proposition 8.2 (Lossless short-time weak estimate). *Let $p > 2$ be the exponent in Theorem 7.1. If $N \geq 1$ and $f_N \in L^2$ has Fourier support in a fixed annulus dilated by N , then for every unit ball $\Omega \subset \mathbb{R}^n$ and every time interval I of length N^{-1} ,*

$$\left\| \sup_{t \in I} |S(t)f_N| \right\|_{L^{p,\infty}(\Omega)} \leq CN^s \|f_N\|_2. \quad (8.4)$$

The constant is independent of the centers of Ω and I .

Proof. Unitarity replaces f_N by $S(t_0)f_N$ and puts the left endpoint of I at zero. Spatial translation also preserves its norm and frequency support. Set $L = \sqrt{N}$, $X = Nx$, and $T = N^2 t$. The normalized bounded-frequency input is $g(\eta) = N^{n/2} \hat{f}_N(N\eta)$, up to the fixed Plancherel normalization. Thus

$$S(t)f_N(x) = N^{n/2} \mathcal{E}g(Nx, N^2 t), \quad \|g\|_2 \asymp \|f_N\|_2.$$

A measurable time graph $T = T(X)$ over the dilated spatial ball carries the measure obtained by pushing forward dX . Its mass in any spacetime ball of radius r is at most Cr^n , regardless of the regularity of the graph. Its native time range is bounded, its unit mass bound has $\chi = 1$, and Theorem 4.1 supplies the tube moment with $D = 1$ and a fixed measure constant.

Apply Theorem 7.1 to the layers in Theorem 8.1 and use the summable threshold splitting there. On the scaled graph the result is

$$\mu\{|\mathcal{E}g| > L^{-e} g_0\} \leq CL^{2n} g_0^{-p} \|g\|_2^p.$$

Returning to the original graph multiplies its mass by $N^{-n} = L^{-2n}$. The amplitude factor is exactly

$$N^{n/2} L^{-e} = N^{(n-e)/2} = N^{n/(2(n+1))} = N^s.$$

This proves the level estimate for every measurable graph. For a finite list of times, choose the first time achieving the maximum; that graph is measurable. Exhaust a countable dense set of times and use the continuity of bounded-frequency extensions to obtain (8.4). At bounded N , the same statement follows directly from the Fourier L^1 bound, so no lower scale is omitted. $\square$

8.2 One time tree for every frequency

The argument below is the common-tree mechanism of the planar companion [24, Section 9]. We give the full calculation in dimension n : its spatial rescaling is the point at which the endpoint exponent enters.

Lemma 8.3 (Dual square sum). *Fix measurable $t : \Omega \rightarrow [0, 1)$ and $w : \Omega \rightarrow \mathbb{C}$ with $|w| \leq 1$, where Ω is a unit ball. Let ψ be a smooth compactly supported annular cutoff. For $N = 2^j$, $j \geq 1$, and a half-open dyadic interval $I \subset [0, 1)$ set*

$$G_N(I)(\xi) = \psi(\xi/N) \int_{\{x \in \Omega : t(x) \in I\}} e^{ix \cdot \xi - it(x)|\xi|^2} w(x) dx, \quad d_j(I) = N^{-2s} \|G_N(I)\|_2^2.$$

Then

$$\sum_{j \geq 1} d_j([0, 1)) \leq C. \quad (8.5)$$

The same t and w are used for every j .

Proof. Write $m_I = |\{x \in \Omega : t(x) \in I\}|$ and let $\mathcal{D}_a$ be the intervals at *absolute* depth a . The pairing kernel for the squared Fourier norm has phase, after $\xi = N\eta$,

$$N(x - y) \cdot \eta - N^2(t(x) - t(y))|\eta|^2.$$

If two intervals in $\mathcal{D}_j$ have indices differing by a sufficiently large fixed constant, then $|t(x) - t(y)| \geq C/N$. Since $|x - y|$ is bounded and η lies in an annulus away from zero, the time term dominates the gradient. Repeated integration by parts gives any prescribed negative power of N for the kernel. Integrate this bound once over the entire set of far pairs; there is no factor equal to the number of pairs of intervals. The other pairs have bounded neighbor multiplicity, and Hilbert-space Cauchy–Schwarz gives

$$d_j([0, 1)) \leq C \sum_{I \in \mathcal{D}_j} d_j(I) + C2^{-2j}. \quad (8.6)$$

Let $J \subset I \in \mathcal{D}_j$ be at relative depth k , where $0 \leq k \leq j$, and put $r = 2^{-k}$. Its time width is r/N . Dualizing Theorem 8.2 and integrating a weak L^p function over a set of mass m_J gives

$$d_j(J) \leq C m_J^{2-2/p}. \quad (8.7)$$

Indeed $\int_A |F| \leq C_p |A|^{1-1/p} \|F\|_{p,\infty}$ for $p > 1$, and duality may be taken over unit L^2 inputs with the annular cutoff.

We also need the spatially localized bound

$$d_j(J) \leq C r^{2s} m_J. \quad (8.8)$$

Partition space into cubes of side r . For pairs outside a fixed number of neighbor offsets, $|x - y| \geq Cr$ whereas $|t(x) - t(y)| \leq r/N$. The spatial term now dominates the phase gradient, giving

$$|K_N(x - y, t(x) - t(y))| \leq C N^n (1 + N|x - y|)^{-n-2}.$$

Its spatial integral is bounded. Thus the normalized contribution of these pairs is at most $C N^{-2s} m_J \leq C r^{2s} m_J$, using $Nr \geq 1$. For neighbor pairs, Cauchy–Schwarz reduces to single-cube energies. On a cube Q of side r , put

$$x = x_Q + rX, \quad t = t_J + r^2 T, \quad \eta = r\xi.$$

The frequency is Nr , the time interval has length $1/(Nr)$, and the squared Fourier norm acquires a factor r^n . If $m_{J,Q} = |Q \cap \Omega \cap \{t \in J\}|$, a fixed cover of the rescaled unit cube by unit balls lets Theorem 8.2 bound this normalized energy by

$$Cr^{n+2s} \left(\frac{m_{J,Q}}{r^n} \right)^{2-2/p} \leq Cr^{2s} m_{J,Q}.$$

The last inequality uses $m_{J,Q} \leq r^n$ and $2 - 2/p > 1$. Summation proves (8.8). This remains valid at $Nr = 1$: the scale-one instance of Theorem 8.2 is the elementary bounded-frequency estimate. We never descend beyond $k = j$. Taking the geometric mean of (8.7) and (8.8) yields

$$d_j(J) \leq C2^{-sk} m_J^\alpha, \quad \alpha = \frac{3}{2} - \frac{1}{p} > 1. \quad (8.9)$$

Starting at each $I \in \mathcal{D}_j$, follow its heavier child for j generations, breaking ties by a fixed convention. Let $H(I, k)$ be the interval after k heavier-child steps, with $H(I, 0) = I$, and let $P(I, k)$ be the lighter child at the k th step; see Figure 2. Thus $H(I, j)$ is the terminal heavy interval. These lighter children and $H(I, j)$ partition I , including any atoms of $t_\# dx$, because intervals are half-open. Additivity of G_N , the triangle inequality, and weighted Cauchy–Schwarz with weights $2^{-sk/2}$ give

$$d_j(I) \leq C \sum_{k=1}^j 2^{-sk/2} m_{P(I,k)}^\alpha + C2^{-sj/2} m_I^\alpha. \quad (8.10)$$

Here the terminal mass was bounded by m_I .

For child masses $a \geq b \geq 0$,

$$(a+b)^\alpha - a^\alpha - b^\alpha \geq (2^\alpha - 2)b^\alpha. \quad (8.11)$$

For $b > 0$, this follows by writing $a/b \geq 1$ and differentiating $(u+1)^\alpha - u^\alpha - 1$; the case $b = 0$ is immediate. Sum the nonnegative deficits over a finite dyadic tree and telescope. Then pass to increasing finite trees. The result bounds the sum of the lighter-child α powers over any set of distinct parents by $m_{[0,1]}^\alpha / (2^\alpha - 2)$.

Fix a relative depth k . As $j \geq k$ and $I \in \mathcal{D}_j$ vary, the parents of $P(I, k)$ are distinct. Their absolute depth is $j + k - 1$, so different j give different levels; for the same j , different initial I have disjoint descendants. This is precisely why the same time tree must be used for all frequencies. Hence

$$\sum_{j \geq k} \sum_{I \in \mathcal{D}_j} m_{P(I,k)}^\alpha \leq C.$$

Also $\sum_{I \in \mathcal{D}_j} m_I^\alpha \leq m_{[0,1]}^\alpha$. Sum (8.10) first at fixed k and then in k ; both $2^{-sk/2}$ and the terminal factors $2^{-sj/2}$ are summable since $s > 0$. Finally sum the errors in (8.6). This proves (8.5). $\square$

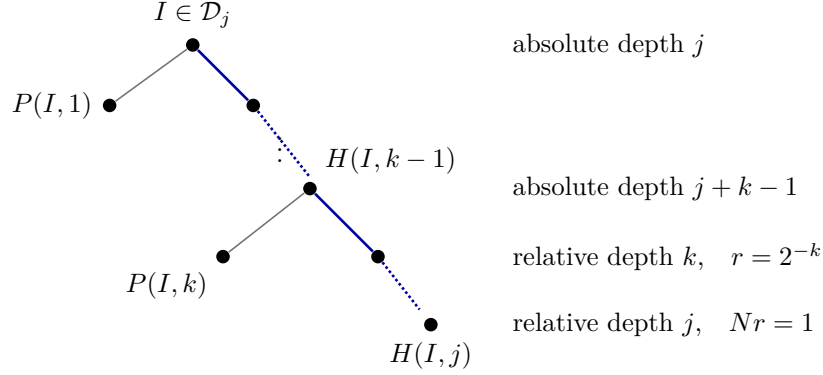


Figure 2: A heavier-child path starting at absolute depth j . Its lighter child at relative depth k has a parent at absolute depth $j + k - 1$. Holding k fixed and changing the frequency index j therefore selects distinct parent vertices in one common tree. The path stops at $k = j$, where the rescaled frequency is $Nr = 1$.

Theorem 8.4 (Local L^1 endpoint maximal estimate). *For compactly supported $\hat{f} \in L^2$ and every unit ball Ω ,*

$$\int_{\Omega} \sup_{0 \leq t < 1} |S(t)f(x)| \, dx \leq C \|f\|_{H^s(\mathbb{R}^n)}. \quad (8.12)$$

The constant is uniform under translation of Ω .

Proof. Take a smooth inhomogeneous Littlewood–Paley decomposition $f = f_{l_0} + \sum_{j \geq 1} f_{2^j}$ with bounded frequency overlap. The low-frequency part satisfies $\sup_{x,t} |S(t)f_{l_0}(x)| \leq C \|f_{l_0}\|_2$. For a fixed measurable graph $t(x)$ and phase $w(x)$, frequency Cauchy–Schwarz and Theorem 8.3 give

$$\begin{aligned} \left| \int_{\Omega} w(x) \sum_j S(t(x)) f_{2^j}(x) \, dx \right| &\leq C \left(\sum_j 2^{2js} \|f_{2^j}\|_2^2 \right)^{1/2} \left(\sum_j 2^{-2js} \|G_{2^j}([0, 1])\|_2^2 \right)^{1/2} \\ &\leq C \|f\|_{H^s}. \end{aligned}$$

Choose the annular cutoff in G_{2^j} equal to one on the corresponding Littlewood–Paley support. The first factor is the Sobolev square sum; no frequency logarithm or ℓ^1 sum has appeared. For a finite time list, select the first maximizer of the full field and its conjugate phase (zero where the field is zero). These are measurable and are common to every frequency. The preceding bound is then the integral of that finite maximum. Increase the lists to a dense countable subset of $[0, 1)$ containing zero. Continuity of the bounded-frequency field and monotone convergence prove (8.12). All kernel estimates depend only on spatial differences, giving the asserted translation uniformity. $\square$

8.3 One representative and all real times

To complete Theorem 1.1, we first construct a representative continuous in time and then remove Gaussian regularization uniformly in time on one full-measure spatial set. The following approximate-identity lemma will be applied to a countable family of error envelopes.

Lemma 8.5 (Gaussian approximation with uniform local integrals). *Let $M \geq 0$ be measurable and $A = \sup_z \int_{B(z,1)} M < \infty$. For*

$$K_a(y) = (4\pi a)^{-n/2} e^{-|y|^2/(4a)}, \quad a > 0,$$

the convolution $K_a * M$ is finite at every point. At every Lebesgue point x of M , it tends to $M(x)$ as $a \downarrow 0$.

Proof. For a fixed $\rho > 0$, cover each annulus $2^\ell \rho \leq |y| < 2^{\ell+1} \rho$ by $C(1 + 2^\ell \rho)^n$ unit balls. Uniformly in the center x ,

$$\int_{|y| \geq \rho} K_a(y) M(x - y) dy \leq C A a^{-n/2} \sum_{\ell \geq 0} (1 + 2^\ell \rho)^n e^{-4^\ell \rho^2 / (4a)}. \quad (8.13)$$

The series is finite for every $a > 0$, and for all sufficiently small a it is at most $C_\rho A a^{-n/2} e^{-\rho^2 / (8a)}$. The integral over a bounded ball is finite by local integrability, which proves the first assertion.

At a Lebesgue point set $H_x(r) = \int_{|y| < r} |M(x - y) - M(x)| dy = o(r^n)$. Given $\varepsilon > 0$, choose ρ so that $H_x(r) \leq \varepsilon r^n$ for $0 < r \leq \rho$. Radial integration by parts gives

$$\int_{|y| < \rho} K_a(y) |M(x - y) - M(x)| dy \leq \varepsilon \left(K_a(\rho) \rho^n + \int_0^\rho (-K'_a(r)) r^n dr \right) \leq C\varepsilon.$$

The complementary integral tends to zero by (8.13), together with the elementary Gaussian tail bound for the constant $|M(x)|$. Letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Proof of Theorem 1.1. Fix $n \geq 3$ and a complex-valued $f \in H^s(\mathbb{R}^n)$. Choose f_j with $\widehat{f_j} \in C_c^\infty$ and $\|f_j - f\|_{H^s} \leq 2^{-j}$. The successive differences satisfy a summable H^s bound. By Theorem 8.4, on every unit ball

$$\int \sum_j \sup_{0 \leq t < 1} |S(t)(f_{j+1} - f_j)(x)| dx < \infty.$$

A countable covering of $\mathbb{R}^n$ therefore gives a common full-measure set E_0 on which $S(t)f_j(x)$ converges uniformly for $0 \leq t < 1$. Call the limit $v(x, t)$ and set $v(x, t) = 0$ off E_0 . For every x , $v(x, \cdot)$ is continuous on $[0, 1)$; it is jointly measurable as a pointwise limit on a measurable set. The measurability of E_0 and of all the suprema just used can equally be read from the countable dense set of times, by continuity.

Fix an arbitrary real $t \in [0, 1)$. Unitarity gives $S(t)f_j \rightarrow S(t)f$ in L^2 , whereas the construction gives pointwise convergence to $v(\cdot, t)$ almost everywhere. Hence $v(\cdot, t)$ is an L^2 representative of $S(t)f$. For each $a > 0$ and $x \in \mathbb{R}^n$, the function $K_a(x - \cdot)$ lies in L^2 . Pairing it with this L^2 class and using Plancherel yields the everywhere-defined identity

$$U_a f(x, t) = (K_a * v(\cdot, t))(x). \quad (8.14)$$

The fixed t was arbitrary. Thus (8.14) holds for every triple (x, a, t) in the indicated ranges; no intersection of time-dependent exceptional sets has been taken. The convolution is absolutely convergent by Cauchy–Schwarz.

Let

$$M_j(x) = \sup_{0 \leq t < 1} |v(x, t) - S(t)f_j(x)|.$$

These functions are measurable because the supremum equals the one over rational times. They tend to zero on E_0 . Uniform convergence there and Fatou's lemma applied to $\sup_t |S(t)(f_\ell - f_j)|$ show that

$$A_j := \sup_z \int_{B(z, 1)} M_j(x) dx \leq C \|f - f_j\|_{H^s}. \quad (8.15)$$

All translated balls have the same constant by Theorem 8.4. The values on the null complement of E_0 do not affect this estimate. Intersect E_0 with the Lebesgue-point sets of the countably many M_j . By Theorem 8.5, at every point of this intersection $K_a * M_j(x) \rightarrow M_j(x)$ for every j .

For every y and every t , the definition of M_j gives the simultaneous inequality $|v(y, t) - S(t)f_j(y)| \leq M_j(y)$. Combining it with (8.14) gives

$$\sup_{0 < t < 1} |U_a f(x, t) - v(x, t)| \leq (K_a * M_j)(x) + M_j(x) + R_j(a), \quad (8.16)$$

where

$$R_j(a) := (2\pi)^{-n} \int |1 - e^{-a|\xi|^2}| |\widehat{f_j}(\xi)| \, d\xi \longrightarrow 0.$$

This last convergence is uniform in x, t since $\widehat{f_j} \in L^1$. First let $a \downarrow 0$ at a point of the common set. The limsup of the left side of (8.16) is at most $2M_j(x)$. Then let $j \rightarrow \infty$. This proves (1.2) for all real times in the interval at once.

Finally, intersect with the Lebesgue points of f and with the full-measure set on which $v(x, 0) = f(x)$, supplied by the fixed-time L^2 identity at zero. On the resulting E_f , $v(x, 0) = f^*(x)$ and time continuity gives $v(x, t) \rightarrow f^*(x)$ as real $t \downarrow 0$. The Gaussian limit in (1.2) is taken first, so these are exactly the asserted iterated limits. $\square$

References

- [1] Saugata Basu. Algorithms in real algebraic geometry: a survey. arXiv:1409.1534v1, 2014. URL <https://arxiv.org/abs/1409.1534v1>. Theorem 2.4 on p. 6 and the connecting-path discussion on p. 37 of this version.
- [2] Jonathan Bennett, Anthony Carbery, and Terence Tao. On the multilinear restriction and Kakeya conjectures. *Acta Mathematica*, 196(2):261–302, 2006. doi: 10.1007/s11511-006-0006-4. URL https://webhomes.maths.ed.ac.uk/~carbery/analysis/papers/Carbery/MRKBct_revised7.pdf. Theorem locators refer to the author manuscript dated April 2006.
- [3] Jean Bourgain. Some new estimates on oscillatory integrals. In Charles Fefferman, Robert Fefferman, and Stephen Wainger, editors, *Essays on Fourier Analysis in Honor of Elias M. Stein*, volume 42 of *Princeton Mathematical Series*, pages 83–112. Princeton University Press, 1995. URL https://omeka.ihes.fr/document/M_91_60.pdf.
- [4] Jean Bourgain. The discretized sum-product and projection theorems. *Journal d’Analyse Mathématique*, 112:193–236, 2010. doi: 10.1007/s11854-010-0028-x.
- [5] Jean Bourgain. On the Schrödinger maximal function in higher dimension. *Proceedings of the Steklov Institute of Mathematics*, 280(1):46–60, 2013. doi: 10.1134/S0081543813010045. URL <https://arxiv.org/abs/1201.3342>.
- [6] Jean Bourgain. A note on the Schrödinger maximal function. *Journal d’Analyse Mathématique*, 130:393–396, 2016. doi: 10.1007/s11854-016-0042-8. URL <https://arxiv.org/abs/1609.05744>.
- [7] Jean Bourgain and Ciprian Demeter. The proof of the l^2 Decoupling Conjecture. *Annals of Mathematics*, 182(1):351–389, 2015. doi: 10.4007/annals.2015.182.1.9. URL <https://annals.math.princeton.edu/2015/182-1/p09>.
- [8] Jean Bourgain and Larry Guth. Bounds on oscillatory integral operators based on multilinear estimates. *Geometric and Functional Analysis*, 21(6):1239–1295, 2011. doi: 10.1007/s00039-011-0140-9. URL <https://arxiv.org/abs/1012.3760v3>. Section locators refer to arXiv:1012.3760v3.

- [9] Lennart Carleson. Some analytic problems related to statistical mechanics. In John J. Benedetto, editor, *Euclidean Harmonic Analysis*, volume 779 of *Lecture Notes in Mathematics*, pages 5–45. Springer, Berlin–Heidelberg, 1980. doi: 10.1007/BFb0087666. URL <https://doi.org/10.1007/BFb0087666>.
- [10] Björn E. J. Dahlberg and Carlos E. Kenig. A note on the almost everywhere behavior of solutions to the Schrödinger equation. In Fulvio Ricci and Guido Weiss, editors, *Harmonic Analysis*, volume 908 of *Lecture Notes in Mathematics*, pages 205–209. Springer, Berlin–Heidelberg, 1982. doi: 10.1007/BFb0093289. URL <https://doi.org/10.1007/BFb0093289>.
- [11] Xiumin Du and Jianhui Li. L^p estimates of the maximal Schrödinger operator in $\mathbb{R}^n$. *Journal of Functional Analysis*, 288(3):110737, 2025. doi: 10.1016/j.jfa.2024.110737. URL <https://arxiv.org/abs/2404.10958v2>. Theorem locators refer to arXiv:2404.10958v2.
- [12] Xiumin Du and Ruixiang Zhang. Sharp L^2 estimates of the Schrödinger maximal function in higher dimensions. *Annals of Mathematics*, 189(3):837–861, 2019. doi: 10.4007/annals.2019.189.3.4. URL <https://arxiv.org/abs/1805.02775v3>. Theorem locators refer to arXiv:1805.02775v3.
- [13] Xiumin Du, Larry Guth, and Xiaochun Li. A sharp Schrödinger maximal estimate in $\mathbb{R}^2$. *Annals of Mathematics*, 186(2):607–640, 2017. doi: 10.4007/annals.2017.186.2.5. URL <https://annals.math.princeton.edu/2017/186-2/p05>.
- [14] Xiumin Du, Larry Guth, Xiaochun Li, and Ruixiang Zhang. Pointwise convergence of Schrödinger solutions and multilinear refined Strichartz estimates. *Forum of Mathematics, Sigma*, 6:1–18, 2018. doi: 10.1017/fms.2018.11. URL <https://arxiv.org/abs/1803.01720v2>. Article e14; theorem locators refer to arXiv:1803.01720v2.
- [15] Xiumin Du, Alex Iosevich, Yumeng Ou, Hong Wang, and Ruixiang Zhang. An improved result for Falconer’s distance set problem in even dimensions. *Mathematische Annalen*, 380:1215–1231, 2021. doi: 10.1007/s00208-021-02170-1. URL <https://arxiv.org/abs/2006.06833v2>. Theorem locators refer to arXiv:2006.06833v2.
- [16] Larry Guth. The endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture. *Acta Mathematica*, 205(2):263–286, 2010. doi: 10.1007/s11511-010-0055-6. URL <https://arxiv.org/abs/0811.2251v2>. Theorem locators refer to arXiv:0811.2251v2.
- [17] Larry Guth. A restriction estimate using polynomial partitioning. *Journal of the American Mathematical Society*, 29(2):371–413, 2016. doi: 10.1090/jams827. URL <https://arxiv.org/abs/1407.1916v3>. Theorem locators refer to arXiv:1407.1916v3.
- [18] Larry Guth and Nets Hawk Katz. Algebraic methods in discrete analogs of the Kakeya problem. *Advances in Mathematics*, 225:2828–2839, 2010. doi: 10.1016/j.aim.2010.05.015. URL <https://arxiv.org/abs/0812.1043v1>. Theorem locators refer to arXiv:0812.1043v1.
- [19] Larry Guth and Nets Hawk Katz. On the Erdős distinct distances problem in the plane. *Annals of Mathematics*, 181(1):155–190, 2015. doi: 10.4007/annals.2015.181.1.2. URL <https://annals.math.princeton.edu/2015/181-1/p02>.
- [20] Larry Guth, Alex Iosevich, Yumeng Ou, and Hong Wang. On Falconer’s distance set problem in the plane. *Inventiones Mathematicae*, 219:779–830, 2020. doi: 10.1007/s00222-019-00917-x. URL <https://arxiv.org/abs/1808.09346v1>. Theorem locators refer to arXiv:1808.09346v1.

- [21] Weikun He. Orthogonal projections of discretized sets. *Journal of Fractal Geometry*, 7(3): 271–317, 2020. doi: 10.4171/JFG/92. URL <https://arxiv.org/abs/1710.00795v2>. Theorem locators refer to arXiv:1710.00795v2.
- [22] Sanghyuk Lee. On pointwise convergence of the solutions to Schrödinger equations in $\mathbb{R}^2$. *International Mathematics Research Notices*, 2006:1–21, 2006. doi: 10.1155/IMRN/2006/32597. Article ID 32597.
- [23] A. Moyua, Ana Vargas, and Luis Vega. Schrödinger maximal function and restriction properties of the fourier transform. *International Mathematics Research Notices*, 1996(16):793–815, 1996. doi: 10.1155/S1073792896000499.
- [24] OpenAI. Endpoint convergence for the planar Schrödinger equation. OpenAI Math Release preprint [OAI:Endpoint-convergence-for-the-planar-Schrodinger-equation-September-24-2026](#), 2026.
- [25] Yucheng Pan, Wenchang Sun, and Jiheng Tan. Pointwise convergence of Schrödinger operators in Bessel potential spaces. arXiv:2605.25833v2, 2026. URL <https://arxiv.org/abs/2605.25833v2>. Version dated 26 May 2026.
- [26] Shalender Singh and Vishnu Priya Singh Parmar. Repairing the refined-decoupling proof of the 5/4 planar pinned Falconer theorem. arXiv:2608.28711v1, 2026. URL <https://arxiv.org/abs/2608.28711v1>. Theorems 1.1–1.2, Proposition 2.1, and Appendix A; planar ambient dimension.
- [27] Peter Sjögren and Per Sjölin. Convergence properties for the time-dependent Schrödinger equation. *Annales Academiæ Scientiarum Fennicæ, Series A I Mathematica*, 14:13–25, 1989. URL <https://www.acadsci.fi/mathematica/Vol14/vol14pp013-025.pdf>.
- [28] Per Sjölin. Regularity of solutions to the Schrödinger equation. *Duke Mathematical Journal*, 55(3):699–715, 1987. doi: 10.1215/S0012-7094-87-05535-9.
- [29] Terence Tao. A sharp bilinear restriction estimate for paraboloids. *Geometric and Functional Analysis*, 13(6):1359–1384, 2003. doi: 10.1007/s00039-003-0449-0. URL <https://arxiv.org/abs/math/0210084>.
- [30] Terence Tao. Sharp bounds for multilinear curved Keakey, restriction and oscillatory integral estimates away from the endpoint. *Mathematika*, 66(2):517–576, 2020. doi: 10.1112/mtk.12029. URL <https://arxiv.org/abs/1907.11342v2>. Theorem locators refer to arXiv:1907.11342v2.
- [31] Terence Tao and Ana Vargas. A bilinear approach to cone multipliers. II. applications. *Geometric and Functional Analysis*, 10(1):216–258, 2000. doi: 10.1007/s000390050007.
- [32] Luis Vega. Schrödinger equations: pointwise convergence to the initial data. *Proceedings of the American Mathematical Society*, 102(4):874–878, 1988. doi: 10.1090/S0002-9939-1988-0934859-0.
- [33] Richard Alexander Wongkew. Volumes of tubular neighbourhoods of real algebraic varieties. *Pacific Journal of Mathematics*, 159(1):177–184, 1993. doi: 10.2140/pjm.1993.159.177. URL <https://msp.org/pjm/1993/159-1/pjm-v159-n1-p10-s.pdf>.